$\mathbf{L}^p$-boundedness of the Bochner-Riesz operator

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Abstract

In this paper, we give a new approach to the Bochner-Riesz summability. As a result, we show that the Bochner-Riesz operator $\mathbf{S}^\delta, 0 < \mathbf{Re}\delta < \frac{1}{2}$ is bounded on $\mathbf{L}^p(\mathbb{R}^n)$ for $\frac{n-1}{2n} \leq \frac{1}{p} \leq \frac{n+1}{2n}$.

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1 Introduction

A classical problem in harmonic analysis is to make precise the sense of Fourier inversion formulae, for given $f \in \mathbf{L}^p(\mathbb{R}^n)$. One natural way of formulating such identities is via the summability method due to Bochner and Riesz. This assertion leads us to study for the $\mathbf{L}^p$-boundedness of Bochner-Riesz operator, defined as

$$\mathbf{S}^\delta f(x) \;=\; \int_{\mathbb{R}^n} e^{2\pi \mathbf{i} x\cdot\xi}\widehat{f}(\xi)\left(1-|\xi|^2\right)^\delta_+ d\xi, \qquad \mathbf{Re}\delta \geq 0. \tag{1. 1}$$

At $\delta = 0$, we revisit on the famous ball multiplier problem. $\mathbf{S}^0$ is bounded only on $\mathbf{L}^2(\mathbb{R}^n)$ proved by Fefferman [7].

⋄ Throughout, $\mathfrak{B} > 0$ is regarded as a generic constant whose value depends on its sub-indices.

⋄ We always write $\mathbf{c} > 0$ for some large constant.

$\mathbf{S}^\delta f$ in (1. 1) can be written as a convolution

$$\begin{aligned} \mathbf{S}^\delta f(x) &= \pi^{-\delta}\Gamma(1+\delta)\int_{\mathbb{R}^n} f(x-u)\Omega^\delta(u)du, \\ \Omega^\delta(u) &= \left(\frac{1}{|u|}\right)^{\frac{n}{2}+\delta} \mathbf{J}_{\frac{n}{2}+\delta}\Big(2\pi|u|\Big) \end{aligned} \tag{1. 2}$$

where Γ and $\mathbf{J}$ denote Gamma and Bessel functions. See (A. 7)-(A. 8).

Moreover, the estimate in (A. 5) implies

$$\left|\Omega^\delta(u)\right| \leq \mathfrak{B}_{\mathbf{Re}\delta}\, e^{\mathbf{c}|\mathbf{Im}\delta|} \left(\frac{1}{1+|u|}\right)^{\frac{n+1}{2}+\mathbf{Re}\delta}.$$

Observe that for $\mathbf{Re}\delta > \frac{n-1}{2}$, the kernel of $\mathbf{S}^\delta$ is an $\mathbf{L}^1$-function in $\mathbb{R}^n$. We thus have

$$\left\|\mathbf{S}^\delta f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\delta}\, e^{\mathbf{c}|\mathbf{Im}\delta|} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad 1 \leq p \leq \infty. \tag{1. 3}$$

When $0 < \delta \leq \frac{n-1}{2}$, a counter example given by Herz [15] shows that $\mathbf{S}^\delta$ is not bounded on $\mathbf{L}^p(\mathbb{R}^n)$ for either $p \leq \frac{2n}{n+1+2\delta}$ or $p \geq \frac{2n}{n-1-2\delta}$.

Bochner-Riesz Conjecture *Let* $\mathbf{S}^\delta$ *defined in (1. 1) for* $0 < \delta \leq \frac{n-1}{2}$. *We have*

$$\left\|\mathbf{S}^\delta f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\delta\, p} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \tag{1. 4}$$

if and only if

$$\frac{n-1-2\delta}{2n} < \frac{1}{p} < \frac{n+1+2\delta}{2n}. \tag{1. 5}$$

In 1972, Carlson and Sjölin [4] proved the conjecture for $n = 2$. A year later, Fefferman [6] observes the restriction estimate:

$$\left\{\int_{\mathbb{S}^{n-1}} \left|\widehat{f}(\xi)\right|^2 d\sigma(\xi)\right\}^{\frac{1}{2}} \leq \mathfrak{B}_p \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad 1 < p < \infty \tag{1. 6}$$

implying $\mathbf{S}^\delta, \delta > 0$ bounded on $\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^n)$. Here, $d\sigma$ denotes the Lebesgue measure on $\mathbb{S}^{n-1}$.

Consequently, the Tomas-Stein restriction theorem implies the $\mathbf{L}^p$-norm inequality in (1. 4) for p belonging to (1. 5) with an extra condition of $p \geq \frac{2n+2}{n-1}$ or $p \leq \frac{2n+2}{n+3}$. See Tomas [23] or Chapter IX of Stein [18]. Today, the connection between Fourier restriction theorem and Bochner-Riesz summability is extensively explored. For instance, Tao [21] proves (1. 4)-(1. 5) implying the Restriction conjecture on $\mathbb{S}^{n-1}$. On the other hand, Carbery [5] shows that such implication can be reversed if the unit sphere is replaced by a paraboloid.

A number of remarkable results have been accomplished in this broad area, for example by Bourgain [1]-[2], Bourgain and Guth [3], Guth [8]-[9], Tao and Vargas [22], Guo-et-al [11], Guo, Wang and Zhang [12], Gressman [13]-[14], Lee [16] and Wolff [25].

Our main result is stated in below.

Theorem One *Let* $\mathbf{S}^\delta$ *defined in (1. 1) for* $0 < \mathbf{Re}\delta < \frac{1}{2}$. *We have*

$$\left\|\mathbf{S}^\delta f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\delta}\, e^{\mathbf{c}|\mathbf{Im}\delta|} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad \frac{n-1}{2n} \leq \frac{1}{p} \leq \frac{n+1}{2n}. \tag{1. 7}$$

From (1. 3) and (1. 7), by applying Stein interpolation theorem [19], we obtain (1. 4)-(1. 5).

In order to prove **Theorem One**, we next introduce another convolution operator, denoted by $\mathbf{I}^\alpha$ for $0 < \mathbf{Re}\alpha < 1$ whose Fourier transform is given in terms of cone multipliers having negative orders. The $\mathbf{L}^p$-boundedness this operator implies **Theorem One**.

Furthermore, we develop two different types of dyadic decomposition respectively on the frequency space and the physical space. Every consisting partial operator will be split into two after a **pairing formulation** between these dyadic decompositions. Each one of them satisfies the desired $\mathbf{L}^p$-estimates.

1.1 Cone multiplier of negative orders

Let $\mathbf{S}^\delta, \mathbf{Re}\delta > 0$ defined in (1. 1). Our key observation is

$$\left(1-|\xi|^2\right)_+^\delta = 2\delta \int_0^1 \left(\frac{1}{\tau^2-|\xi|^2}\right)_+^{1-\delta} \tau d\tau, \qquad \mathbf{Re}\delta > 0.$$

Let $0 < \mathbf{Re}\delta < \frac{1}{2}$. The cone multiplier $\left(\frac{1}{\tau^2-|\xi|^2}\right)_+^{1-\delta}$ having a negative order has been previously investigated by Gelfand and Shilov [10].

Denote $1-\delta = \alpha$ for $\alpha \in \mathbb{C}$. Λ^α is a distribution defined in $\mathbb{R}^{n+1}$ by analytic continuation from

$$\mathbf{Re}\alpha > \tfrac{n-1}{2}, \qquad \Lambda^\alpha(x,t) = \pi^{\alpha-\frac{n+1}{2}}\Gamma^{-1}\left(\alpha-\tfrac{n-1}{2}\right)\left(\frac{1}{t^2-|x|^2}\right)_+^{\frac{n+1}{2}-\alpha}.$$

For $\frac{1}{2} < \mathbf{Re}\alpha < 1$, the Fourier transform of Λ^α agrees with the function

$$\widehat{\Lambda}^\alpha(\xi,\tau) = \pi^{\frac{n-1}{2}-2\alpha}\Gamma(\alpha)\left\{\left(\frac{1}{\tau^2-|\xi|^2}\right)_-^\alpha - \sin\pi\left(\alpha-\tfrac{1}{2}\right)\left(\frac{1}{\tau^2-|\xi|^2}\right)_+^\alpha\right\} \tag{1. 8}$$

whenever $|\tau| \neq |\xi|$. See appendix B.

Let $\varphi \in C_o^\infty(\mathbb{R})$ be a smooth cut-off function such that $\varphi(t) = 1$ if $|t| \leq 1$ and $\varphi(t) = 0$ for $|t| > 2$. Define

$$\widehat{\phi}(\xi) = \varphi\left(\frac{2}{3}|\xi|\right) - \varphi\left(3|\xi|\right). \tag{1. 9}$$

Note that $\mathbf{supp}\widehat{\phi} \subset \left\{\xi \in \mathbb{R}^n : \frac{1}{3} < |\xi| \leq 3\right\}$ and $\widehat{\phi}(\xi) = 1$ for $\frac{2}{3} < |\xi| < \frac{3}{2}$.

In order to prove **Theorem One**, we consider

$$\mathbf{I}^\alpha f(x) = \int_{\mathbb{R}^n} e^{2\pi \mathbf{i} x\cdot\xi}\widehat{f}(\xi)\widehat{\phi}(\xi)\left\{\int_0^1 \widehat{\Lambda}^\alpha(\xi,\tau)\tau d\tau\right\} d\xi. \tag{1. 10}$$

Theorem Two *Let* $\mathbf{I}^\alpha$ *defined in (1. 8)-(1. 10) for* $\frac{1}{2} < \mathbf{Re}\alpha < 1$. *We have*

$$\left\|\mathbf{I}^\alpha f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad \frac{n-1}{2n} \leq \frac{1}{p} \leq \frac{n+1}{2n}. \tag{1. 11}$$

Remark 1.1 **Theorem Two** *implies* **Theorem One**.

As investigated by Strichartz [20], Ω^α is a distribution defined in $\mathbb{R}^n$ by analytic continuation from

$$\mathbf{Re}\alpha > \tfrac{n-1}{2}, \qquad \Omega^\alpha(x) = \pi^{\alpha-\frac{n+1}{2}}\Gamma^{-1}\left(\alpha - \tfrac{n-1}{2}\right)\left(\frac{1}{1-|x|^2}\right)_+^{\frac{n+1}{2}-\alpha}.$$

Equivalently, it can be defined by

$$\begin{aligned}\widehat{\Omega}^\alpha(\xi) &= \left(\frac{1}{|\xi|}\right)^{\frac{n}{2}-\left[\frac{n+1}{2}-\alpha\right]} \mathbf{J}_{\frac{n}{2}-\left[\frac{n+1}{2}-\alpha\right]}\Big(2\pi|\xi|\Big)\\ &= \left(\frac{1}{|\xi|}\right)^{\alpha-\frac{1}{2}} \mathbf{J}_{\alpha-\frac{1}{2}}\Big(2\pi|\xi|\Big), \qquad \alpha\in\mathbb{C}.\end{aligned}$$

In section 3, we show that $\mathbf{I}^\alpha$ defined in (1. 10) for $\frac{1}{2} < \mathbf{Re}\alpha < 1$ can be expressed as

$$\begin{aligned}\mathbf{I}^\alpha f(x) &= \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\widehat{\phi}(\xi)\left\{\int_{\mathbb{R}} e^{-2\pi\mathbf{i}r}\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1}dr\right\}d\xi,\\ \omega(r) &= e^{2\pi\mathbf{i}r}\int_0^1 e^{-2\pi\mathbf{i}\tau r}\tau d\tau = \frac{-1}{2\pi\mathbf{i}}\frac{1}{r} - \frac{1}{4\pi^2r^2}\left[1-e^{-2\pi\mathbf{i}r}\right].\end{aligned} \tag{1. 12}$$

Moreover, define

$$\begin{aligned}\mathbf{I}^\alpha_< f(x) &= \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\widehat{\mathbf{P}}^\alpha_<(\xi)d\xi,\\ \widehat{\mathbf{P}}^\alpha_<(\xi) &= \widehat{\phi}(\xi)\int_{-1}^1 e^{-2\pi\mathbf{i}r}\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1}dr\end{aligned} \tag{1. 13}$$

and

$$\begin{aligned}\mathbf{I}^\alpha_j f(x) &= \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\widehat{\mathbf{P}}^\alpha_j(\xi)d\xi,\\ \widehat{\mathbf{P}}^\alpha_j(\xi) &= \widehat{\phi}(\xi)\int_{2^{j-1}\leq|r|<2^j} e^{-2\pi\mathbf{i}r}\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1}dr, \qquad j>0.\end{aligned} \tag{1. 14}$$

Later, we shall see $\mathbf{P}^\alpha_< \in \mathbf{L}^1(\mathbb{R}^n)$. Hence that $\mathbf{I}^\alpha_<$ is bounded on $\mathbf{L}^p(\mathbb{R}^n)$ for $1\leq p\leq\infty$.

Our aim is to show

$$\left\|\mathbf{I}^\alpha_j f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\varepsilon j} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad \frac{n-1}{2n} \leq \frac{1}{p} \leq \frac{n+1}{2n} \tag{1. 15}$$

for every $j>0$ and some $\varepsilon = \varepsilon(\mathbf{Re}\alpha) > 0$.

Given $\frac{1}{2} < \mathbf{Re}\alpha < 1$, $\mathbf{I}_j^{\alpha\ z}, 0 \leq \mathbf{Re} z \leq 1$ is a family of analytic operators which will be defined explicitly in section 4. In particular, we have $\mathbf{I}_j^{\alpha\ \frac{1}{n}} f(x) = \mathbf{I}_j^\alpha f(x)$.

Ideally, we hope to establish (1. 15) by using complex interpolation as follows. Suppose

$$\left\|\mathbf{I}_j^{\alpha\ 0+\mathbf{iIm}z} f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\varepsilon j} \left\|f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \tag{1. 16}$$

and

$$\left\|\mathbf{I}_j^{\alpha\ 1+\mathbf{iIm}z} f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\varepsilon j} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad 1 \leq p \leq \infty \tag{1. 17}$$

for every $j > 0$ and some $\varepsilon = \varepsilon(\mathbf{Re}\alpha) > 0$.

We obtain (1. 15) by applying Stein interpolation theorem [19] and taking into account for $\frac{n-1}{2n} = 0\left(\frac{1}{n}\right) + \frac{1}{2}\left(\frac{n-1}{n}\right)$ and $\frac{n+1}{2n} = 1\left(\frac{1}{n}\right) + \frac{1}{2}\left(\frac{n-1}{n}\right)$.

Unfortunately, the $\mathbf{L}^2$-estimate in (1. 16) is NOT true. The best result can be concluded is

$$\left\|\mathbf{I}_j^{\alpha\ 0+\mathbf{iIm}z} f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{j/2} 2^{-\varepsilon j} \left\|f\right\|_{\mathbf{L}^2(\mathbb{R}^n)}, \qquad j > 0. \tag{1. 18}$$

This is consequence of which the Fourier multiplier regarding every $\mathbf{I}_j^{\alpha\ 0+\mathbf{iIm}z}$ has its norm bounded by

$$\mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}-\varepsilon}, \qquad |\xi| \approx 1 \pm 2^{-j}.$$

To obtain the $\mathbf{L}^p$-estimate in (1. 15), we first write $\mathbf{I}_j^\alpha$ into a group of partial operators.

Given $\frac{1}{2} < \mathbf{Re}\alpha < 1$, denote $0 < \sigma = \sigma(\mathbf{Re}\alpha) < \frac{1}{2}$ for some constant which can be chosen sufficiently small. In particular, we shall find $\sigma \longrightarrow 0$ as $\mathbf{Re}\alpha \longrightarrow 1$.

For every $j > 0$, assert $\lambda_m \in [2^{j-1}, 2^j]$ where $\lambda_0 = 2^{j-1}$, $\lambda_M = 2^j$ and

$$2^{\sigma j-1} \leq \lambda_m - \lambda_{m-1} < 2^{\sigma j}. \tag{1. 19}$$

Remark 1.2 *There are at most a constant multiple of* $2^{(1-\sigma)j}$ *many* λ_m*'s inside* $[2^{j-1}, 2^j]$.

Let $\mathbf{I}_j^\alpha$ defined in (1. 14). Consider

$$\begin{gathered}\mathbf{I}_j^\alpha f(x) = \sum_{m=1}^M \mathbf{I}_{j\ m}^\alpha f(x), \qquad \mathbf{I}_{j\ m}^\alpha f(x) = \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi} \widehat{f}(\xi) \widehat{\mathbf{P}}_{j\ m}^\alpha(\xi) d\xi, \\ \widehat{\mathbf{P}}_{j\ m}^\alpha(\xi) = \widehat{\phi}(\xi) \int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r} \widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1} dr.\end{gathered} \tag{1. 20}$$

We claim

$$\left\|\mathbf{I}_{j\ m}^\alpha f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}\, 2^{-\varepsilon j} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad \frac{n-1}{2n} \leq \frac{1}{p} \leq \frac{n+1}{2n} \tag{1. 21}$$

for every $j > 0$, $m = 1, \ldots, M$ and some $\varepsilon = \varepsilon(\mathbf{Re}\alpha) > 0$.

Clearly, (1. 21) and **Remark 1.2** together with Minkowski inequality imply (1. 15).

1.2 A pairing formulation

First, we introduce an ingenious dyadic decomposition initially constructed by Fefferman [6] and later refined by Seeger, Sogge and Stein [17] in their study of Fourier integral operators. Given $j > 0$, $\left\{\xi_j^\nu\right\}_\nu$ is a collection of points almost equally distributed on $\mathbb{S}^{n-1}$ having a grid length between $2^{-j/2-1}$ and $2^{-j/2}$:

(1) $|\xi_j^\mu - \xi_j^\nu| \geq 2^{-j/2-1}$ for every $\xi_j^\mu \neq \xi_j^\nu$.

(2) For any $u \in \mathbb{S}^{n-1}$, there is a ξ_j^ν in the open set $\{\xi \in \mathbb{S}^{n-1} : |\xi - u| < 2^{-j/2+1}\}$.

Let $\varphi \in C_o^\infty(\mathbb{R})$ be a smooth cut-off function such that $\varphi(t) = 1$ if $|t| \leq 1$ and $\varphi(t) = 0$ for $|t| > 2$. Define

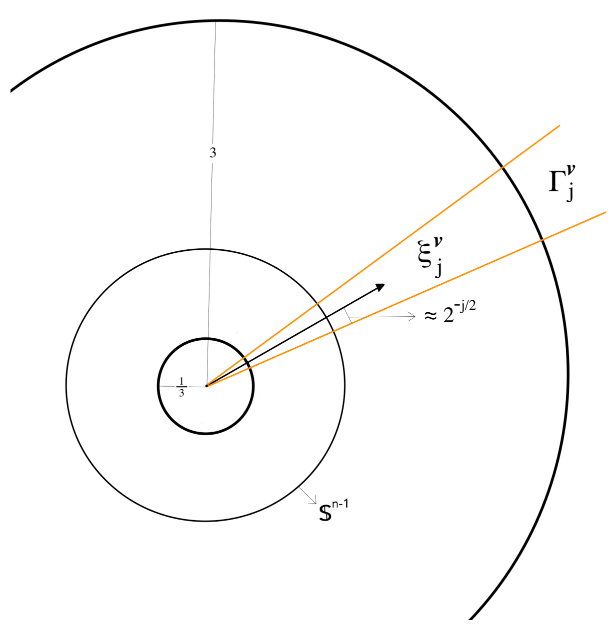
$$\varphi_j^\nu(\xi) = \frac{\varphi\left[2^{j/2}\left|\frac{\xi}{|\xi|} - \xi_j^\nu\right|\right]}{\sum_{\nu:\xi_j^\nu \in \mathbb{S}^{n-1}} \varphi\left[2^{j/2}\left|\frac{\xi}{|\xi|} - \xi_j^\nu\right|\right]} \tag{1. 22}$$

whose support is contained in the cone

$$\Gamma_j^\nu = \left\{\xi \in \mathbb{R}^n : \left|\frac{\xi}{|\xi|} - \xi_j^\nu\right| < 2^{-j/2+1}\right\}. \tag{1. 23}$$

Observe that by the construction in **(1)-(2)**, the support of $\sum_{\nu:\xi_j^\nu \in \mathbb{S}^{n-1}} \varphi\left[2^{j/2}\left|\frac{\xi}{|\xi|} - \xi_j^\nu\right|\right]$ is $\mathbb{R}^n$.

Remark 1.3 *There are at most a constant multiple of* $2^{\left(\frac{n-1}{2}\right)j}$ *many* ξ_j^ν *s.*



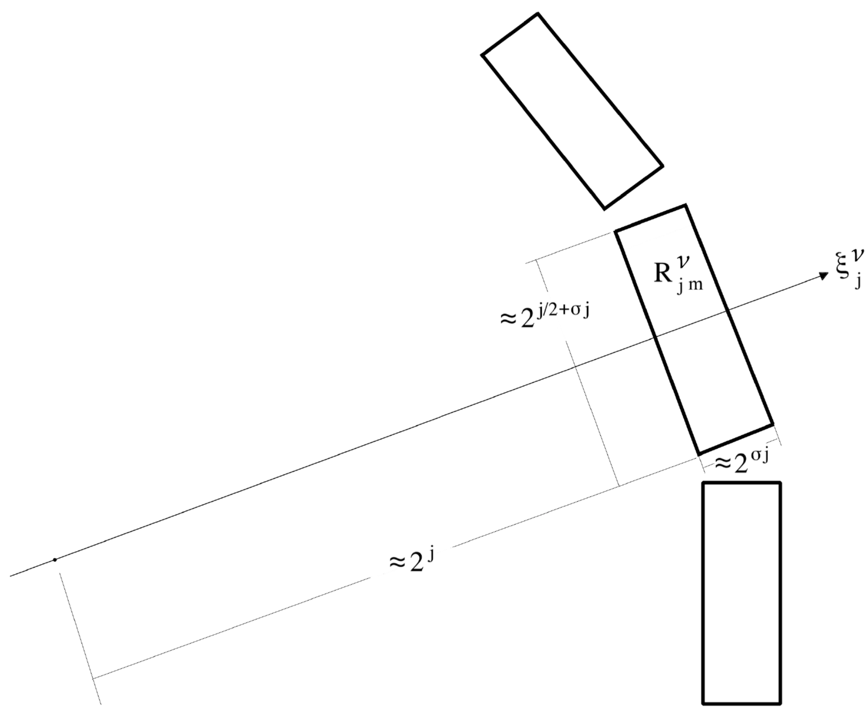
Given ξ_j^ν, $\mathbf{L}_\nu$ is an $n\times n$-orthogonal matrix with $\det \mathbf{L}_\nu = 1$ and $\mathbf{L}_\nu^T\xi_j^\nu = (1,0)^T \in \mathbb{R}\times\mathbb{R}^{n-1}$. Define

$$
\begin{aligned}
\Psi^\alpha_{j\,m}(x) &= \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} \Psi^{\alpha\,\nu}_{j\,m}(x),\\
\Psi^{\alpha\,\nu}_{j\,m}(x) &= \varphi\left[2^{-\sigma j-1}\left|\lambda_m - \left(\mathbf{L}_\nu^T x\right)_1\right|\right]\prod_{i=2}^n \varphi\left[2^{-[\frac{1}{2}+\sigma]j}\left|\left(\mathbf{L}_\nu^T x\right)_i\right|\right].
\end{aligned}
\tag{1. 24}
$$

Each $\Psi^{\alpha\,\nu}_{j\,m}$ is supported in the rectangle

$$
\mathrm{R}^\nu_{j\,m} = \left\{x\in\mathbb{R}^n\colon \lambda_m - 2^{\sigma j+2} < \left(\mathbf{L}_\nu^T x\right)_1 < \lambda_m + 2^{\sigma j+2},\ \left|\left(\mathbf{L}_\nu^T x\right)_i\right| < 2^{[\frac{1}{2}+\sigma]j+1},\ i=2,\ldots,n\right\}. \tag{1. 25}
$$

Moreover, $\mathrm{R}^\mu_{j\,m}\cap\mathrm{R}^\nu_{j\,m}=\emptyset$ if $\left|\xi_j^\mu - \xi_j^\nu\right| \geq \mathbf{c}2^{-j/2}2^{\sigma j}$ for $\mathbf{c}$ large.



For every $j>0$ and $m=1,\ldots,M$, we assert

$$
\begin{aligned}
\widehat{\mathbf{P}}^\alpha_{j\,m}(\xi) &= \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} \widehat{\mathbf{P}}^{\alpha\,\nu}_{j\,m}(\xi),\\
\widehat{\mathbf{P}}^{\alpha\,\nu}_{j\,m}(\xi) &= \varphi_j^\nu(\xi)\widehat{\phi}(\xi)\int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi \mathbf{i} r}\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1}dr.
\end{aligned}
\tag{1. 26}
$$

Recall $\mathbf{I}^\alpha_{j\,m} f = f * \mathbf{P}^\alpha_{j\,m}$ from (1. 20). We have

$$\begin{aligned}\mathbf{I}^\alpha_{j\,m} f(x) &= \int_{\mathbb{R}^n} e^{2\pi \mathbf{i} x\cdot\xi}\widehat{f}(\xi)\widehat{\mathbf{P}}^\alpha_{j\,m}(\xi)d\xi,\\ \widehat{\mathbf{P}}^\alpha_{j\,m}(\xi) &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \widehat{\mathbf{P}}^{\alpha\,\nu}_{j\,m}(\xi).\end{aligned}$$

Let $\Psi^\alpha_{j\,m}$ and $\Psi^{\alpha\,\nu}_{j\,m}$ defined in (1. 24). Consider

$$\begin{aligned}\mathbf{II}^\alpha_{j\,m} f(x) &= \int_{\mathbb{R}^n} f(x-u)\mathbf{U}^\alpha_{j\,m}(u)du\\ \mathbf{U}^\alpha_{j\,m}(x) &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \mathbf{U}^{\alpha\,\nu}_{j\,m}(x), \qquad \mathbf{U}^{\alpha\,\nu}_{j\,m}(x) = \mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\left[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\right]\end{aligned} \tag{1. 27}$$

and

$$\begin{aligned}\mathbf{III}^\alpha_{j\,m} f(x) &= \int_{\mathbb{R}^n} f(x-u)\mathbf{V}^\alpha_{j\,m}(u)du\\ \mathbf{V}^\alpha_{j\,m}(x) &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \mathbf{V}^{\alpha\,\nu}_{j\,m}(x), \qquad \mathbf{V}^{\alpha\,\nu}_{j\,m}(x) = \mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\Psi^{\alpha\,\nu}_{j\,m}(x).\end{aligned} \tag{1. 28}$$

From (1. 27)-(1. 28), we find

$$\begin{aligned}\mathbf{U}^\alpha_{j\,m}(x)+\mathbf{V}^\alpha_{j\,m}(x) &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \mathbf{U}^{\alpha\,\nu}_{j\,m}(x)+\mathbf{V}^{\alpha\,\nu}_{j\,m}(x)\\ &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\left[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\right]+\mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\Psi^{\alpha\,\nu}_{j\,m}(x)\\ &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\\ &= \mathbf{P}^\alpha_{j\,m}(x)\end{aligned}$$

and therefore

$$\mathbf{I}^\alpha_{j\,m} f(x) = \mathbf{II}^\alpha_{j\,m} f(x)+\mathbf{III}^\alpha_{j\,m} f(x). \tag{1. 29}$$

In section 4, we will explicitly define

$$\mathbf{I}^{\alpha\,z}_{j\,m} f(x) = \mathbf{II}^{\alpha\,z}_{j\,m} f(x)+\mathbf{III}^{\alpha\,z}_{j\,m} f(x), \qquad 0\le \mathbf{Re}z\le 1$$

of which $\mathbf{II}^{\alpha\,\ell\,z}_{j\,m}$ and $\mathbf{III}^{\alpha\,\ell\,z}_{j\,m}$ are two families of analytic operators.

In particular,

$$\mathbf{II}^{\alpha\,\frac{1}{n}}_{j\,m} f(x) = \mathbf{II}^\alpha_{j\,m} f(x), \qquad \mathbf{III}^{\alpha\,\frac{1}{n}}_{j\,m} f(x) = \mathbf{III}^\alpha_{j\,m} f(x).$$

• Given $j>0$, $m=1,\ldots,M$, we have

$$\left\|\mathbf{II}_{j\,m}^{\alpha\;0+\mathbf{i}\mathbf{Im}z}f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{j/2}2^{-\varepsilon j}\, \left\|f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \tag{1. 30}$$

and

$$\left\|\mathbf{III}_{j\,m}^{\alpha\;0+\mathbf{i}\mathbf{Im}z}f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j}\, \left\|f\right\|_{\mathbf{L}^2(\mathbb{R}^n)} \tag{1. 31}$$

for some $\varepsilon=\varepsilon(\mathbf{Re}\alpha)>0$.

Observe that at $\mathbf{Re}z=0$, $\mathbf{III}_{j\,m}^{\alpha\,z}$ is a *good* partial operator of $\mathbf{I}_{j\,m}^{\alpha\,z}$ whose $\mathbf{L}^2$-norm is bound by $\mathfrak{B}_{\mathbf{Re}\alpha}e^{\mathbf{c}|\mathbf{Im}\alpha|}2^{-(1-\sigma)j}2^{-\varepsilon j}$. On the other hand, $\mathbf{II}_{j\,m}^{\alpha\,z}$ is the *bad* one, satisfying (1. 30).

• Given $j>0$, $m=1,\ldots,M$, we find

$$\left\|\mathbf{II}_{j\,m}^{\alpha\;1+\mathbf{i}\mathbf{Im}z}f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{N\;\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-Nj}\,2^{-\varepsilon j}\, \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad 1\leq p\leq\infty \tag{1. 32}$$

and

$$\left\|\mathbf{III}_{j\,m}^{\alpha\;1+\mathbf{i}\mathbf{Im}z}f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j}\, \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad 1\leq p\leq\infty \tag{1. 33}$$

for every $N\geq 0$ and some $\varepsilon=\varepsilon(\mathbf{Re}\alpha)>0$.

At $\mathbf{Re}z=1$, $\mathbf{III}_{j\,m}^{\alpha\,z}$ is a *bad* partial operator of $\mathbf{I}_{j\,m}^{\alpha\,z}$. The best can be concluded is that it satisfies (1. 33). On the other hand, $\mathbf{II}_{j\,m}^{\alpha\,z}$ becomes the *good* one whose $\mathbf{L}^p$-norm is bounded by $\mathfrak{B}_{N\;\mathbf{Re}\alpha}e^{\mathbf{c}|\mathbf{Im}\alpha|}2^{-(1-\sigma)j}2^{-Nj}2^{-\varepsilon j}$ for every $N\geq 0$.

Recall $\mathbf{II}_{j\,m}^{\alpha\;\frac{1}{n}}=\mathbf{II}_{j\,m}^{\alpha}$ and $\mathbf{III}_{j\,m}^{\alpha\;\frac{1}{n}}=\mathbf{III}_{j\,m}^{\alpha}$. Take into account for $\frac{n-1}{2n}=0\left(\frac{1}{n}\right)+\frac{1}{2}\left(\frac{n-1}{n}\right)$ and $\frac{n+1}{2n}=1\left(\frac{1}{n}\right)+\frac{1}{2}\left(\frac{n-1}{n}\right)$.

From (1. 30) and (1. 32) with N chosen sufficiently large, we have

$$\begin{gathered}\left\|\mathbf{II}_{j\,m}^{\alpha}f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j}\, \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)},\\ \frac{n-1}{2n}\leq\frac{1}{p}\leq\frac{n+1}{2n}\end{gathered} \tag{1. 34}$$

by applying Stein interpolation theorem [19].

From (1. 31) and (1. 33), Stein interpolation theorem [19] again implies

$$\begin{gathered}\left\|\mathbf{III}_{j\,m}^{\alpha}f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j}\, \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)},\\ \frac{n-1}{2n}\leq\frac{1}{p}\leq\frac{n+1}{2n}.\end{gathered} \tag{1. 35}$$

Recall (1. 29). By putting together (1. 34)-(1. 35), we obtain (1. 21) as desired.

The remaining paper is organized as follows. In section 2, we show **Theorem Two** implying **Theorem One**. In section 3, after a formal derivation of $\mathbf{I}_{j\,m}^{\alpha}=\mathbf{II}_{j\,m}^{\alpha}+\mathbf{III}_{j\,m}^{\alpha}$, we introduce two end-point estimates associated with $\mathbf{Re}z=0$ and $\mathbf{Re}z=1$ in (1. 30)-(1. 33), stated as **Proposition One** and **Proposition Two**. By using these results, we finish the proof of **Theorem Two** in section 4. We prove **Proposition One** in section 5. Section 6 is devoted to certain preliminaries regarding **Proposition Two**. We prove **Proposition Two** in section 7. Two appendices added in the end for the sake of self-containedness.

2 Theorem Two implies Theorem One

Let $0 < \mathbf{Re}\delta < \frac{1}{2}$. Consider

$$\mathbf{S}_\psi^\delta f(x) = \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\widehat{\psi}(\xi)\left(1-|\xi|^2\right)_+^\delta d\xi \tag{2. 1}$$

where $\widehat{\psi}(\xi) = \left[\widehat{\phi}(\xi)\right]^2$ and $\widehat{\phi}$ is defined in (1. 9).

Observe that $\left[1-\widehat{\psi}(\xi)\right]\left(1-|\xi|^2\right)_+^\delta$ is a $\mathbf{L}^p$-Fourier multiplier for $1 < p < \infty$. In order to prove **Theorem One**, it is suffice to show

$$\left\|\mathbf{S}_\psi^\delta f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \leq \mathfrak{B}_{\mathbf{Re}\delta}\, e^{\mathbf{c}|\mathbf{Im}\delta|}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad \frac{n-1}{2n} \leq \frac{1}{p} \leq \frac{n+1}{2n}. \tag{2. 2}$$

Let $1-\delta = \alpha \in \mathbb{C}$. We define

$$\begin{aligned}
\mathbf{m}_+^\alpha(\xi) &= \widehat{\phi}(\xi)\int_0^1\left(\frac{1}{\tau^2-|\xi|^2}\right)_+^\alpha \tau d\tau \\
&= \widehat{\phi}(\xi)\begin{cases}\displaystyle\int_{|\xi|}^1\left(\frac{1}{\tau^2-|\xi|^2}\right)^\alpha \tau d\tau, & |\xi|<1 \\ 0 & |\xi|\geq 1\end{cases} \\
&= \frac{1}{2}(1-\alpha)^{-1}\widehat{\phi}(\xi)\left(1-|\xi|^2\right)_+^{1-\alpha}
\end{aligned} \tag{2. 3}$$

and

$$\begin{aligned}
\mathbf{m}_-^\alpha(\xi) &= \widehat{\phi}(\xi)\int_0^1\left(\frac{1}{\tau^2-|\xi|^2}\right)_-^\alpha \tau d\tau \\
&= (-1)^{-\alpha}\widehat{\phi}(\xi)\begin{cases}\displaystyle\int_0^{|\xi|}\left(\frac{1}{\tau^2-|\xi|^2}\right)^\alpha \tau d\tau, & |\xi|\leq 1 \\ \displaystyle\int_0^1\left(\frac{1}{\tau^2-|\xi|^2}\right)^\alpha \tau d\tau, & |\xi|>1\end{cases} \\
&= \frac{1}{2}(1-\alpha)^{-1}\widehat{\phi}(\xi)\left[|\xi|^{2(1-\alpha)}-\left(1-|\xi|^2\right)_-^{1-\alpha}\right].
\end{aligned} \tag{2. 4}$$

Recall $\widehat{\Lambda}^\alpha(\xi,\tau)$ given at (1. 8). From (2. 3)-(2. 4), we find

$$\widehat{\phi}(\xi)\int_0^1\widehat{\Lambda}^\alpha(\xi,\tau)\tau d\tau = \pi^{\frac{n-1}{2}-2\alpha}\Gamma(\alpha)\left\{\mathbf{m}_-^\alpha(\xi)-\sin\pi\left(\alpha-\frac{1}{2}\right)\mathbf{m}_+^\alpha(\xi)\right\}. \tag{2. 5}$$

Moreover, define

$$\mathbf{m}^\alpha(\xi) = \widehat{\phi}(\xi)\left\{-\left(1-|\xi|^2\right)_-^{1-\alpha}-\sin\pi\left(\alpha-\frac{1}{2}\right)\left(1-|\xi|^2\right)_+^{1-\alpha}\right\}. \tag{2. 6}$$

From (2. 3)-(2. 4), (2. 5) and (2. 6), we have

$$\widehat{\phi}(\xi)\int_0^1\widehat{\Lambda}^\alpha(\xi,\tau)\tau d\tau = \pi^{\frac{n-2}{2}-2\alpha}\frac{1}{2}(1-\alpha)^{-1}\Gamma(\alpha)\left[\mathbf{m}^\alpha(\xi)+\widehat{\phi}(\xi)|\xi|^{2(1-\alpha)}\right]. \tag{2. 7}$$

Remark 2.1 *Let*

$$\mathbf{T}_z f(x) \;=\; \int_{\mathbb{R}^n} e^{2\pi \mathbf{i} x\cdot\xi}\widehat{f}(\xi)\mathbf{m}^z(\xi)d\xi, \qquad z\in\mathbb{C}.$$

We call $\mathbf{m}^z(\xi)$ *a restricted* $\mathbf{L}^p$*-Fourier multiplier if*

$$\left\|\mathbf{T}_z f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} \;\leq\; \mathfrak{B}_{\mathbf{Re}z}\, e^{\mathbf{c}|\mathbf{Im}z|}\,\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad \frac{n-1}{2n} \;\leq\; \frac{1}{p} \;\leq\; \frac{n+1}{2n}.$$

Recall $\mathbf{I}^\alpha$ defined in (1. 10). Let $\frac{1}{2} < \mathbf{Re}\alpha < 1$. **Theorem Two** states that $\mathbf{I}^\alpha f$ satisfies the $\mathbf{L}^p$-norm inequality in (1. 11). Therefore, the left-hand-side of the equation in (2. 7) is a restricted $\mathbf{L}^p$-Fourier multiplier. Because $\widehat{\phi}(\xi)|\xi|^{2(1-\alpha)}$ is a $\mathbf{L}^p$-Fourier multiplier for $1<p<\infty$, we find $\mathbf{m}^\alpha(\xi)$ as a restricted $\mathbf{L}^p$-Fourier multiplier.

Consider

$$\widehat{\phi}(\xi)\mathbf{m}^\alpha(\xi) \;=\; \widehat{\psi}(\xi)\left\{-\left(1-|\xi|^2\right)_-^{1-\alpha} - \sin\pi\left(\alpha-\frac{1}{2}\right)\left(1-|\xi|^2\right)_+^{1-\alpha}\right\}$$

and

$$\mathbf{m}^{\frac{1}{2}+\frac{\alpha}{2}}(\xi) \;=\; \widehat{\phi}(\xi)\left\{-\left(1-|\xi|^2\right)_-^{\frac{1}{2}-\frac{\alpha}{2}} - \sin\pi\left(\frac{\alpha}{2}\right)\left(1-|\xi|^2\right)_+^{\frac{1}{2}-\frac{\alpha}{2}}\right\}.$$

Clearly, both $\widehat{\phi}(\xi)\mathbf{m}^\alpha(\xi)$ and $\mathbf{m}^{\frac{1}{2}+\frac{\alpha}{2}}(\xi)$ are restricted $\mathbf{L}^p$-Fourier multipliers. Furthermore,

$$\left[\mathbf{m}^{\frac{1}{2}+\frac{\alpha}{2}}(\xi)\right]^2 \;=\; \widehat{\psi}(\xi)\left\{\left(1-|\xi|^2\right)_-^{1-\alpha} + \sin^2\pi\left(\frac{\alpha}{2}\right)\left(1-|\xi|^2\right)_+^{1-\alpha}\right\} \tag{2. 8}$$

is indeed another restricted $\mathbf{L}^p$-Fourier multiplier.

By adding $\left[\mathbf{m}^{\frac{1}{2}+\frac{\alpha}{2}}(\xi)\right]^2$ and $\widehat{\phi}(\xi)\mathbf{m}^\alpha(\xi)$ together, we obtain

$$\widehat{\psi}(\xi)\left[\sin^2\pi\left(\frac{\alpha}{2}\right) - \sin\pi\left(\alpha-\frac{1}{2}\right)\right]\left(1-|\xi|^2\right)_+^{1-\alpha}. \tag{2. 9}$$

A direct computation shows

$$\sin^2\pi\left(\frac{\alpha}{2}\right) \;=\; \frac{1}{2}-\frac{1}{4}\left[e^{-\pi\mathbf{Im}\alpha}+e^{\pi\mathbf{Im}\alpha}\right]\cos\pi\mathbf{Re}\alpha - \frac{\mathbf{i}}{4}\left[e^{-\pi\mathbf{Im}\alpha}-e^{\pi\mathbf{Im}\alpha}\right]\sin\pi\mathbf{Re}\alpha,$$

$$\sin\pi\left(\alpha-\frac{1}{2}\right) \;=\; -\frac{1}{2}\left[e^{-\pi\mathbf{Im}\alpha}+e^{\pi\mathbf{Im}\alpha}\right]\cos\pi\mathbf{Re}\alpha - \frac{\mathbf{i}}{2}\left[e^{-\pi\mathbf{Im}\alpha}-e^{\pi\mathbf{Im}\alpha}\right]\sin\pi\mathbf{Re}\alpha.$$

Consequently, we have

$$\sin^2\pi\left(\frac{\alpha}{2}\right) - \sin\pi\left(\alpha-\frac{1}{2}\right) \neq 0, \qquad 0<\mathbf{Re}\alpha<1. \tag{2. 10}$$

From (2. 9) and (2. 10), we conclude that $\widehat{\psi}(\xi)\left(1-|\xi|^2\right)_+^{1-\alpha}$ is a restricted $\mathbf{L}^p$-Fourier multiplier as desired.

3 End-point estimates regarding $\mathbf{II}^\alpha_{j\,m}$ and $\mathbf{III}^\alpha_{j\,m}$

Recall $\mathbf{I}^\alpha$ defined in (1. 10). First, we show $\mathbf{I}^\alpha f$ for $\frac{1}{2} < \mathbf{Re}\alpha < 1$ can be expressed as (1. 12). Ω^α is a distribution defined in $\mathbb{R}^n$ by analytic continuation from

$$\mathbf{Re}\alpha > \tfrac{n-1}{2}, \qquad \Omega^\alpha(x) \;=\; \pi^{\alpha-\frac{n+1}{2}}\Gamma^{-1}\left(\alpha-\tfrac{n-1}{2}\right)\left(\frac{1}{1-|x|^2}\right)_+^{\frac{n+1}{2}-\alpha}.$$

Equivalently, it can be defined by

$$\begin{aligned}\widehat{\Omega}^\alpha(\xi) \;&=\; \left(\frac{1}{|\xi|}\right)^{\frac{n}{2}-\left[\frac{n+1}{2}-\alpha\right]}\mathbf{J}_{\frac{n}{2}-\left[\frac{n+1}{2}-\alpha\right]}\Big(2\pi|\xi|\Big)\\ &=\; \left(\frac{1}{|\xi|}\right)^{\alpha-\frac{1}{2}}\mathbf{J}_{\alpha-\frac{1}{2}}\Big(2\pi|\xi|\Big), \qquad \alpha\in\mathbb{C}.\end{aligned} \tag{3. 1}$$

See (A. 7)-(A. 8). By using the integral formula of Bessel functions in (A. 1), we find

$$\widehat{\Omega}^\alpha(\xi) \;=\; \pi^{\alpha-1}\Gamma(\alpha)\int_{-1}^{1} e^{2\pi\mathbf{i}|\xi|s}(1-s^2)^{\alpha-1}ds. \tag{3. 2}$$

On the other hand, Λ^α is a distribution defined in $\mathbb{R}^{n+1}$ by analytic continuation from

$$\mathbf{Re}\alpha > \tfrac{n-1}{2}, \qquad \Lambda^\alpha(x,t) \;=\; \pi^{\alpha-\frac{n+1}{2}}\Gamma^{-1}\left(\alpha-\tfrac{n-1}{2}\right)\left(\frac{1}{t^2-|x|^2}\right)_+^{\frac{n+1}{2}-\alpha}. \tag{3. 3}$$

Let h be a Schwartz function defined in $\mathbb{R}^{n+1}$. We have

$$\begin{aligned}h*\Lambda^\alpha(x,t) \;&=\; \pi^{\alpha-\frac{n+1}{2}}\Gamma^{-1}\left(\alpha-\tfrac{n-1}{2}\right)\iint_{|u|<|r|} h(x-u,t-r)\left(\frac{1}{r^2-|u|^2}\right)^{\frac{n+1}{2}-\alpha}dudr\\ &=\; \iint_{\mathbb{R}^{n+1}} h(x-u,t-r)\Omega^\alpha\left(\frac{u}{r}\right)|r|^{2\alpha-1-n}dudr\\ &=\; \int_{\mathbb{R}}\left\{\int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{h}(\xi,t-r)\widehat{\Omega}^\alpha(r\xi)d\xi\right\}|r|^{2\alpha-1}dr\end{aligned} \tag{3. 4}$$

whenever $\mathbf{Re}\alpha > \frac{n-1}{2}$. Here, $\widehat{h}(\cdot,t-r)$ is the Fourier transform of $h(x,t-r)$ in $x\in\mathbb{R}^n$.

Remark 3.1 *By the principle of analytic continuation, we must have (3. 4) hold for every* $\mathbf{Re}\alpha > 0$.

Let $\frac{1}{2} < \mathbf{Re}\alpha < 1$. From (3. 4), we find

$$h*\Lambda^\alpha(x,t) \;=\; \lim_{N\longrightarrow\infty}\iint_{\mathbb{R}^{n+1}} e^{2\pi\mathbf{i}\left[x\cdot\xi+t\tau\right]}\widehat{h}(\xi,\tau)\left\{\int_{-N}^{N} e^{-2\pi\mathbf{i}\tau r}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr\right\}d\xi d\tau.$$

By using the asymptotic expansion of Bessel functions in (A. 2)-(A. 4), together with (3. 2), we write

$$
\begin{aligned}
\int_{\mathbb{R}} e^{-2\pi \mathbf{i}\tau r}\widehat{\Omega}^{\alpha}(r\xi)|r|^{2\alpha-1}dr &= \int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\left(\frac{1}{|r\xi|}\right)^{\alpha-\frac{1}{2}} \mathbf{J}_{\alpha-\frac{1}{2}}\Big(2\pi|r\xi|\Big)|r|^{2\alpha-1}dr \qquad \text{by (3. 1)} \\
&= \frac{1}{\pi}\int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\left(\frac{1}{|r\xi|}\right)^{\alpha}\cos\left[2\pi|r\xi|-\frac{\pi}{2}\alpha\right]|r|^{2\alpha-1}dr \\
&+ \int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\mathcal{E}^{\alpha}\left(2\pi|r\xi|\right)|r|^{2\alpha-1}dr,
\end{aligned}
\tag{3. 5}
$$

$$
\Big|\mathcal{E}^{\alpha}(\rho)\Big| \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\begin{cases} \rho^{-\mathbf{Re}\alpha}, & 0<\rho\leq 1, \\ \rho^{-\mathbf{Re}\alpha-1}, & \rho>1. \end{cases}
$$

Consider $\frac{1}{10}<|\xi|\leq 10$. The norm estimate for $\mathcal{E}^{\alpha}$ implies

$$
\left|\int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\mathcal{E}^{\alpha}\left(2\pi|r\xi|\right)|r|^{2\alpha-1}dr\right| \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}.
$$

Because of Euler's formulae, we replace the cosine function in (3. 5) with $e^{-\mathbf{i}\left(\frac{\pi}{2}\right)\alpha}e^{2\pi\mathbf{i}|r\xi|}$ or $e^{\mathbf{i}\left(\frac{\pi}{2}\right)\alpha}e^{-2\pi\mathbf{i}|r\xi|}$. By integration by parts *w.r.t* r, we find

$$
\begin{aligned}
&\left|\int_{2^{j-1}}^{2^j} e^{2\pi\mathbf{i}\left[|\xi|-\tau\right]r}\left(\frac{1}{|r\xi|}\right)^{\alpha}|r|^{2\alpha-1}dr\right| \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\begin{cases} 2^{\mathbf{Re}\alpha j}, & 2^j\leq \Big||\tau|-|\xi|\Big|^{-1}, \\ \left|\frac{1}{|\tau|-|\xi|}\right|2^{\left[\mathbf{Re}\alpha-1\right]j}, & 2^j> \Big||\tau|-|\xi|\Big|^{-1}. \end{cases}
\end{aligned}
\tag{3. 6}
$$

For symmetry reason, (3. 6) further implies

$$
\begin{aligned}
&\left|\int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\left(\frac{1}{|r\xi|}\right)^{\alpha}\cos\left[2\pi|r\xi|-\frac{\pi}{2}\alpha\right]|r|^{2\alpha-1}dr\right| \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\sum_{2^j\leq ||\tau|-|\xi||^{-1}} 2^{\mathbf{Re}\alpha j} + \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\sum_{2^j> ||\tau|-|\xi||^{-1}}\left|\frac{1}{|\tau|-|\xi|}\right|2^{\left[\mathbf{Re}\alpha-1\right]j} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\left|\frac{1}{|\tau|-|\xi|}\right|^{\mathbf{Re}\alpha}.
\end{aligned}
\tag{3. 7}
$$

On the other hand, the Fourier transform of Λ^{α} defined by analytic continuation from (3. 3) agrees with the function

$$
\widehat{\Lambda}^{\alpha}(\xi,\tau) = \pi^{\frac{n-1}{2}-2\alpha}\Gamma(\alpha)\left\{\left(\frac{1}{\tau^2-|\xi|^2}\right)_{-}^{\alpha} - \sin\pi\left(\alpha-\tfrac{1}{2}\right)\left(\frac{1}{\tau^2-|\xi|^2}\right)_{+}^{\alpha}\right\}
$$

whenever $|\tau|\neq|\xi|$. See appendix B.

From (3. 4) to (3. 7), by applying Lebesgue's dominated convergence theorem, we have

$$
\begin{aligned}
h * \Lambda^\alpha(0,0) &= \lim_{N\longrightarrow\infty} \iint_{\mathbb{R}^{n+1}} \widehat{h}(\xi,\tau)\left\{\int_{-N}^{N} e^{-2\pi\mathbf{i}\tau r}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr\right\}d\xi d\tau \\
&= \iint_{\mathbb{R}^{n+1}} \widehat{h}(\xi,\tau)\left\{\int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr\right\}d\xi d\tau \qquad (3.\ 8) \\
&= \iint_{\mathbb{R}^{n+1}} \widehat{h}(\xi,\tau)\widehat{\Lambda}^\alpha(\xi,\tau)d\xi d\tau
\end{aligned}
$$

for every h defined in $\mathbb{R}^{n+1}$ whose Fourier transform is supported in the cylindrical region: $\left\{(\xi,\tau)\in\mathbb{R}^n\times\mathbb{R}\colon \frac{1}{10}<|\xi|\leq 10\right\}$.

Let $\varphi\in C_o^\infty(\mathbb{R})$ such that $\varphi(t)=1$ if $|t|\leq 1$ and $\varphi(t)=0$ if $|t|>2$. Denote $\mathbf{c}_\varphi^{-1}=\int_{\mathbb{R}}\varphi(t)dt$. Given $(\eta,s)\in\mathbb{R}^n\times\mathbb{R}$ for $|s|\neq|\eta|$ and $\frac{1}{3}\leq|\eta|\leq 3$, we consider a family of *good kernels*: $\widehat{h_\varepsilon}(\xi,\tau)=\mathbf{c}_\varphi\varepsilon^{-(n+1)}\varphi\left[\varepsilon^{-1}\sqrt{|\xi-\eta|^2+(\tau-s)^2}\right]$, $0<\varepsilon<1/10$.

Replace $\widehat{h}(\xi,\tau)$ by $\widehat{h_\varepsilon}(\xi,\tau)$ in (3. 8). By taking $\varepsilon\longrightarrow 0$, we find

$$
\widehat{\Lambda}^\alpha(\xi,\tau) = \int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr, \qquad |\tau|\neq|\xi|, \qquad \tfrac{1}{3}<|\xi|\leq 3. \qquad (3.\ 9)
$$

By applying Lebesgue's dominated convergence theorem again, we have

$$
\widehat{\mathbf{I}^\alpha f}(\xi) = \widehat{f}(\xi)\widehat{\mathbf{P}}^\alpha(\xi),
$$

$$
\begin{aligned}
\widehat{\mathbf{P}}^\alpha(\xi) &= \widehat{\phi}(\xi)\int_0^1 \widehat{\Lambda}^\alpha(\xi,\tau)\tau d\tau \\
&= \widehat{\phi}(\xi)\int_{0<\tau<1,\ \tau\neq|\xi|}\left\{\int_{\mathbb{R}} e^{-2\pi\mathbf{i}\tau r}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr\right\}\tau d\tau \qquad \text{by (3. 9)} \\
&= \widehat{\phi}(\xi)\lim_{N\longrightarrow\infty}\int_{0<\tau<1,\ \tau\neq|\xi|}\left\{\int_{-N}^{N} e^{-2\pi\mathbf{i}\tau r}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr\right\}\tau d\tau \qquad (3.\ 10) \\
&= \widehat{\phi}(\xi)\int_{\mathbb{R}}\left\{\int_0^1 e^{-2\pi\mathbf{i}\tau r}\tau d\tau\right\}\widehat{\Omega}^\alpha(r\xi)|r|^{2\alpha-1}dr \\
&= \widehat{\phi}(\xi)\int_{\mathbb{R}} e^{-2\pi\mathbf{i}r}\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1}dr,
\end{aligned}
$$

$$
\omega(r) = e^{2\pi\mathbf{i}r}\int_0^1 e^{-2\pi\mathbf{i}\tau r}\tau d\tau = \frac{-1}{2\pi\mathbf{i}}\frac{1}{r}-\frac{1}{4\pi^2r^2}\left[1-e^{-2\pi\mathbf{i}r}\right].
$$

Note that $\widehat{\phi}$ defined in (1. 9) has $\mathbf{supp}\widehat{\phi}\subset\left\{\xi\in\mathbb{R}^n\colon \frac{1}{3}<|\xi|\leq 3\right\}$.

Consider

$$\widehat{\mathbf{P}}^{\alpha}(\xi) \;=\; \widehat{\mathbf{P}}^{\alpha}_{<}(\xi)+\sum_{j>0}\widehat{\mathbf{P}}^{\alpha}_{j}(\xi),$$

$$\widehat{\mathbf{P}}^{\alpha}_{<}(\xi) \;=\; \widehat{\phi}(\xi)\int_{-1}^{1}e^{-2\pi\mathbf{i}r}\widehat{\Omega}^{\alpha}(r\xi)\omega(r)|r|^{2\alpha-1}dr, \tag{3. 11}$$

$$\widehat{\mathbf{P}}^{\alpha}_{j}(\xi) \;=\; \widehat{\phi}(\xi)\int_{2^{j-1}\leq|r|<2^{j}}e^{-2\pi\mathbf{i}r}\widehat{\Omega}^{\alpha}(r\xi)\omega(r)|r|^{2\alpha-1}dr, \qquad j>0.$$

Furthermore, we assert

$$\widehat{\mathbf{P}}^{\alpha}_{j}(\xi) \;=\; \sum_{m=1}^{M}\widehat{\mathbf{P}}^{\alpha}_{j\,m}(\xi),$$

$$\widehat{\mathbf{P}}^{\alpha}_{j\,m}(\xi) \;=\; \widehat{\phi}(\xi)\int_{\lambda_{m-1}\leq|r|<\lambda_{m}}e^{-2\pi\mathbf{i}r}\widehat{\Omega}^{\alpha}(r\xi)\omega(r)|r|^{2\alpha-1}dr, \tag{3. 12}$$

$$\lambda_m\in[2^{j-1},2^{j}],\qquad \lambda_0=2^{j-1},\ \lambda_M=2^{j}\ \text{ and }\ 2^{\sigma j-1}\;\leq\;\lambda_m-\lambda_{m-1}<2^{\sigma j}$$

where $0<\sigma=\sigma(\mathbf{Re}\alpha)<\frac{1}{2}$ can be chosen sufficiently small.

By using (3. 2), we find

$$\begin{aligned}\mathbf{P}^{\alpha}_{<}(x) \;&=\; \int_{\mathbb{R}^n}e^{2\pi\mathbf{i}x\cdot\xi}\widehat{\phi}(\xi)\left\{\int_{-1}^{1}e^{-2\pi\mathbf{i}r}\widehat{\Omega}^{\alpha}(r\xi)\omega(r)|r|^{2\alpha-1}dr\right\}d\xi\\ &=\; \pi^{\alpha-1}\Gamma(\alpha)\\ &\quad\int_{\mathbb{R}^n}e^{2\pi\mathbf{i}x\cdot\xi}\widehat{\phi}(\xi)\left\{\int_{-1}^{1}e^{-2\pi\mathbf{i}r}\left\{\int_{-1}^{1}e^{2\pi\mathbf{i}|r\xi|s}(1-s^2)^{\alpha-1}ds\right\}\omega(r)|r|^{2\alpha-1}dr\right\}d\xi.\end{aligned} \tag{3. 13}$$

An N-fold integration by parts *w.r.t* ξ inside (3. 13) shows $\left|\mathbf{P}^{\alpha}_{<}(x)\right|\;\leq\;\mathfrak{B}_{N\ \mathbf{Re}\alpha}e^{\mathbf{c}|\mathbf{Im}\alpha|}\left(\frac{1}{1+|x|}\right)^{N}$. Hence that $\mathbf{P}^{\alpha}_{<}$ is an $\mathbf{L}^1$-function in $\mathbb{R}^n$.

Given $j>0$, $\left\{\xi^{\nu}_{j}\right\}_{\nu}$ is a collection of points almost equally distributed on $\mathbb{S}^{n-1}$ having a grid length between $2^{-j/2-1}$ and $2^{-j/2}$: **(1)** $|\xi^{\mu}_{j}-\xi^{\nu}_{j}|\geq 2^{-j/2-1}$ for every $\xi^{\mu}_{j}\neq\xi^{\nu}_{j}$. **(2)** For any $u\in\mathbb{S}^{n-1}$, there is a ξ^{ν}_{j} in the open set $\{\xi\in\mathbb{S}^{n-1}\colon|\xi-u|<2^{-j/2+1}\}$.

Let $\varphi\in C^{\infty}_{o}(\mathbb{R})$ be a smooth cut-off function such that $\varphi(t)=1$ if $|t|\leq 1$ and $\varphi(t)=0$ if $|t|>2$. Recall (1. 22)-(1. 23). We have

$$\varphi^{\nu}_{j}(\xi) \;=\; \frac{\varphi\left[2^{j/2}\left|\frac{\xi}{|\xi|}-\xi^{\nu}_{j}\right|\right]}{\sum_{\nu:\xi^{\nu}_{j}\in\mathbb{S}^{n-1}}\varphi\left[2^{j/2}\left|\frac{\xi}{|\xi|}-\xi^{\nu}_{j}\right|\right]}, \tag{3. 14}$$

$$\mathbf{supp}\varphi^{\nu}_{j}\;\subset\;\Gamma^{\nu}_{j}\;=\;\left\{\xi\in\mathbb{R}^n\colon\left|\frac{\xi}{|\xi|}-\xi^{\nu}_{j}\right|<2^{-j/2+1}\right\}.$$

Let
$$\widehat{\mathbf{P}}{}^{\alpha\,\nu}_{j\,m}(\xi) \;=\; \varphi^{\nu}_{j}(\xi)\widehat{\phi}(\xi)\int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\widehat{\Omega}^{\alpha}(r\xi)\omega(r)|r|^{2\alpha-1}dr. \tag{3. 15}$$
We write
$$\widehat{\mathbf{P}}{}^{\alpha}_{j\,m}(\xi) \;=\; \sum_{\nu:\xi^{\nu}_{j}\in\mathbb{S}^{n-1}} \widehat{\mathbf{P}}{}^{\alpha\,\nu}_{j\,m}(\xi) \tag{3. 16}$$
and
$$\mathbf{I}^{\alpha}_{j\,m}f(x) \;=\; \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\widehat{\mathbf{P}}{}^{\alpha}_{j\,m}(\xi)d\xi \tag{3. 17}$$
for every $j>0$ and $m=1,2,\ldots,M$.

Given $\xi^{\nu}_{j}\in\mathbb{S}^{n-1}$, $\mathbf{L}_{\nu}$ is an $n\times n$-orthogonal matrix with $\det\mathbf{L}_{\nu}=1$. Moreover, we require $\mathbf{L}^{T}_{\nu}\xi^{\nu}_{j}=(1,0)^{T}\in\mathbb{R}\times\mathbb{R}^{n-1}$. Recall (1. 24)-(1. 25). We have
$$\Psi^{\alpha}_{j\,m}(x) \;=\; \sum_{\nu:\xi^{\nu}_{j}\in\mathbb{S}^{n-1}} \Psi^{\alpha\,\nu}_{j\,m}(x),$$
$$\Psi^{\alpha\,\nu}_{j\,m}(x) \;=\; \varphi\left[2^{-\sigma j-1}\left|\lambda_m-\left(\mathbf{L}^{T}_{\nu}x\right)_1\right|\right]\prod_{i=2}^{n}\varphi\left[2^{-\left[\frac{1}{2}+\sigma\right]j}\left|\left(\mathbf{L}^{T}_{\nu}x\right)_i\right|\right],$$
$$\mathbf{supp}\Psi^{\alpha\,\nu}_{j\,m} \;\subset\; \mathrm{R}^{\nu}_{j\,m},$$
$$\mathrm{R}^{\nu}_{j\,m} \;=\; \left\{x\in\mathbb{R}^n\colon \lambda_m-2^{\sigma j+2}<\left(\mathbf{L}^{T}_{\nu}x\right)_1<\lambda_m+2^{\sigma j+2},\;\left|\left(\mathbf{L}^{T}_{\nu}x\right)_i\right|<2^{\left[\frac{1}{2}+\sigma\right]j+1},\; i=2,\ldots,n\right\}. \tag{3. 18}$$
Consider
$$\mathbf{I}^{\alpha}_{j\,m}f(x) \;=\; \mathbf{II}^{\alpha}_{j\,m}f(x)+\mathbf{III}^{\alpha}_{j\,m}f(x) \tag{3. 19}$$
of which
$$\begin{aligned}\mathbf{II}^{\alpha}_{j\,m}f(x) \;&=\; \int_{\mathbb{R}^n} f(x-u)\mathbf{U}^{\alpha}_{j\,m}(u)du,\\ \mathbf{U}^{\alpha}_{j\,m}(x) \;=\; \sum_{\nu:\xi^{\nu}_{j}\in\mathbb{S}^{n-1}}\mathbf{U}^{\alpha\,\nu}_{j\,m}(x),\qquad &\mathbf{U}^{\alpha\,\nu}_{j\,m}(x) \;=\; \mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\left[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\right]\end{aligned} \tag{3. 20}$$
and
$$\begin{aligned}\mathbf{III}^{\alpha}_{j\,m}f(x) \;&=\; \int_{\mathbb{R}^n} f(x-u)\mathbf{V}^{\alpha}_{j\,m}(u)du,\\ \mathbf{V}^{\alpha}_{j\,m}(x) \;=\; \sum_{\nu:\xi^{\nu}_{j}\in\mathbb{S}^{n-1}}\mathbf{V}^{\alpha\,\nu}_{j\,m}(x),\qquad &\mathbf{V}^{\alpha\,\nu}_{j\,m}(x) \;=\; \Psi^{\alpha\,\mu}_{j\,m}(x)\mathbf{P}^{\alpha\,\nu}_{j\,m}(x)\end{aligned} \tag{3. 21}$$
for every $j>0$ and $m=1,2,\ldots,M$.

Next, we begin to introduce the two end-point estimates regarding $\mathbf{II}^{\alpha}_{j\,m}$ and $\mathbf{III}^{\alpha}_{j\,m}$.

3.1 $\sharp\mathbf{U}_{j\,m}^{\alpha\,\beta}$ and $\sharp\mathbf{V}_{j\,m}^{\alpha\,\beta}$

$\sharp\Omega^\alpha$ is a distribution defined in $\mathbb{R}^n$ by analytic continuation from

$$\sharp\Omega^\alpha(x) \;=\; \pi^{\alpha-\frac{n+2}{2}}\Gamma^{-1}\left(\alpha-\tfrac{n}{2}\right)\left(\frac{1}{1-|x|^2}\right)_+^{\frac{n+2}{2}-\alpha}, \qquad \mathbf{Re}\alpha > \tfrac{n}{2}. \tag{3. 22}$$

Equivalently, it can be defined by

$$\begin{aligned}\sharp\widehat{\Omega}^\alpha(\xi) \;&=\; \left(\frac{1}{|\xi|}\right)^{\frac{n}{2}-\left[\frac{n+2}{2}-\alpha\right]}\mathbf{J}_{\frac{n}{2}-\left[\frac{n+2}{2}-\alpha\right]}\Big(2\pi|\xi|\Big)\\ &=\; \left(\frac{1}{|\xi|}\right)^{\alpha-1}\mathbf{J}_{\alpha-1}\Big(2\pi|\xi|\Big), \qquad \alpha\in\mathbb{C}.\end{aligned} \tag{3. 23}$$

Let $\mathbf{Re}\alpha \geq \left[\frac{2n}{2n-1}\right]\mathbf{Re}\beta$ and $\frac{2n-1}{4n} < \mathbf{Re}\beta < \frac{2n-1}{2n-2}$. Define

$$\sharp\widehat{\mathbf{P}}_j^{\alpha\,\beta}(\xi) \;=\; \widehat{\phi}(\xi)\int_{2^{j-1}\leq|r|<2^j} e^{-2\pi\mathbf{i}r}\sharp\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr, \qquad j>0. \tag{3. 24}$$

Remark 3.2 *Given $T>0$, we have*

$$\left|\sum_{j>T}\sharp\widehat{\mathbf{P}}_j^{\alpha\,\beta}(\xi)\right| \;\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\;\mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}-\varepsilon} 2^{-T\varepsilon} \tag{3. 25}$$

for some $\varepsilon=\varepsilon(\mathbf{Re}\alpha,\mathbf{Re}\beta)>0$.

Consider

$$\begin{aligned}\sharp\widehat{\mathbf{P}}_j^{\alpha\,\beta}(\xi) \;&=\; \sum_{m=1}^M \sharp\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\beta}(\xi),\\ \sharp\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\beta}(\xi) \;&=\; \widehat{\phi}(\xi)\int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\sharp\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr,\\ \lambda_m\in[2^{j-1},2^j], \qquad &\lambda_0=2^{j-1},\ \lambda_M=2^j \text{ and } 2^{\sigma j-1}\;\leq\;\lambda_m-\lambda_{m-1}<2^{\sigma j}\end{aligned} \tag{3. 26}$$

where $0<\sigma=\sigma(\mathbf{Re}\alpha,\mathbf{Re}\beta)<\frac{1}{2}$ can be chosen sufficiently small.

Let φ_j^ν defined in (3. 14). Assert

$$\begin{aligned}\sharp\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\beta}(\xi) \;&=\; \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}\sharp\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\beta\,\nu}(\xi),\\ \sharp\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\beta\,\nu}(\xi) \;&=\; \varphi_j^\nu(\xi)\widehat{\phi}(\xi)\int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\sharp\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr.\end{aligned} \tag{3. 27}$$

For every $j > 0$ and $m = 1, \dots, M$, we define

$$ {}^\sharp\mathbf{U}^{\alpha\,\beta}_{j\,m}(x) \;=\; \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} {}^\sharp\mathbf{U}^{\alpha\,\beta\,\nu}_{j\,m}(x), \qquad {}^\sharp\mathbf{U}^{\alpha\,\beta\,\nu}_{j\,m}(x) \;=\; {}^\sharp\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\Big[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\Big], \tag{3. 28} $$

$$ {}^\sharp\mathbf{V}^{\alpha\,\beta}_{j\,m}(x) \;=\; \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} {}^\sharp\mathbf{V}^{\alpha\,\beta\,\nu}_{j\,m}(x), \qquad {}^\sharp\mathbf{V}^{\alpha\,\beta\,\nu}_{j\,m}(x) \;=\; {}^\sharp\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\Psi^{\alpha\,\nu}_{j\,m}(x). \tag{3. 29} $$

Proposition One *Let* $\mathbf{Re}\alpha \geq \left[\frac{2n}{2n-1}\right]\mathbf{Re}\beta$ *and* $\frac{2n-1}{4n} < \mathbf{Re}\beta < \frac{2n-1}{2n-2}$. *We have*

$$ \left|{}^\sharp\widehat{\mathbf{U}}^{\alpha\,\beta}_{j\,m}(\xi)\right| \;\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\;\mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j}2^{j/2}2^{-\varepsilon j} \tag{3. 30} $$

and

$$ \left|{}^\sharp\widehat{\mathbf{V}}^{\alpha\,\beta}_{j\,m}(\xi)\right| \;\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\;\mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j} \tag{3. 31} $$

for some $\varepsilon = \varepsilon(\mathbf{Re}\alpha, \mathbf{Re}\beta) > 0$.

3.2 ${}^\flat\mathbf{U}^{\alpha\,\beta}_{j\,m}$ and ${}^\flat\mathbf{V}^{\alpha\,\beta}_{j\,m}$

${}^\flat\Omega^\alpha$ is a distribution defined in $\mathbb{R}^n$ by analytic continuation from

$$ {}^\flat\Omega^\alpha(x) \;=\; \pi^{\alpha-1}\Gamma^{-1}(\alpha)\left(\frac{1}{1-|x|^2}\right)_+^{1-\alpha}, \qquad \mathbf{Re}\alpha > 0. \tag{3. 32} $$

Equivalently, it can be defined by

$$ \begin{aligned} {}^\flat\widehat{\Omega}^\alpha(\xi) \;&=\; \left(\frac{1}{|\xi|}\right)^{\frac{n}{2}-(1-\alpha)} \mathbf{J}_{\frac{n}{2}-(1-\alpha)}\Big(2\pi|\xi|\Big) \\ &=\; \left(\frac{1}{|\xi|}\right)^{\frac{n-1}{2}+\alpha-\frac{1}{2}} \mathbf{J}_{\frac{n-1}{2}+\alpha-\frac{1}{2}}\Big(2\pi|\xi|\Big), \qquad \alpha\in\mathbb{C}. \end{aligned} \tag{3. 33} $$

Let $\mathbf{Re}\alpha > 0$ and $0 < \mathbf{Re}\beta < \frac{1}{2}$. Define

$$ {}^\flat\widehat{\mathbf{P}}^{\alpha\,\beta}_{j}(\xi) \;=\; \widehat{\phi}(\xi)\int_{2^{j-1}\leq|r|<2^j} e^{-2\pi\mathbf{i}r}\,{}^\flat\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr, \qquad j > 0. \tag{3. 34} $$

Consider

$$ \begin{aligned} {}^\flat\widehat{\mathbf{P}}^{\alpha\,\beta}_{j}(\xi) \;&=\; \sum_{m=1}^M {}^\flat\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}(\xi), \\ {}^\flat\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}(\xi) \;&=\; \widehat{\phi}(\xi)\int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\,{}^\flat\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr, \end{aligned} \tag{3. 35} $$

$$ \lambda_m\in[2^{j-1},2^j], \qquad \lambda_0 = 2^{j-1},\ \lambda_M = 2^j \ \text{ and } \ 2^{\sigma j-1} \;\leq\; \lambda_m-\lambda_{m-1} < 2^{\sigma j} $$

where $0 < \sigma = \sigma(\mathbf{Re}\alpha) < \frac{1}{2}$ can be chosen sufficiently small.

Let φ_j^ν defined in (3. 14). Assert

$$
\begin{aligned}
{}^\flat\widehat{\mathbf{P}}^{\alpha\ \beta}_{j\ m}(\xi) &= \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} {}^\flat\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\xi),\\
{}^\flat\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\xi) &= \varphi_j^\nu(\xi)\widehat{\phi}(\xi)\int_{\lambda_{m-1}\le|r|<\lambda_m} e^{-2\pi\mathbf{i}r}{}^\flat\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr.
\end{aligned}
\tag{3. 36}
$$

For every $j>0$, $m=1,\ldots,M$ and $\ell=1,\ldots,L$, we define

$$
{}^\flat\mathbf{U}^{\alpha\ \beta}_{j\ m}(x) = \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} {}^\flat\mathbf{U}^{\alpha\ \beta\ \nu}_{j\ m}(x), \qquad {}^\flat\mathbf{U}^{\alpha\ \beta\ \nu}_{j\ m}(x) = {}^\flat\mathbf{P}^{\alpha\ \beta\ \nu}_{j\ m}(x)\Big[1-\Psi^{\alpha\ \nu}_{j\ m}(x)\Big], \tag{3. 37}
$$

$$
{}^\flat\mathbf{V}^{\alpha\ \beta}_{j\ m}(x) = \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} {}^\flat\mathbf{V}^{\alpha\ \beta\ \nu}_{j\ m}(x), \qquad {}^\flat\mathbf{V}^{\alpha\ \beta\ \nu}_{j\ m}(x) = {}^\flat\mathbf{P}^{\alpha\ \beta\ \nu}_{j\ m}(x)\Psi^{\alpha\ \nu}_{j\ m}(x). \tag{3. 38}
$$

Proposition Two *Let* $\mathbf{Re}\alpha>0$ *and* $0<\mathbf{Re}\beta<\frac{1}{2}$. *We have*

$$
\int_{\mathbb{R}^n}\left|{}^\flat\mathbf{U}^{\alpha\ \beta}_{j\ m}(x)\right|dx \le \mathfrak{B}_{N\ \mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-Nj}2^{-\varepsilon j}, \qquad N\ge 0 \tag{3. 39}
$$

and

$$
\int_{\mathbb{R}^n}\left|{}^\flat\mathbf{V}^{\alpha\ \beta}_{j\ m}(x)\right|dx \le \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j} \tag{3. 40}
$$

for some $\varepsilon=\varepsilon(\mathbf{Re}\alpha)>0$.

4 Proof of Theorem Two

Recall $\mathbf{I}^\alpha f$ defined in (3. 10) and $\widehat{\mathbf{P}}^\alpha_<$, $\widehat{\mathbf{P}}^\alpha_j$ and $\widehat{\mathbf{P}}^\alpha_{j\ m}$ defined in (3. 11) and (3. 12). We have

$$
\begin{aligned}
\mathbf{I}^\alpha f(x) &= \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\left\{\widehat{\mathbf{P}}^\alpha_<(\xi)+\sum_{0<j\le T}\sum_{m=1}^M \widehat{\mathbf{P}}^\alpha_{j\ m}(\xi)+\sum_{j>T}\widehat{\mathbf{P}}^\alpha_j(\xi)\right\}d\xi\\
&= \mathbf{I}^\alpha_< f(x)+\sum_{0<j\le T}\sum_{m=1}^M \mathbf{I}^\alpha_{j\ m}f(x)+\mathbf{R}^\alpha_T f(x);
\end{aligned}
\tag{4. 1}
$$

$$
\mathbf{R}^\alpha_T f(x) = \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\widehat{f}(\xi)\widehat{\phi}(\xi)\left\{\int_{|r|\ge 2^T} e^{-2\pi\mathbf{i}r}\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\alpha-1}dr\right\}d\xi
$$

where $\widehat{\phi}$ is defined in (1. 9) and $\mathbf{supp}\widehat{\phi}\subset\left\{\xi\in\mathbb{R}^n\colon \frac{1}{3}<|\xi|\le 3\right\}$.

The kernel of $\mathbf{I}^\alpha_<$ as shown in (3. 13) is an $\mathbf{L}^1$-function in $\mathbb{R}^n$. From now on, we focus on $\mathbf{I}^\alpha_{j\ m}f, 0<j\le T, m=1,\ldots,M$ and $\mathbf{R}^\alpha_T f$.

Let $\frac{1}{2} < \mathbf{Re}\alpha < 1$. We can find $\mathbf{a}_i = \mathbf{a}_i(\mathbf{Re}\alpha), \mathbf{b}_i = \mathbf{b}_i(\mathbf{Re}\alpha), i = 1, 2$ such that

$$\begin{gathered} \mathbf{a}_1 > 0, \qquad 0 < \mathbf{b}_1 < \frac{1}{2}, \qquad \mathbf{a}_2 \geq \left[\frac{2n}{2n-1}\right]\mathbf{b}_2, \qquad \frac{2n-1}{4n} < \mathbf{b}_2 < \frac{2n-1}{2n-2}, \\ \mathbf{Re}\alpha \ = \ \frac{\mathbf{a}_1}{n} + \mathbf{a}_2\left(\frac{n-1}{n}\right) \ = \ \frac{\mathbf{b}_1}{n} + \mathbf{b}_2\left(\frac{n-1}{n}\right). \end{gathered} \tag{4. 2}$$

As $\mathbf{Re}\alpha \longrightarrow 1$, we necessarily have $\mathbf{b}_1 \longrightarrow \frac{1}{2},\ \mathbf{b}_2 \longrightarrow \frac{2n-1}{2n-2}$ and $\mathbf{a}_1 \longrightarrow 0,\ \mathbf{a}_2 \longrightarrow \frac{n}{n-1}$.

Let $0 \leq \mathbf{Re}z \leq 1$. We define

$$\begin{aligned} \widehat{\Omega}^{\alpha\ z}(\xi) \ &= \ \left(\frac{1}{|\xi|}\right)^{\mathbf{a}_1 z + \mathbf{a}_2(1-z) - \frac{1}{2} + \left(\frac{n-1}{2}\right)z - \frac{1}{2}(1-z) + \mathbf{i}\mathbf{Im}\alpha} \mathbf{J}_{\mathbf{a}_1 z + \mathbf{a}_2(1-z) - \frac{1}{2} + \left(\frac{n-1}{2}\right)z - \frac{1}{2}(1-z) + \mathbf{i}\mathbf{Im}\alpha}(2\pi|\xi|) \\ &= \ \pi^{\mathbf{a}_1 z + \mathbf{a}_2(1-z) - 1 + \left(\frac{n-1}{2}\right)z - \frac{1}{2}(1-z) + \mathbf{i}\mathbf{Im}\alpha} \Gamma^{-1}\left(\mathbf{a}_1 z + \mathbf{a}_2(1-z) + \left(\frac{n-1}{2}\right)z - \frac{1}{2}(1-z) + \mathbf{i}\mathbf{Im}\alpha\right) \\ &\quad \int_{-1}^{1} e^{2\pi\mathbf{i}|\xi|s}(1-s^2)^{\mathbf{a}_1 z + \mathbf{a}_2(1-z) - 1 + \left(\frac{n-1}{2}\right)z - \frac{1}{2}(1-z) + \mathbf{i}\mathbf{Im}\alpha} ds \qquad \text{by (A. 1)} \end{aligned} \tag{4. 3}$$

and

$$\widehat{\mathbf{P}}_j^{\alpha\ z}(\xi) \ = \ e^{\left[z - \frac{1}{n}\right]^2} \widehat{\phi}(\xi) \int_{2^{j-1} \leq |r| < 2^j} e^{-2\pi\mathbf{i}r} \widehat{\Omega}_z^{\alpha\ \mathbf{a}}(r\xi)\omega(r)|r|^{2\left[\mathbf{b}_1 z + \mathbf{b}_2(1-z)\right] - 1 + 2\mathbf{i}\mathbf{Im}\alpha} dr, \qquad j > 0. \tag{4. 4}$$

Let $\lambda_m \in [2^{j-1}, 2^j]$: $\lambda_0 = 2^{j-1}$, $\lambda_M = 2^j$ and $2^{\sigma j - 1} \leq \lambda_m - \lambda_{m-1} < 2^{\sigma j}$ for which $0 < \sigma = \sigma(\mathbf{Re}\alpha) < \frac{1}{2}$ can be chosen sufficiently small. We consider

$$\begin{gathered} \widehat{\mathbf{P}}_j^{\alpha\ z}(\xi) \ = \ \sum_{m=1}^{M} \widehat{\mathbf{P}}_{j\ m}^{\alpha\ z}(\xi), \\ \widehat{\mathbf{P}}_{j\ m}^{\alpha\ z}(\xi) \ = \ e^{\left[z - \frac{1}{n}\right]^2} \widehat{\phi}(\xi) \int_{\lambda_{m-1} \leq |r| < \lambda_m} e^{-2\pi\mathbf{i}r} \widehat{\Omega}_z^{\alpha\ \mathbf{a}}(r\xi)\omega(r)|r|^{2\left[\mathbf{b}_1 z + \mathbf{b}_2(1-z)\right] - 1 + 2\mathbf{i}\mathbf{Im}\alpha} dr. \end{gathered} \tag{4. 5}$$

Observe that $\widehat{\mathbf{P}}_{j\ m}^{\alpha\ z}(\xi)$ is analytic for $0 \leq \mathbf{Re}z \leq 1$. In particular, $\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \frac{1}{n}}(\xi) = \widehat{\mathbf{P}}_{j\ m}^{\alpha}(\xi)$.

Define

$$\mathbf{I}_{j\ m}^{\alpha\ z} f(x) \ = \ \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi} \widehat{f}(\xi) \widehat{\mathbf{P}}_{j\ m}^{\alpha\ z}(\xi) d\xi \tag{4. 6}$$

for every $j > 0$, $m = 1, 2, \ldots, M$ and

$$\mathbf{R}_T^{\alpha\ z} f(x) \ = \ \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi} \widehat{f}(\xi) \sum_{j > T} \widehat{\mathbf{P}}_j^{\alpha\ z}(\xi) d\xi, \qquad T > 0. \tag{4. 7}$$

Given $f \in \mathbf{L}^p(\mathbb{R}^n)$, $g \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^n)$ to be simple functions and $p = \frac{2n}{n+1}$, we define

$$f_z(x) \ = \ \mathbf{sign} f(x)|f(x)|^{\left[\frac{1-z}{2} + z\right]p}, \qquad g_z(x) \ = \ \mathbf{sign} g(x)|g(x)|^{\left[1 - \frac{1-z}{2} - z\right]\frac{p}{p-1}} \tag{4. 8}$$

for $0 \leq \mathbf{Re}z \leq 1$.

Remark 4.1 *Note that* $f_{0+\mathbf{iIm}z} \in \mathbf{L}^2(\mathbb{R}^n)$, $g_{0+\mathbf{iIm}z} \in \mathbf{L}^2(\mathbb{R}^n)$ *and* $f_{1+\mathbf{iIm}z} \in \mathbf{L}^1(\mathbb{R}^n)$, $g_{1+\mathbf{iIm}z} \in \mathbf{L}^\infty(\mathbb{R}^n)$. *At* $z = \frac{1}{n}$, *we find* $f_z = f$ *and* $g_z = g$. *Without loss of generality, assume*

$$\|f\|_{\mathbf{L}^p(\mathbb{R}^n)} = \|g\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^n)} = 1.$$

We claim

$$\int_{\mathbb{R}^n} \mathbf{R}_T^{\alpha\ z} f_z(x) g_z(x) dx \longrightarrow 0, \qquad T \longrightarrow \infty \tag{4. 9}$$

for $0 \leq \mathbf{Re} z \leq 1$.

First, at $\mathbf{Re} z = 0$, we find $\widehat{\mathbf{P}}_j^{\alpha\ 0+\mathbf{iIm}z}(\xi) = e^{\left[\mathbf{iIm}z - \frac{1}{n}\right]^2} \sharp \widehat{\mathbf{P}}_j^{\mathbf{A}_2\ \mathbf{B}_2}(\xi)$ as defined in (3. 24) where $\mathbf{A}_2 = \mathbf{a}_2 + \mathbf{i}\left[\mathbf{a}_1 - \mathbf{a}_2 + \left(\frac{n-1}{2}\right) + \frac{1}{2}\right] \mathbf{Im} z + \mathbf{iIm}\alpha$ and $\mathbf{B}_2 = \mathbf{b}_2 + \mathbf{i}\left[\mathbf{b}_1 - \mathbf{b}_2\right] \mathbf{Im} z + \mathbf{iIm}\alpha$.

Note that $\mathbf{a}_2 \geq \left[\frac{2n}{2n-1}\right] \mathbf{b}_2$ and $\frac{2n-1}{4n} < \mathbf{b}_2 < \frac{2n-1}{2n-2}$ as shown in (4. 2). Recall **Remark 3.2**. We have

$$\begin{aligned} \left| \sum_{j>T} \widehat{\mathbf{P}}_j^{\alpha\ 0+\mathbf{iIm}z}(\xi) \right| &\leq \mathfrak{B}_{\mathbf{a}_2\ \mathbf{b}_2}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, e^{\mathbf{c}|\mathbf{Im}z|}\, e^{-[\mathbf{Im}z]^2} \left| \frac{1}{1-|\xi|} \right|^{\frac{1}{2}-\varepsilon} 2^{-T\varepsilon} \\ &\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \left| \frac{1}{1-|\xi|} \right|^{\frac{1}{2}-\varepsilon} 2^{-T\varepsilon}, \qquad 0 \leq \mathbf{Re} z \leq 1 \end{aligned} \tag{4. 10}$$

for some $\varepsilon = \varepsilon(\mathbf{a}_2, \mathbf{b}_2) = \varepsilon(\mathbf{Re}\alpha) > 0$.

As defined in (4. 4), $\mathbf{supp}\widehat{\mathbf{P}}_j^{\alpha\ z} = \mathbf{supp}\widehat{\phi} \subset \left\{\xi \in \mathbb{R}^n : \frac{1}{3} < |\xi| \leq 3\right\}$. From (4. 7), by using (4. 10) and Schwartz inequality, we find

$$\begin{aligned} \left| \mathbf{R}_T^{\alpha\ 0+\mathbf{iIm}z} f_{0+\mathbf{iIm}z}(x) \right| &\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \left\| \widehat{f}_{0+\mathbf{iIm}z} \right\|_{\mathbf{L}^2(\mathbb{R}^n)} \left\{ \int_{|\xi|\leq 3} \left| \frac{1}{1-|\xi|} \right|^{1-2\varepsilon} d\xi \right\}^{\frac{1}{2}} 2^{-T\varepsilon} \\ &\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-T\varepsilon} \left\| f_{0+\mathbf{iIm}z} \right\|_{\mathbf{L}^2(\mathbb{R}^n)} \qquad \text{by Plancherel theorem.} \end{aligned} \tag{4. 11}$$

Next, at $\mathbf{Re} z = 1$, we have

$$\begin{aligned} \left| \widehat{\Omega}^{\alpha\ 1+\mathbf{iIm}z}(r\xi) \right| &\leq \left(\frac{1}{|r\xi|} \right)^{\mathbf{a}_1 - \frac{1}{2} + \frac{n-1}{2}} \left| \mathbf{J}_{\mathbf{a}_1(1+\mathbf{iIm}z) + \mathbf{ia}_2\mathbf{Im}z - \frac{1}{2} + \left(\frac{n-1}{2}\right)(1+\mathbf{iIm}z) - \frac{\mathbf{i}}{2}\mathbf{Im}z + \mathbf{iIm}\alpha} (2\pi|r\xi|) \right| \\ &\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, e^{\mathbf{c}|\mathbf{Im}z|} \left(\frac{1}{1+|r\xi|} \right)^{\mathbf{a}_1 + \frac{n-1}{2}} \qquad \text{by (A. 5).} \end{aligned} \tag{4. 12}$$

Recall $\omega(r)$ from (3. 10). We find

$$\omega(r) = e^{2\pi \mathbf{i} r} \int_0^1 e^{-2\pi \mathbf{i} \tau r} \tau d\tau = \frac{-1}{2\pi \mathbf{i}} \frac{1}{r} - \frac{1}{4\pi^2 r^2} \left[1 - e^{-2\pi \mathbf{i} r}\right]$$

implying

$$|\omega(r)| \leq \mathfrak{B}\left[1 + |r|\right]^{-1}. \tag{4. 13}$$

Denote χ to be an indicator function. From (4. 4), we have

$$
\begin{aligned}
\left|\sum_{j>T}\widehat{\mathbf{P}}_j^{\alpha~1+\mathbf{iIm}z}(\xi)\right| &= \left|e^{\left[z-\frac{1}{n}\right]^2}\widehat{\phi}(\xi)\int_{|r|\geq 2^T} e^{-2\pi \mathbf{i} r}\widehat{\Omega}^{\alpha~1+\mathbf{iIm}z}(r\xi)\omega(r)|r|^{2\left[\mathbf{b}_1 z+\mathbf{b}_2(1-z)\right]-1+2\mathbf{iIm}\alpha}dr\right| \\
&\leq \mathfrak{B} e^{-\left[\mathbf{Im}z\right]^2}\int_{|r|\geq 2^T}\left|\widehat{\Omega}^{\alpha~\mathbf{a}}_{1+\mathbf{iIm}z}(r\xi)\chi_{\left\{\frac{1}{3}<|\xi|\leq 3\right\}}(\xi)\right||r|^{2\mathbf{b}_1-2}dr \qquad \text{by (4. 13)} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, e^{\mathbf{c}|\mathbf{Im}z|}\, e^{-\left[\mathbf{Im}z\right]^2}\int_{|r|\geq 2^T}\left(\frac{1}{1+|r|}\right)^{\mathbf{a}_1+\frac{n-1}{2}}|r|^{2\mathbf{b}_1-2}dr \qquad \text{by (4. 12)} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\left(\frac{1}{1+2^T}\right)^{\mathbf{a}_1+\frac{n-1}{2}} 2^{T(2\mathbf{b}_1-1)} \qquad \mathbf{a}_1>0,\ 0<\mathbf{b}_1<\tfrac{1}{2} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-T\left(\frac{n-1}{2}\right)}.
\end{aligned}
\tag{4. 14}
$$

Let $\mathbf{R}_T^{\alpha~z}$ defined (4. 7). By using (4. 14), we find

$$
\begin{aligned}
\left|\mathbf{R}_T^{\alpha~1+\mathbf{iIm}z} f_{1+\mathbf{iIm}z}(x)\right| &\leq \int_{|\xi|\leq 3}\left|\widehat{f}_{1+\mathbf{iIm}z}(\xi)\right|\left|\sum_{j>T}\widehat{\mathbf{P}}_j^{\alpha~1+\mathbf{iIm}z}(\xi)\right|d\xi \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-T\left(\frac{n-1}{2}\right)}\left\|\widehat{f}_{1+\mathbf{iIm}z}\right\|_{\mathbf{L}^\infty(\mathbb{R}^n)} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-T\left(\frac{n-1}{2}\right)}\left\|f_{1+\mathbf{iIm}z}\right\|_{\mathbf{L}^1(\mathbb{R}^n)}.
\end{aligned}
\tag{4. 15}
$$

Consider f_z, g_z given in (4. 8) and **Remark 4.1**. From (4. 11) and (4. 15), by applying the Three-Line lemma, we obtain

$$
\left|\mathbf{R}_T^{\alpha~z} f_z(x)\right| \leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-T\varepsilon}, \qquad 0\leq \mathbf{Re}z\leq 1. \tag{4. 16}
$$

This estimate further implies (4. 9). Consequently, we have

$$
\begin{aligned}
\int_{\mathbb{R}^n}\sum_{j>0}\mathbf{I}_j^{\alpha~z} f_z(x)g_z(x)dx &= \lim_{T\longrightarrow\infty}\sum_{0<j\leq T}\int_{\mathbb{R}^n}\mathbf{I}_j^{\alpha~z} f_z(x)g_z(x)dx \\
&= \lim_{T\longrightarrow\infty}\sum_{0<j\leq T}\sum_{m=1}^{M}\int_{\mathbb{R}^n}\mathbf{I}_{j~m}^{\alpha~z} f_z(x)g_z(x)dx
\end{aligned}
\tag{4. 17}
$$

for $0\leq \mathbf{Re}z\leq 1$.

Given $j>0$, $\left\{\xi_j^\nu\right\}_\nu$ is a collection of points almost equally distributed on $\mathbb{S}^{n-1}$ having a grid length between $2^{-j/2-1}$ and $2^{-j/2}$. See **(1)-(2)** above (1. 22)-(1. 23).

Let φ_j^ν defined in (3. 14) and $\Psi_{j\,m}^{\alpha}, \Psi_{j\,m}^{\alpha\,\mu}$ defined in (3. 18). Consider

$$\begin{aligned}
\widehat{\mathbf{P}}_{j\,m}^{\alpha\,z}(\xi) &= \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} \widehat{\mathbf{P}}_{j\,m}^{\alpha\,\nu\,z}(\xi),\\
\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\nu\,z}(\xi) &= e^{\left[z-\frac{1}{n}\right]^2}\widehat{\phi}(\xi)\varphi_j^\nu(\xi)\int_{\lambda_{m-1}\le|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\widehat{\Omega}^{\alpha\,z}(r\xi)\omega(r)|r|^{2\left[\mathbf{b}_1 z+\mathbf{b}_2(1-z)\right]-1+2\mathbf{Im}\alpha}dr.
\end{aligned} \tag{4. 18}$$

For every $j>0$ and $m=1,\ldots,M$, we define

$$\begin{aligned}
\mathbf{II}_{j\,m}^{\alpha\,z}f(x) &= \int_{\mathbb{R}^n} f(x-u)\mathbf{U}_{j\,m}^{\alpha\,z}(u)du,\\
\mathbf{U}_{j\,m}^{\alpha\,z}(x) &= \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}\mathbf{U}_{j\,m}^{\alpha\,\nu\,z}(x), \qquad \mathbf{U}_{j\,m}^{\alpha\,\nu\,z}(x) = \mathbf{P}_{j\,m}^{\alpha\,\nu\,z}(x)\left[1-\Psi_{j\,m}^{\alpha\,\nu}(x)\right]
\end{aligned} \tag{4. 19}$$

and

$$\begin{aligned}
\mathbf{III}_{j\,m}^{\alpha\,z}f(x) &= \int_{\mathbb{R}^n} f(x-u)\mathbf{V}_{j\,m}^{\alpha\,z}(u)du,\\
\mathbf{V}_{j\,m}^{\alpha\,z}(x) &= \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}\mathbf{V}_{j\,m}^{\alpha\,\nu\,z}(x), \qquad \mathbf{V}_{j\,m}^{\alpha\,\nu\,z}(x) = \mathbf{P}_{j\,m}^{\alpha\,\nu\,z}(x)\Psi_{j\,m}^{\alpha\,\nu}(x).
\end{aligned} \tag{4. 20}$$

In particular, at $z=\frac{1}{n}$, we find $\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\nu\,\frac{1}{n}}(\xi)=\widehat{\mathbf{P}}_{j\,m}^{\alpha\,\nu}(\xi)$ and $\mathbf{II}_{j\,m}^{\alpha\,\frac{1}{n}}f(x)=\mathbf{II}_j^\alpha f(x)$, $\mathbf{III}_{j\,m}^{\alpha\,\frac{1}{n}}f(x)=\mathbf{III}_j^\alpha f(x)$ as defined in (3. 20)-(3. 21).

Let $\mathbf{Re}z=0$. From (4. 19)-(4. 20), we have $\mathbf{II}_{j\,m}^{\alpha\,0+\mathbf{iIm}z}f(x)=e^{\left[\mathbf{iIm}z-\frac{1}{n}\right]^2}f*{}^\sharp\mathbf{U}_{j\,m}^{\mathbf{A}_2\,\mathbf{B}_2}(x)$ and $\mathbf{III}_{j\,m}^{\alpha\,0+\mathbf{iIm}z}f(x)=e^{\left[\mathbf{iIm}z-\frac{1}{n}\right]^2}f*{}^\sharp\mathbf{V}_j^{\mathbf{A}_2\,\mathbf{B}_2}f(x)$ of which ${}^\sharp\mathbf{U}_{j\,m}^{\mathbf{A}_2\,\mathbf{B}_2}$, ${}^\sharp\mathbf{V}_{j\,m}^{\mathbf{A}_2\,\mathbf{B}_2}$ are defined in (3. 28)-(3. 29) with $\mathbf{A}_2=\mathbf{a}_2+\mathbf{i}\left[\mathbf{a}_1-\mathbf{a}_2+\left(\frac{n-1}{2}\right)+\frac{1}{2}\right]\mathbf{Im}z+\mathbf{iIm}\alpha$ and $\mathbf{B}_2=\mathbf{b}_2+\mathbf{i}\left[\mathbf{b}_1-\mathbf{b}_2\right]\mathbf{Im}z+\mathbf{iIm}\alpha$.

Recall $\mathbf{a}_2\ge\left[\frac{2n}{2n-1}\right]\mathbf{b}_2$, $\frac{2n-1}{4n}<\mathbf{b}_2<\frac{2n-1}{2n-2}$ in (4. 2). By applying (3. 30) in **Proposition One** together with Plancherel theorem and Schwartz inequality, we find

$$\begin{aligned}
&\int_{\mathbb{R}^n}\mathbf{II}_{j\,m}^{\alpha\,0+\mathbf{iIm}z}f_{0+\mathbf{iIm}z}(x)g_{0+\mathbf{iIm}z}(x)dx\\
&\le \mathfrak{B}_{\mathbf{a}_2\,\mathbf{b}_2}\,e^{\mathbf{c}|\mathbf{Im}\alpha|}\,2^{-(1-\sigma)j}2^{j/2}2^{-\varepsilon j}\,\left\|f_{0+\mathbf{iIm}z}\right\|_{\mathbf{L}^2(\mathbb{R}^n)}\left\|g_{0+\mathbf{iIm}z}\right\|_{\mathbf{L}^2(\mathbb{R}^n)}\\
&\le \mathfrak{B}_{\mathbf{Re}\alpha}\,e^{\mathbf{c}|\mathbf{Im}\alpha|}\,2^{-(1-\sigma)j}2^{j/2}2^{-\varepsilon j}
\end{aligned} \tag{4. 21}$$

for some $\varepsilon=\varepsilon(\mathbf{a}_2,\mathbf{b}_2)=\varepsilon(\mathbf{Re}\alpha)>0$.

On the other hand, by using (3. 31) in **Proposition One**, we have

$$\int_{\mathbb{R}^n}\mathbf{III}_{j\,m}^{\alpha\,0+\mathbf{iIm}z}f_{0+\mathbf{iIm}z}(x)g_{0+\mathbf{iIm}z}(x)dx \le \mathfrak{B}_{\mathbf{Re}\alpha}\,e^{\mathbf{c}|\mathbf{Im}\alpha|}\,2^{-(1-\sigma)j}2^{-\varepsilon j}. \tag{4. 22}$$

Let $\mathbf{Re}z=1$. From (4. 19)-(4. 20), we find $\mathbf{II}_{j\,m}^{\alpha\,1+\mathbf{iIm}z}f(x)=e^{\left[\mathbf{iIm}z+\frac{n-1}{n}\right]^2}f*{}^\flat\mathbf{U}_{j\,m}^{\mathbf{A}_1\,\mathbf{B}_1}(x)$ and $\mathbf{III}_{j\,m}^{\alpha\,1+\mathbf{iIm}z}f(x)=e^{\left[\mathbf{iIm}z+\frac{n-1}{n}\right]^2}f*{}^\flat\mathbf{V}_{j\,m}^{\mathbf{A}_1\,\mathbf{B}_1}f(x)$ where ${}^\flat\mathbf{U}_{j\,m}^{\mathbf{A}_1\,\mathbf{B}_1}$, ${}^\flat\mathbf{V}_{j\,m}^{\mathbf{A}_1\,\mathbf{B}_1}$ are defined in (3. 37)-(3. 38) with $\mathbf{A}_1=\mathbf{a}_1+\mathbf{i}\left[\mathbf{a}_1-\mathbf{a}_2+\left(\frac{n-1}{2}\right)+\frac{1}{2}\right]\mathbf{Im}z+\mathbf{iIm}\alpha$ and $\mathbf{B}_1=\mathbf{b}_1+\mathbf{i}\left[\mathbf{b}_1-\mathbf{b}_2\right]\mathbf{Im}z+\mathbf{iIm}\alpha$.

Note that $\mathbf{a}_1 > 0, 0 < \mathbf{b}_1 < \frac{1}{2}$ in (4. 2). By applying (3. 39) in **Proposition Two** and using Hölder inequality, we have

$$
\begin{aligned}
&\int_{\mathbb{R}^n} \mathbf{II}^{\alpha}_{j\,m}{}^{1+\mathbf{iIm}z} f_{1+\mathbf{iIm}z}(x) g_{1+\mathbf{iIm}z}(x) dx \\
&\leq\ \mathfrak{B}_{N\ \mathbf{a}_1\ \mathbf{b}_1}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j} 2^{-jN} 2^{-j\varepsilon}\ \left\| f_{1+\mathbf{iIm}z}\right\|_{\mathbf{L}^1(\mathbb{R}^n)} \left\| g_{1+\mathbf{iIm}z}\right\|_{\mathbf{L}^\infty(\mathbb{R}^n)} \\
&\leq\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, 2^{-(1-\sigma)j} 2^{-Nj} 2^{-\varepsilon j}, \qquad N \geq 0
\end{aligned}
\tag{4. 23}
$$

for some $\varepsilon = \varepsilon(\mathbf{a}_1, \mathbf{b}_1) = \varepsilon(\mathbf{Re}\alpha) > 0$.

On the other hand, by using (3. 40) in **Proposition Two**, we find

$$
\int_{\mathbb{R}^n} \mathbf{III}^{\alpha}_{j\,m}{}^{1+\mathbf{iIm}z} f_{1+\mathbf{iIm}z}(x) g_{1+\mathbf{iIm}z}(x) dx\ \leq\ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j} 2^{-\varepsilon j}. \tag{4. 24}
$$

Recall **Remark 4.1**. At $z = \frac{1}{n}$, $f_z = f \in \mathbf{L}^p(\mathbb{R}^n)$ and $g_z = g \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^n)$ which are simple functions and $p = \frac{2n}{n+1}$. From (4. 21) and (4. 23) with $N \geq \frac{n-1}{2}$, the Three-Line lemma implies

$$
\begin{aligned}
\int_{\mathbb{R}^n} \mathbf{II}^{\alpha}_{j} f(x) g(x) dx\ &=\ \sum_{m=1}^{M} \int_{\mathbb{R}^n} \mathbf{II}^{\alpha}_{j\,m} f(x) g(x) dx \\
&\leq\ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\varepsilon j}.
\end{aligned}
\tag{4. 25}
$$

Recall **Remark 1.2**. There are at most a constant multiple of $2^{(1-\sigma)j}$ many $1 \leq m \leq M$.

From (4. 22) and (4. 24), by applying the Three-Line lemma again, we have

$$
\begin{aligned}
\int_{\mathbb{R}^n} \mathbf{III}^{\alpha}_{j} f(x) g(x) dx\ &=\ \sum_{m=1}^{M} \int_{\mathbb{R}^n} \mathbf{III}^{\alpha}_{j\,m} f(x) g(x) dx \\
&\leq\ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\varepsilon j}.
\end{aligned}
\tag{4. 26}
$$

A standard argument extends (4. 25)-(4. 26) to every $f \in \mathbf{L}^p(\mathbb{R}^n)$ and $g \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^n)$ with $\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)} = \left\|g\right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^n)} = 1$.

Recall (4. 17) . Take into account $\mathbf{I}^{\alpha}_j = \mathbf{II}^{\alpha}_j + \mathbf{III}^{\alpha}_j$, $j > 0$. By summing over every $j > 0$ and using (4. 25)-(4. 26), we conclude

$$
\left\|\mathbf{I}^{\alpha} f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}\ \leq\ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad p = \frac{2n}{n+1}. \tag{4. 27}
$$

Lastly, the adjoint operator of $\mathbf{I}^{\alpha}$ defined in (1. 10) can be simply given by $\mathbf{I}^{\overline{\alpha}}$ where $\overline{\alpha}$ is the complex conjugate of α. Clearly, it satisfies all above estimates. By duality, we obtain

$$
\left\|\mathbf{I}^{\alpha} f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}\ \leq\ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^n)}, \qquad p = \frac{2n}{n-1}. \tag{4. 28}
$$

The $\mathbf{L}^p$-norm inequality in (1. 11) follows by applying Riesz-Thorin interpolation theorem.

5 Proof of Proposition One

Recall (3. 26). Let $\mathbf{Re}\alpha \geq \left[\frac{2n}{2n-1}\right]\mathbf{Re}\beta$ and $\frac{2n-1}{4n} < \mathbf{Re}\beta < \frac{2n-1}{2n-2}$. We have

$$
{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi) \;=\; \sum_{m=1}^{M} \widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi),
$$

$$
{}^\sharp\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi) \;=\; \widehat{\phi}(\xi) \int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\,{}^\sharp\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr
$$

where ${}^\sharp\widehat{\Omega}^\alpha$ is defined in (3. 23).

Moreover, $\widehat{\phi}$ is defined in (1. 9). ω is given in (3. 10). $\lambda_m \in [2^{j-1}, 2^j]$: $\lambda_0 = 2^{j-1}$, $\lambda_M = 2^j$ and $2^{\sigma j-1} \leq \lambda_m - \lambda_{m-1} < 2^{\sigma j}$ for which $0 < \sigma = \sigma(\mathbf{Re}\alpha, \mathbf{Re}\beta) < \frac{1}{2}$ can be chosen sufficiently small.

Lemma One *Given $j > 0$ and $m = 1, 2, \ldots, M$, we have*

$$
\left|{}^\sharp\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi)\right| \;\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{j/2}2^{-\varepsilon j}, \qquad |1-|\xi|| \leq 2^{-j} \tag{5. 1}
$$

and

$$
\left|{}^\sharp\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi)\right| \;\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j}\left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}} 2^{-\varepsilon j}, \qquad |1-|\xi|| > 2^{-j} \tag{5. 2}
$$

for some $\varepsilon = \varepsilon(\mathbf{Re}\alpha, \mathbf{Re}\beta) > 0$.

Remark 5.1 *As required for later estimates, we choose $\sigma = \sigma(\mathbf{Re}\alpha, \mathbf{Re}\beta)$ to satisfy $n\sigma \leq \frac{1}{2}\varepsilon$.*

Recall **Remark 1.2**. There are at most a constant multiple of $2^{(1-\sigma)j}$ many $\lambda_m \in [2^{j-1}, 2^j]$. Consequently, by applying (5. 1) and (5. 2), we find

$$
\begin{aligned}
\left|{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right| \;&\leq\; \sum_{m=1}^{M}\left|{}^\sharp\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi)\right| \\
&\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{j/2}2^{-\varepsilon j}, \qquad |1-|\xi|| \leq 2^{-j}
\end{aligned} \tag{5. 3}
$$

and

$$
\begin{aligned}
\left|{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right| \;&\leq\; \sum_{m=1}^{M}\left|{}^\sharp\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi)\right| \\
&\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}} 2^{-\varepsilon j}, \qquad |1-|\xi|| > 2^{-j}.
\end{aligned} \tag{5. 4}
$$

Recall **Remark 3.2**. Consider

$$\left|\sum_{j>T} {}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right| \ \le \ \sum_{j>T:|1-|\xi||\le 2^{-j}} \left|{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right| + \sum_{j>T:|1-|\xi||>2^{-j}} \left|{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right|.$$

We have

$$\begin{aligned}
\sum_{j>T:|1-|\xi||\le 2^{-j}} \left|{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right| \ &\le\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} \sum_{j>T:|1-|\xi||\le 2^{-j}} 2^{j/2}2^{-\varepsilon j} \qquad \text{by (5. 3)} \\
&\le\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}-\frac{\varepsilon}{2}} \sum_{j>T} 2^{-\left(\frac{\varepsilon}{2}\right)j} \qquad (5.\ 5) \\
&\le\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}-\frac{\varepsilon}{2}} 2^{-T\varepsilon/2}
\end{aligned}$$

and

$$\begin{aligned}
\sum_{j>T:|1-|\xi||>2^{-j}} \left|{}^\sharp\widehat{\mathbf{P}}_j^{\alpha\ \beta}(\xi)\right| \ &\le\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha}e^{\mathbf{cIm}\beta} \sum_{j>T:|1-|\xi||>2^{-j}} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}} 2^{-\varepsilon j} \qquad \text{by (5. 4)} \\
&\le\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha}e^{\mathbf{cIm}\beta} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}-\frac{\varepsilon}{2}} \sum_{j>T} 2^{-\left(\frac{\varepsilon}{2}\right)j} \qquad (5.\ 6) \\
&\le\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha}e^{\mathbf{cIm}\beta} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}-\frac{\varepsilon}{2}} 2^{-T\varepsilon/2}.
\end{aligned}$$

From (5. 5)-(5. 6), we conclude (3. 25).

5.1 Proof of Lemma One

Recall ${}^\sharp\Omega^\alpha$ defined in (3. 22)-(3. 23). We have

$$\begin{aligned}
{}^\sharp\widehat{\mathbf{P}}_{j\ m}^{\alpha\ \beta}(\xi) \ &=\ \widehat{\phi}(\xi) \int_{\lambda_{m-1}\le|r|<\lambda_m} e^{-2\pi \mathbf{i} r}\, {}^\sharp\Omega^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr \\
&=\ \widehat{\phi}(\xi) \int_{\lambda_{m-1}\le|r|<\lambda_m} e^{-2\pi \mathbf{i} r} \left(\frac{1}{|r\xi|}\right)^{\alpha-1} \mathbf{J}_{\alpha-1}\left(2\pi|r\xi|\right)\omega(r)|r|^{2\beta-1}dr.
\end{aligned}$$

Note that $\mathbf{supp}\widehat{\phi} \subset \left\{\xi\in\mathbb{R}^n\colon \frac{1}{3} < |\xi| \le 3\right\}$ and

$$\omega(r) \ = \ \frac{-1}{2\pi\mathbf{i}}\frac{1}{r} - \frac{1}{4\pi^2r^2}\left[1-e^{-2\pi\mathbf{i}r}\right]$$

as shown in (3. 10).

For symmetry reason, we consider $r > 0$ only. Define

$$ {}^{\sharp}\widehat{\mathbf{Q}}_{j\ m}^{\alpha\ \beta}(\xi) ~=~ \widehat{\phi}(\xi) \int_{\lambda_{m-1}}^{\lambda_m} e^{-2\pi \mathbf{i} r} \left(\frac{1}{r|\xi|} \right)^{\alpha-1} \mathbf{J}_{\alpha-1} \left(2\pi r|\xi| \right) r^{2\beta-2} dr \tag{5. 7} $$

and

$$ {}^{\sharp}\widehat{\mathbf{R}}_{j\ m}^{\alpha\ \beta}(\xi) ~=~ \widehat{\phi}(\xi) \int_{\lambda_{m-1}}^{\lambda_m} e^{-2\pi \mathbf{i} r} \left[1 - e^{-2\pi \mathbf{i} r} \right] \left(\frac{1}{r|\xi|} \right)^{\alpha-1} \mathbf{J}_{\alpha-1} \left(2\pi r|\xi| \right) r^{2\beta-3} dr. \tag{5. 8} $$

Because $\mathbf{Re}\alpha \geq \left[\frac{2n}{2n-1} \right] \mathbf{Re}\beta$ and $\frac{2n-1}{4n} < \mathbf{Re}\beta < \frac{2n-1}{2n-2}$, we find

$$ \mathbf{Re}\alpha ~\geq~ \left[\frac{2n}{2n-1} \right] \mathbf{Re}\beta ~>~ \frac{1}{2}, \qquad 2\mathbf{Re}\beta - \mathbf{Re}\alpha ~\leq~ \left[\frac{2n-2}{2n-1} \right] \mathbf{Re}\beta ~<~ 1. \tag{5. 9} $$

By using the norm estimate of Bessel functions in (A. 5), we have

$$ \begin{aligned} \left| {}^{\sharp}\Omega^{\alpha}(r\xi) \right| ~&=~ \left(\frac{1}{r|\xi|} \right)^{\mathbf{Re}\alpha-1} \left| \mathbf{J}_{\alpha-1} \left(2\pi r|\xi| \right) \right| \\ &\leq~ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \left\{ \frac{1}{1+|r\xi|} \right\}^{\mathbf{Re}\alpha-\frac{1}{2}}. \end{aligned} $$

This further implies

$$ \begin{aligned} \left| {}^{\sharp}\widehat{\mathbf{R}}_{j}^{\alpha\ \beta}(\xi) \right| ~&\leq~ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \int_{\lambda_{m-1}}^{\lambda_m} \left\{ \frac{1}{1+r} \right\}^{\mathbf{Re}\alpha-\frac{1}{2}} r^{2\mathbf{Re}\beta-3} dr \qquad \left(\tfrac{1}{3} < |\xi| \leq 3 \right) \\ &\leq~ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \int_{\lambda_{m-1}}^{\lambda_m} r^{2\mathbf{Re}\beta-\mathbf{Re}\alpha-2-\frac{1}{2}} dr \\ &\leq~ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-2-\frac{1}{2}\right]j} \left| \lambda_m - \lambda_{m-1} \right| \\ &\leq~ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-2-\frac{1}{2}\right]j} 2^{\sigma j} \\ &=~ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j} 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-1\right]j} 2^{-j/2}, \qquad 2\mathbf{Re}\beta - \mathbf{Re}\alpha - 1 < 0. \end{aligned} \tag{5. 10} $$

Next, we aim to show

$$ \left| {}^{\sharp}\widehat{\mathbf{Q}}_{j\ m}^{\alpha\ \beta\ \ell}(\xi) \right| ~\leq~ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{C}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j} 2^{j/2} 2^{-j\varepsilon}, \qquad |1-|\xi|| \leq 2^{-j} \tag{5. 11} $$

and

$$ \left| {}^{\sharp}\widehat{\mathbf{Q}}_{j\ m}^{\alpha\ \beta\ \ell}(\xi) \right| ~\leq~ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{C}|\mathbf{Im}\alpha|} e^{\mathbf{C}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j} \left| \frac{1}{1-|\xi|} \right|^{\frac{1}{2}} 2^{-j\varepsilon}, \qquad |1-|\xi|| > 2^{-j} \tag{5. 12} $$

for some $\varepsilon = \varepsilon(\mathbf{Re}\alpha, \mathbf{Re}\beta) > 0$.

Later, it should be clear that the same estimates regarding ${}^{\sharp}\widehat{\mathbf{Q}}_{j\ m}^{\alpha\ \beta}$ and ${}^{\sharp}\widehat{\mathbf{R}}_{j\ m}^{\alpha\ \beta}$ remain to be true for $r \in [-\lambda_m, -\lambda_{m-1}]$. Therefore, we conclude (5. 1) and (5. 2) from (5. 10) and (5. 11)-(5. 12).

By using the asymptotic expansion of Bessel functions in (A. 2), we write

$$ {}^{\sharp}\widehat{\mathbf{Q}}_{j\ m}^{\alpha\ \beta}(\xi) \;=\; \widehat{\phi}(\xi)\int_{\lambda_{m-1}}^{\lambda_m} e^{-2\pi\mathbf{i}r}\Big[{}^{\sharp}\mathfrak{S}^{\alpha}\left(r|\xi|\right)+{}^{\sharp}\mathcal{E}^{\alpha}\left(r|\xi|\right)\Big]r^{2\beta-2}dr \tag{5. 13}$$

where

$$ {}^{\sharp}\mathfrak{S}^{\alpha}\left(\rho\right) \;=\; \frac{1}{\pi}\left(\frac{1}{\rho}\right)^{\alpha-\frac{1}{2}}\cos\left[2\pi\rho-\frac{\pi}{2}\alpha+\frac{\pi}{4}\right], \qquad \rho>0 \tag{5. 14}$$

and

$$ \left|{}^{\sharp}\mathcal{E}^{\alpha}\left(\rho\right)\right| \;\leq\; \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\begin{cases}\rho^{-\mathbf{Re}\alpha+\frac{1}{2}}, & 0<\rho\leq 1,\\ \rho^{-\mathbf{Re}\alpha-\frac{1}{2}}, & \rho>1.\end{cases} \tag{5. 15}$$

By using (5. 15), we find

$$\begin{aligned} \left|\widehat{\phi}(\xi)\int_{\lambda_{m-1}}^{\lambda_m} e^{-2\pi\mathbf{i}r\,\sharp}\mathcal{E}^{\alpha}\left(r|\xi|\right)r^{2\beta-2}dr\right| \;&\leq\; \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\int_{\lambda_{m-1}}^{\lambda_m}\left(\frac{1}{r}\right)^{\mathbf{Re}\alpha+\frac{1}{2}} r^{2\mathbf{Re}\beta-2}dr \\ &\leq\; \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-2-\frac{1}{2}\right]j}2^{\sigma j} \\ &=\; \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j}2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-1\right]j}2^{-j/2}. \end{aligned} \tag{5. 16}$$

Because of Euler's formulae, we replace the cosine function in (5. 14) with $e^{\pm 2\pi\mathbf{i}\rho}$ multiplied by $\mathfrak{B}_{\mathbf{Re}\alpha}e^{\mathbf{c}|\mathbf{Im}\alpha|}$. Assert

$$\begin{aligned} {}^{\sharp}\widehat{\mathbf{A}}_{j\ m}^{\alpha\ \beta}(\xi) \;&=\; \widehat{\phi}(\xi)\int_{\lambda_{m-1}}^{\lambda_m} e^{2\pi\mathbf{i}\left[|\xi|-1\right]r}\left(\frac{1}{r|\xi|}\right)^{\alpha-\frac{1}{2}} r^{2\beta-2}dr \\ &=\; |\xi|^{\frac{1}{2}-\alpha}\widehat{\phi}(\xi)\int_{\lambda_{m-1}}^{\lambda_m} e^{2\pi\mathbf{i}\left[|\xi|-1\right]r}r^{2\beta-\alpha+\frac{1}{2}-2}dr. \end{aligned} \tag{5. 17}$$

Suppose $|1-|\xi||\leq 2^{-j}$. We have

$$\begin{aligned} \left|{}^{\sharp}\widehat{\mathbf{A}}_{j\ m}^{\alpha\ \beta}(\xi)\right| \;&\leq\; \mathfrak{B}_{\mathbf{Re}\alpha}\int_{\lambda_{m-1}}^{\lambda_m} r^{2\mathbf{Re}\beta-\mathbf{Re}\alpha+\frac{1}{2}-2}dr \\ &\leq\; \mathfrak{B}_{\mathbf{Re}\alpha}\, 2^{j/2}2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-2\right]j}2^{\sigma j} \\ &=\; \mathfrak{B}_{\mathbf{Re}\alpha}\, 2^{-(1-\sigma)j}2^{j/2}2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-1\right]j}. \end{aligned} \tag{5. 18}$$

On the other hand, suppose $|1-|\xi||>2^{-j}$. By integration by parts *w.r.t* r in (5. 17), we find

$$\begin{aligned} {}^{\sharp}\widehat{\mathbf{A}}_{j\ m}^{\alpha\ \beta}(\xi) \;&=\; \frac{1}{2\pi\mathbf{i}}|\xi|^{\frac{1}{2}-\alpha}\widehat{\phi}(\xi)\frac{1}{|\xi|-1}e^{2\pi\mathbf{i}\left[|\xi|-1\right]r}r^{2\beta-\alpha+\frac{1}{2}-2}\Bigg|_{\lambda_{m-1}}^{\lambda_m} \\ &\quad -\frac{1}{2\pi\mathbf{i}}\left[2\beta-\alpha-\tfrac{3}{2}\right]|\xi|^{\frac{1}{2}-\alpha}\widehat{\phi}(\xi)\frac{1}{|\xi|-1}\int_{\lambda_{m-1}}^{\lambda_m} e^{2\pi\mathbf{i}\left[|\xi|-1\right]r}r^{2\beta-\alpha-\frac{1}{2}-2}dr. \end{aligned} \tag{5. 19}$$

From (5. 19), by using the mean-value theorem on the boundary term, we have

$$\begin{aligned}
\left|{}^\sharp\widehat{\mathbf{A}}^{\alpha\,\beta}_{j\,m}(\xi)\right| &\leq \mathfrak{B}_{\mathbf{Re}\alpha\,\mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} e^{\mathbf{cIm}\beta} \left|\frac{1}{1-|\xi|}\right| 2^{j\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-\frac{1}{2}-2\right]} |\lambda_m-\lambda_{m-1}| \\
&+ \mathfrak{B}_{\mathbf{Re}\alpha\,\mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} e^{\mathbf{cIm}\beta} \left|\frac{1}{1-|\xi|}\right| \int_{\lambda_{m-1}}^{\lambda_m} r^{2\mathbf{Re}\beta-\mathbf{Re}\alpha-\frac{1}{2}-2} dr \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha\,\mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} e^{\mathbf{cIm}\beta} \left|\frac{1}{1-|\xi|}\right| 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-\frac{1}{2}-2\right]j} 2^{\sigma j} \qquad (5.\ 20)\\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha\,\mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} e^{\mathbf{cIm}\beta}\, 2^{-(1-\sigma)j} \left|\frac{1}{1-|\xi|}\right| 2^{-j/2} 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-1\right]j} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha\,\mathbf{Re}\beta}\, e^{\mathbf{cIm}\alpha} e^{\mathbf{cIm}\beta}\, 2^{-(1-\sigma)j} \left|\frac{1}{1-|\xi|}\right|^{\frac{1}{2}} 2^{\left[2\mathbf{Re}\beta-\mathbf{Re}\alpha-1\right]j}.
\end{aligned}$$

Our estimates in (5. 18)-(5. 20) remain to be true if $e^{2\pi\mathbf{i}r|\xi|}$ in (5. 17) is replaced by $e^{-2\pi\mathbf{i}r|\xi|}$. Together with (5. 16), we conclude (5. 11)-(5. 12). This finishes the proof of **Lemma One**.

5.2 Size of ${}^\sharp\widehat{\mathbf{U}}^{\alpha\,\beta}_{j\,m}(\xi)$ and ${}^\sharp\widehat{\mathbf{V}}^{\alpha\,\beta}_{j\,m}(\xi)$

We begin to prove (3. 30) and (3. 31) in **Proposition One**. Recall (3. 27). We have

$$\begin{aligned}
{}^\sharp\widehat{\mathbf{P}}^{\alpha\,\beta\,\nu}_{j\,m}(\xi) &= \varphi^\nu_j(\xi)\widehat{\phi}(\xi) \int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\,{}^\sharp\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1} dr \\
&= \varphi^\nu_j(\xi)\,{}^\sharp\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}(\xi)
\end{aligned}$$

where φ^ν_j is defined in (3. 14). Moreover,

$${}^\sharp\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}(\xi) = \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\sharp\widehat{\mathbf{P}}^{\alpha\,\beta\,\nu}_{j\,m}(\xi) = {}^\sharp\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}(\xi) \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \varphi^\nu_j(\xi).$$

Remark 5.2 *For φ^ν_j defined in (3. 14), it is easy to see*

$$0\leq\varphi^\nu_j(\xi)\leq 1, \qquad \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \varphi^\nu_j(\xi) = 1.$$

By applying (5. 1)-(5. 2) in **Lemma One**, we find

$$\left|{}^\sharp\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}(\xi)\right| \leq \mathfrak{B}_{\mathbf{Re}\alpha\,\mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j} 2^{j/2} 2^{-\varepsilon j}. \qquad (5.\ 21)$$

Recall $\Psi^{\alpha}_{j\,m}, \Psi^{\alpha\,\nu}_{j\,m}$ defined in (3. 18) and $^\sharp\mathbf{U}^{\alpha\,\beta}_{j\,m}, {^\sharp}\mathbf{V}^{\alpha\,\beta}_{j\,m}$ defined in (3. 28)-(3. 29). We find

$$\begin{aligned}
{^\sharp}\mathbf{U}^{\alpha\,\beta}_{j\,m}(x)+{^\sharp}\mathbf{V}^{\alpha\,\beta}_{j\,m}(x) &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {^\sharp}\mathbf{U}^{\alpha\,\beta}_{j\,m}(x)+{^\sharp}\mathbf{V}^{\alpha\,\beta}_{j\,m}(x)\\
&= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {^\sharp}\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\left[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\right]+{^\sharp}\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\Psi^{\alpha\,\nu}_{j\,m}(x)\\
&= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {^\sharp}\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x) \;=\; {^\sharp}\mathbf{P}^{\alpha\,\beta}_{j\,m}(x).
\end{aligned}$$

Hence, (3. 30) can be obtained by putting together (3. 31) and (5. 21).

Our task is left to show

$$\left|{^\sharp}\widehat{\mathbf{V}}^{\alpha\,\beta}_{j\,m}(\xi)\right| \le \mathfrak{B}_{\mathbf{Re}\alpha\;\mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\,2^{-(1-\sigma)j}2^{-j\varepsilon},\qquad \varepsilon=\varepsilon(\mathbf{Re}\alpha,\mathbf{Re}\beta)>0. \tag{5. 22}$$

Let ${^\sharp}\mathbf{V}^{\alpha\,\beta}_{j\,m}, {^\sharp}\mathbf{V}^{\alpha\,\beta\,\nu}_{j\,m}$ defined in (3. 29). We have

$$\begin{aligned}
{^\sharp}\widehat{\mathbf{V}}^{\alpha\,\beta}_{j\,m}(\xi) &= \int_{\mathbb{R}^n} e^{-2\pi\mathbf{i}x\cdot\xi}\left\{\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \Psi^{\alpha\,\nu}_{j\,m}(x){^\sharp}\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\right\}dx\\
&= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbb{R}^n}\widehat{\Psi}^{\alpha\,\nu}_{j\,m}(\xi-\zeta){^\sharp}\widehat{\mathbf{P}}^{\alpha\,\beta\,\nu}_{j\,m}(\zeta)d\zeta.
\end{aligned}\tag{5. 23}$$

On the other hand, for $\Psi^{\alpha\,\nu}_{j\,m}$ defined in (3. 18), we find

$$\widehat{\Psi}^{\alpha\,\nu}_{j\,m}(\xi-\zeta) = \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot(\xi-\zeta)}\varphi\left[2^{-\sigma j-1}\left|\lambda_m-\left(\mathbf{L}^T_\nu x\right)_1\right|\right]\prod_{i=2}^n\varphi\left[2^{-\left[\frac{1}{2}+\sigma\right]j}\left|\left(\mathbf{L}^T_\nu x\right)_i\right|\right]dx. \tag{5. 24}$$

$\mathbf{L}_\nu$ is an $n\times n$-orthogonal matrix with $\det\mathbf{L}_\nu=1$. Moreover, $\mathbf{L}^T_\nu\xi^\nu_j=(1,0)^T\in\mathbb{R}\times\mathbb{R}^{n-1}$. Write $\zeta=\mathbf{L}_\nu\eta$ and $x=\mathbf{L}_\nu u$ inside (5. 23)-(5. 24). We have

$$\widehat{\Psi}^{\alpha\,\nu}_{j\,m}(\xi-\mathbf{L}_\mu\eta) = \int_{\mathbb{R}^n} e^{-2\pi\mathbf{i}\left[\mathbf{L}^T_\nu\xi-\eta\right]\cdot u}\varphi\left[2^{-\sigma j-1}\left|\lambda_m-u_1\right|\right]\prod_{i=2}^n\varphi\left[2^{-\left[\frac{1}{2}+\sigma\right]j}|u_i|\right]du \tag{5. 25}$$

and

$${^\sharp}\widehat{\mathbf{P}}^{\alpha\,\beta\,\nu}_{j\,m}(\mathbf{L}_\nu\eta) = \varphi^\nu_j(\mathbf{L}_\nu\eta)\widehat{\phi}(\mathbf{L}_\nu\eta)\int_{\lambda_{m-1}\le|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\,{^\sharp}\widehat{\Omega}^\alpha(r\mathbf{L}_\nu\eta)\omega(r)|r|^{2\beta-1}dr.$$

Recall $\varphi\in C^\infty_o(\mathbb{R})$ such that $\varphi(t)=1$ if $|t|\le1$ and $\varphi(t)=0$ for $|t|>2$. We find

$$\begin{aligned}
&\mathbf{vol\ supp}\left\{\varphi\left[2^{-\sigma j-1}|\lambda_m-u_1|\right]\prod_{i=2}^n\varphi\left[2^{-\left[\frac{1}{2}+\sigma\right]j}|u_i|\right]\right\}\\
&\lesssim 2^{(n-1)\left[\frac{1}{2}+\sigma\right]j}2^{\sigma j} = 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}.
\end{aligned}\tag{5. 26}$$

From (5. 25) and (5. 26), we have

$$\left|\widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi - \mathbf{L}_\mu\eta)\right| \lesssim 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}. \tag{5. 27}$$

Observe that

$$\begin{aligned}
&\partial_{u_1}\varphi\left[2^{-\sigma j-1}\left|\lambda_m \pm u_1\right|\right] = 0, \qquad \text{if} \qquad |\lambda_m \pm u_1| < 2^{\sigma j+1},\\
&\partial_{u_i}\varphi\left[2^{-\left[\frac{1}{2}+\sigma\right]j}\left|u_i\right|\right] = 0 \qquad \text{if} \qquad |u_i| < 2^{\left[\frac{1}{2}+\sigma\right]j}, \ \ i = 2,\ldots,n.
\end{aligned}$$

A direct computation shows

$$\begin{aligned}
&\left|\partial^N_{u_1}\varphi\left[2^{-\sigma j-2}\left|\lambda_m \pm u_1\right|\right]\right| \leq \mathfrak{B}_N\, 2^{-N\sigma j},\\
&\left|\partial^N_{u_i}\varphi\left[2^{-\left[\frac{1}{2}+\sigma\right]j}\left|u_i\right|\right]\right| \leq \mathfrak{B}_N\, 2^{-N\left[\frac{1}{2}+\sigma\right]j}, \qquad i = 2,\ldots,n
\end{aligned} \tag{5. 28}$$

for every $N \geq 0$.

Case 1 Suppose $|\xi| \leq \frac{1}{10}$ or $|\xi| > 10$. For $\frac{1}{3} < |\eta| \leq 3$, we either have $\left|\left[\mathbf{L}^T_\mu\xi - \eta\right]_1\right| > \frac{1}{5\sqrt{n}}$ or $\left|\left[\mathbf{L}^T_\mu\xi - \eta\right]_i\right| > \frac{1}{5\sqrt{n}}$ for some $i = 2,\ldots,n$.

By integration by parts *w.r.t* u_1 or u_i inside (5. 25) and using (5. 26)-(5. 28), we find

$$\left|\widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi - \mathbf{L}_\mu\eta)\right| \leq \mathfrak{B}_N\, 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}\, 2^{-N\sigma j}, \qquad N \geq 0.$$

By choosing N sufficiently large depending on $\sigma = \sigma(\mathbf{Re}\alpha, \mathbf{Re}\beta) > 0$, we obtain

$$\left|\widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi - \mathbf{L}_\mu\eta)\right| \leq \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}. \tag{5. 29}$$

Case 2 Let $\frac{1}{10} < |\xi| \leq 10$. We must have

$$\xi \in \Gamma^\mu_j = \left\{\xi \in \mathbb{R}^n \colon \left|\frac{\xi}{|\xi|} - \xi^\mu_j\right| < 2^{-j/2+1}\right\} \qquad \text{for some } \ \xi^\mu_j \in \mathbb{S}^{n-1}.$$

Lemma 5.1 *Let* $\frac{1}{10} < |\xi| \leq 10$ *and* $\frac{1}{3} < |\eta| \leq 3$. *Suppose* $\left|\xi^\mu_j - \xi^\nu_j\right| \geq \mathbf{c}2^{-j/2}$ *for some* $\mathbf{c} > 0$ *large. Either we have* $\left|\left[\mathbf{L}^T_\nu\xi - \eta\right]_1\right| \approx 2$ *or* $\left|\left[\mathbf{L}^T_\nu\xi - \eta\right]_\imath\right| \gtrsim 2^{-j/2}$ *for some* $\imath \in \{2,\ldots,n\}$ *whenever* $\mathbf{L}_\nu\eta \in \Gamma^\nu_j$.

Proof Note that $\mathbf{L}^T_\nu\xi^\nu_j = (1,0)^T \in \mathbb{R}\times\mathbb{R}^{n-1}$. Consider $\mathbf{L}_\nu\eta \in \Gamma^\nu_j$. We have $\left|\mathbf{L}^T_\nu\xi^\nu_j - (1,0)^T\right| = 0$. By using the triangle inequality, we find

$$\begin{aligned}
\frac{|\eta_i|}{|\eta|} \leq \left|\frac{\eta}{|\eta|} - (1,0)^T\right| &\leq \left|\frac{\eta}{|\eta|} - \mathbf{L}^T_\nu\xi^\nu_j\right| + \left|\mathbf{L}^T_\nu\xi^\nu_j - (1,0)^T\right|\\
&\leq 2^{-j/2+1}, \qquad i = 2,\ldots,n.
\end{aligned} \tag{5. 30}$$

On the other hand, recall $\left|\xi_j^\mu - \xi_j^\nu\right| \geq \mathbf{c}2^{-j/2}$ for some $\mathbf{c}$ large. Let $\xi \in \Gamma_j^\mu$. By using the triangle inequality, we find

$$\begin{aligned}
\left|\frac{\mathbf{L}_\nu^T \xi}{|\xi|} - (1,0)^T\right| &= \left|\frac{\xi}{|\xi|} - \xi_j^\nu\right| \\
&\geq \left|\xi_j^\mu - \xi_j^\nu\right| - \left|\frac{\xi}{|\xi|} - \xi_j^\mu\right| \\
&\geq \mathbf{c}2^{-j/2} - 2^{-j/2+1} \\
&= [\mathbf{c} - 2]\, 2^{-j/2}.
\end{aligned} \tag{5. 31}$$

Suppose $\left|\left(\frac{\mathbf{L}_\nu^T \xi}{|\xi|}\right)_1 - 1\right| \leq \left[\sum_{i=2}^n \left(\frac{\mathbf{L}_\nu^T \xi}{|\xi|}\right)_i^2\right]^{\frac{1}{2}}$. The estimate in (5. 31) further implies

$$\begin{aligned}
\left|\frac{\left(\mathbf{L}_\nu^T \xi\right)_\imath}{|\xi|}\right| &\geq \frac{1}{\sqrt{2}\sqrt{n-1}} \left|\frac{\mathbf{L}_\nu^T \xi}{|\xi|} - (1,0)^T\right| \\
&\geq \frac{1}{\sqrt{2}\sqrt{n-1}} [\mathbf{c} - 2]\, 2^{-j/2} \qquad \text{for some} \qquad \imath \in \{2, \ldots, n\}.
\end{aligned} \tag{5. 32}$$

Suppose $\left|\left(\frac{\mathbf{L}_\nu^T \xi}{|\xi|}\right)_1 - 1\right| > \left[\sum_{i=2}^n \left(\frac{\mathbf{L}_\nu^T \xi}{|\xi|}\right)_i^2\right]^{\frac{1}{2}}$. As a geometric fact, $\frac{\mathbf{L}_\nu^T \xi}{|\xi|}$ belongs to the semi-sphere opposite to $(1,0)^T$. We either have $\left(\frac{\mathbf{L}_\nu^T \xi}{|\xi|}\right)_1 \approx -1$ or $\left|\left(\frac{\mathbf{L}_\nu^T \xi}{|\xi|}\right)_\imath\right| > \frac{1}{\sqrt{2}\sqrt{n-1}} [\mathbf{c} - 2]\, 2^{-j/2}$ for some $\imath \in \{2, \ldots, n\}$ as (5. 32).

By taking into account $\frac{1}{3} < |\eta| \leq 3$ and $\frac{1}{10} < |\xi| \leq 10$ in (5. 30) and (5. 31)-(5. 32), we find

$$|\eta_\imath| \leq 3\, 2^{-j/2+1}, \qquad \left|\left(\mathbf{L}_\nu^T \xi\right)_\imath\right| \geq \frac{1}{10\sqrt{2}\sqrt{n-1}} [\mathbf{c} - 2]\, 2^{-j/2}.$$

Consequently, $\left|[\mathbf{L}_\nu^T \xi - \eta]_\imath\right| \gtrsim 2^{-j/2}$ provided that $\mathbf{c}$ is large. $\square$

Let $\xi \in \Gamma_j^\mu$ and $\left|\xi_j^\mu - \xi_j^\nu\right| \geq \mathbf{c}2^{-j/2}$ for some $\mathbf{c}$ large. By applying **Lemma 5.1** and using the estimates in (5. 26)-(5. 28), an N-fold integration by parts either *w.r.t* u_1 or $u_\imath$ inside (5. 25) shows

$$\begin{aligned}
\left|\widehat{\Psi}^{\alpha\ \mu}_{j\ m}(\xi - \mathbf{L}_\mu \eta)\right| &\leq \mathfrak{B}_N\, 2^{\left(\frac{n-1}{2}\right)j} 2^{n\sigma j} \left[2^{-N\sigma j} + 2^{N\left(\frac{1}{2}\right)j} 2^{-N\left[\frac{1}{2}+\sigma\right]j}\right] \\
&\leq \mathfrak{B}_N\, 2^{\left(\frac{n-1}{2}\right)j} 2^{n\sigma j}\, 2^{-N\sigma j} \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}
\end{aligned} \tag{5. 33}$$

for N sufficiently large depending on $\sigma = \sigma(\mathbf{Re}\alpha, \mathbf{Re}\beta) > 0$.

Now, we assert

$$\int_{\mathbb{R}^n} \widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi - \mathbf{L}_\mu \eta) {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\mu \eta) d\eta \ = \ \sum_{k=0}^{\infty} \int_{\mathcal{S}_k} \widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi - \mathbf{L}_\mu \eta) {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\mu \eta) d\eta,$$
$$\mathcal{S}_k \ = \ \left\{ \eta \in \mathbb{R}^n : 2^{-k} \leq \big|1 - |\eta|\big| < 2^{-k+1} \right\}. \tag{5. 34}$$

Observe that

$$\mathbf{vol}\left\{ \mathcal{S}_k \cap \mathbf{supp} {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m} \right\} \ \lesssim \ 2^{-\left(\frac{n-1}{2}\right) j} 2^{-k}. \tag{5. 35}$$

Recall ${}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\mu \eta) = \varphi^\nu_j(\mathbf{L}_\mu \eta) {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta}_{j\ m}(\mathbf{L}_\mu \eta)$ where $0 \leq \varphi^\nu_j(\mathbf{L}_\mu \eta) \leq 1$. See **Remark 5.2**.

For $j \leq k$, we have $\Big|1 - |\eta|\Big| \leq 2^{-k+1} \leq 2^{-j+1}$. By applying (5. 1) in **Lemma One**, we find

$$\begin{aligned} \left| {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\mu \eta) \right| \ &\leq \ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j} 2^{j/2} 2^{-\varepsilon j} \\ &\leq \ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-(1-\sigma)j} 2^{k/2} 2^{-\varepsilon j}. \end{aligned} \tag{5. 36}$$

For $j > k$, we have $\Big|1 - |\eta|\Big| > 2^{-k} > 2^{-j}$. By applying (5. 2) in **Lemma One**, we find

$$\begin{aligned} \left| {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j}(\mathbf{L}_\mu \eta) \right| \ &\leq \ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j} \left| \frac{1}{1 - |\eta|} \right|^{\frac{1}{2}} 2^{-\varepsilon j} \\ &\leq \ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j} 2^{k/2} 2^{-\varepsilon j}. \end{aligned} \tag{5. 37}$$

From (5. 35) and (5. 36)-(5. 37), we obtain

$$\begin{aligned} &\sum_{k=0}^{\infty} \int_{\mathcal{S}_k} \left| {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\mu \eta) \right| d\eta \\ &\leq \ \sum_{k=0}^{\infty} \left| {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j}(\mathbf{L}_\mu \eta) \right| \mathbf{vol}\left\{ \mathcal{S}_k \cap \mathbf{supp} {}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m} \right\} \\ &\leq \ \sum_{k=0}^{\infty} \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} e^{\mathbf{c}|\mathbf{Im}\beta|} 2^{-(1-\sigma)j} 2^{k/2} 2^{-\varepsilon j}\ 2^{-\left(\frac{n-1}{2}\right) j} 2^{-k} \\ &= \ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j} 2^{-\left(\frac{n-1}{2}\right) j} 2^{-\varepsilon j} \sum_{k=0}^{\infty} 2^{-k/2} \\ &\leq \ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j} 2^{-\left(\frac{n-1}{2}\right) j} 2^{-\varepsilon j}. \end{aligned} \tag{5. 38}$$

Recall **Remark 1.3**. There are at most a constant multiple of $2^{\left(\frac{n-1}{2}\right) j}$ many $\xi^\nu_j \in \mathbb{S}^{n-1}$.

Consider $|\xi|\leq\frac{1}{10}$ or $|\xi|>10$. From (5. 23) and (5. 34), we have

$$
\begin{array}{lll}
\left|{}^\sharp\widehat{\mathbf{V}}^{\alpha\ \beta}_{j\ m}(\xi)\right| & \leq & \displaystyle\sum_{\nu:\xi_j^\nu\in\mathbb{Z}^{n-1}}\sum_{k=0}^\infty\left|\int_{\mathcal{S}_k}\widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi-\mathbf{L}_\nu\eta){}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\nu\eta)d\eta\right| \\[2ex]
& \leq & \displaystyle\mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}\sum_{k=0}^\infty\int_{\mathcal{S}_k}\left|{}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\mu\eta)\right|d\eta \qquad \text{by (5. 29)} \\[2ex]
& \leq & \displaystyle\mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}2^{-(1-\sigma)j}2^{-\left(\frac{n-1}{2}\right)j}2^{-\varepsilon j} \qquad \text{by (5. 38)} \\[2ex]
& \leq & \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{\left(\frac{n-1}{2}\right)j}\, 2^{-(1-\sigma)j}2^{-\left(\frac{n-1}{2}\right)j}2^{-\varepsilon j} \qquad \text{by } \textbf{Remark 1.3} \\[2ex]
& = & \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j}2^{-\varepsilon j}.
\end{array}
\tag{5. 39}
$$

Let $\frac{1}{10}<|\xi|\leq 10$ and $\xi\in\Gamma_j^\mu$ for some $\xi_j^\mu\in\mathbb{S}^{n-1}$. From (5. 23) and (5. 34), we have

$$
\begin{array}{l}
\left|{}^\sharp\widehat{\mathbf{V}}^{\alpha\ \beta}_{j\ m}(\xi)\right| \leq \displaystyle\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}\sum_{k=0}^\infty\left|\int_{\mathcal{S}_k}\widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi-\mathbf{L}_\nu\eta){}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\nu\eta)d\eta\right| \\[2ex]
= \displaystyle\left[\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1},\left|\xi_j^\mu-\xi_j^\nu\right|\geq\mathbf{c}2^{-j/2}}+\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1},\left|\xi_j^\mu-\xi_j^\nu\right|\leq\mathbf{c}2^{-j/2}}\right]\sum_{k=0}^\infty\left|\int_{\mathcal{S}_k}\widehat{\Psi}^{\alpha\ \nu}_{j\ m}(\xi-\mathbf{L}_\nu\eta){}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\nu\eta)d\eta\right| \quad \text{for some } \mathbf{c} \text{ large} \\[2ex]
\leq \displaystyle\mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1},\left|\xi_j^\mu-\xi_j^\nu\right|\geq\mathbf{c}2^{-j/2}}\sum_{k=0}^\infty\int_{\mathcal{S}_k}\left|{}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\nu\eta)\right|d\eta \qquad \text{by (5. 33)} \\
+\ \displaystyle\mathfrak{B}\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1},\left|\xi_j^\mu-\xi_j^\nu\right|\leq\mathbf{c}2^{-j/2}}2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}\sum_{k=0}^\infty\int_{\mathcal{S}_k}\left|{}^\sharp\widehat{\mathbf{P}}^{\alpha\ \beta\ \nu}_{j\ m}(\mathbf{L}_\nu\eta)\right|d\eta \qquad \text{by (5. 27)} \\[2ex]
\leq \displaystyle\mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1},\left|\xi_j^\mu-\xi_j^\nu\right|\geq\mathbf{c}2^{-j/2}}2^{-(1-\sigma)j}2^{-\left(\frac{n-1}{2}\right)j}2^{-\varepsilon j} \\
+\ \displaystyle\mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1},\left|\xi_j^\mu-\xi_j^\nu\right|\leq\mathbf{c}2^{-j/2}}2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}\,2^{-(1-\sigma)j}2^{-\left(\frac{n-1}{2}\right)j}2^{-\varepsilon j} \qquad \text{by (5. 38)} \\[2ex]
\leq \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{\left(\frac{n-1}{2}\right)j}\, 2^{-(1-\sigma)j}2^{-\left(\frac{n-1}{2}\right)j}2^{-\varepsilon j} \\
+\ \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}\, 2^{-(1-\sigma)j}2^{-\left(\frac{n-1}{2}\right)j}2^{-\varepsilon j} \qquad \text{by } \textbf{Remark 1.3} \\[2ex]
\leq \mathfrak{B}_{\mathbf{Re}\alpha\ \mathbf{Re}\beta}e^{\mathbf{c}|\mathbf{Im}\alpha|}e^{\mathbf{c}|\mathbf{Im}\beta|}\, 2^{-(1-\sigma)j}2^{n\sigma j}2^{-\varepsilon j}.
\end{array}
\tag{5. 40}
$$

We obtain (5. 22) provided that $n\sigma\leq\frac{1}{2}\varepsilon$ as discussed in **Remark 5.1**.

6 On the $\mathbf{L}^1$-norm of ${}^\flat\mathbf{U}^{\alpha\,\beta}_{j\,m}$ and ${}^\flat\mathbf{V}^{\alpha\,\beta}_{j\,m}$

Recall ${}^\flat\Omega^\alpha$ defined in (3. 32)-(3. 33) and ${}^\flat\widehat{\mathbf{P}}^{\alpha\,\beta}_{j\,m}$ defined in (3. 36) for $\mathbf{Re}\alpha > 0$ and $0 < \mathbf{Re}\beta < \frac{1}{2}$. Given $j > 0$ and $m = 1, 2, \ldots, M$, we have

$$
{}^\flat\mathbf{P}^{\alpha\,\beta}_{j\,m}(x) = \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x),
$$

$$
{}^\flat\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x) = \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\left\{\int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\varphi^\nu_j(\xi)\widehat{\phi}(\xi){}^\flat\widehat{\Omega}^\alpha(r\xi)\omega(r)|r|^{2\beta-1}dr\right\}d\xi,
$$

$$
{}^\flat\widehat{\Omega}^\alpha(\xi) = \left(\frac{1}{|\xi|}\right)^{\frac{n-1}{2}+\mathbf{Re}\alpha-\frac{1}{2}} \mathbf{J}_{\frac{n-1}{2}+\alpha-\frac{1}{2}}\Big(2\pi|\xi|\Big).
$$

$\lambda_m \in [2^{j-1}, 2^j]$: $\lambda_0 = 2^{j-1}$, $\lambda_M = 2^j$ and $2^{\sigma j-1} \leq \lambda_m - \lambda_{m-1} < 2^{\sigma j}$ for some $0 < \sigma = \sigma(\mathbf{Re}\alpha) < \frac{1}{2}$ which can be chosen sufficiently small.

φ^ν_j is defined in (3. 14) whose support is contained in $\Gamma^\nu_j = \left\{\xi \in \mathbb{R}^n\colon \left|\frac{\xi}{|\xi|} - \xi^\nu_j\right| < 2^{-j/2+1}\right\}$.

Recall **Remark 1.3**. There are at most a constant multiple of $2^{\left(\frac{n-1}{2}\right)j}$ many $\xi^\nu_j \in \mathbb{S}^{n-1}$.

$\widehat{\phi}$ is defined in (1. 9) and $\mathbf{supp}\widehat{\phi} \subset \left\{\xi \in \mathbb{R}^n\colon \frac{1}{3} < |\xi| \leq 3\right\}$. Moreover, $\omega(r)$ given in (3. 10) satisfies $|\omega(r)| \lesssim \Big[1 + |r|\Big]^{-1}$ as shown in (4. 13).

By using the norm estimate of Bessel functions in (A. 5), we find

$$
\begin{aligned}
\left|{}^\flat\widehat{\Omega}^\alpha(r\xi)\right| &= \left(\frac{1}{|r\xi|}\right)^{\frac{n-1}{2}+\mathbf{Re}\alpha-\frac{1}{2}} \left|\mathbf{J}_{\frac{n-1}{2}+\alpha-\frac{1}{2}}\Big(2\pi|r\xi|\Big)\right| \\
&\leq \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\left(\frac{1}{1+|r\xi|}\right)^{\frac{n-1}{2}+\mathbf{Re}\alpha}.
\end{aligned}
\tag{6. 1}
$$

Fubini's theorem allows us to write

$$
{}^\flat\mathbf{P}^{\alpha\,\beta}_{j\,m}(x) = \int_{\lambda_{m-1}\leq|r|<\lambda_m} e^{-2\pi\mathbf{i}r}\,{}^\flat\mathbf{P}^\alpha_{r\,j}(x)\omega(r)|r|^{2\beta-1}dr \tag{6. 2}
$$

for which

$$
\begin{aligned}
{}^\flat\mathbf{P}^\alpha_{r\,j}(x) &= \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{P}^{\alpha\,\nu}_{r\,j}(x) \\
{}^\flat\mathbf{P}^{\alpha\,\nu}_{r\,j}(x) &= \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\varphi^\nu_j(\xi)\widehat{\phi}(\xi){}^\flat\widehat{\Omega}^\alpha(r\xi)d\xi.
\end{aligned}
\tag{6. 3}
$$

Recall $\Psi^{\alpha}_{j\,m}$, $\Psi^{\alpha\,\nu}_{j\,m}$ defined in (3. 18) and ${}^\flat\mathbf{U}^{\alpha\,\beta}_{j\,m}$, ${}^\flat\mathbf{V}^{\alpha\,\beta}_{j\,m}$ defined in (3. 37)-(3. 38). We have

$$
{}^\flat\mathbf{U}^{\alpha\,\beta}_{j\,m}(x) ~=~ \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{U}^{\alpha\,\beta\,\nu}_{j\,m}(x), \qquad {}^\flat\mathbf{U}^{\alpha\,\beta\,\nu}_{j\,m}(x) ~=~ {}^\flat\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\Big[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\Big],
$$

$$
{}^\flat\mathbf{V}^{\alpha\,\beta}_{j\,m}(x) ~=~ \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{V}^{\alpha\,\beta\,\nu}_{j\,m}(x), \qquad {}^\flat\mathbf{V}^{\alpha\,\beta\,\nu}_{j\,m}(x) ~=~ {}^\flat\mathbf{P}^{\alpha\,\beta\,\nu}_{j\,m}(x)\Psi^{\alpha\,\nu}_{j\,m}(x).
$$

In order to prove (3. 39)-(3. 40) in **Proposition Two**, it is suffice to show

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left|{}^\flat\mathbf{P}^{\alpha\,\nu}_{r\,j}(x)\right|\Big[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\Big]dx ~\leq~ \mathfrak{B}_{N\,\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-Nj}2^{-\varepsilon j}, \qquad N\geq 0 \tag{6. 4}
$$

and

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \Psi^{\alpha\,\nu}_{j\,m}(x)\left|{}^\flat\mathbf{P}^{\alpha\,\nu}_{r\,j}(x)\right|dx ~\leq~ \mathfrak{B}_{\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\varepsilon j} \tag{6. 5}
$$

for some $\varepsilon=\varepsilon(\mathbf{Re}\alpha)>0$ whenever $\lambda_{m-1}\leq|r|<\lambda_m$.

This is an immediate consequence of using Minkowski integral inequality together with $|\omega(r)|\leq\mathfrak{B}\Big[1+|r|\Big]^{-1}$, $0<\mathbf{Re}\beta<\frac{1}{2}$ and $|\lambda_m-\lambda_{m-1}|\leq 2^{\sigma j}$.

Let ${}^\flat\mathbf{P}^{\alpha\,\nu}_{r\,j}$ defined in (6. 3). We write

$$
\begin{aligned}
{}^\flat\mathbf{P}^{\alpha\,\nu}_{r\,j}(x) ~&=~ \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{|r\xi|}\right)^{\frac{n-1}{2}+\alpha-\frac{1}{2}} \mathbf{J}_{\frac{n-1}{2}+\alpha-\frac{1}{2}}\Big(2\pi|r\xi|\Big)d\xi \\
&=~ {}^\flat\mathbf{Q}^{\alpha\,\nu}_{r\,j}(x)+\sum_{k=1}^{L}{}^\flat\mathbf{Q}^{\alpha\,\nu}_{r\,j\,\ell}(x)+{}^\flat\mathbf{R}^{\alpha\,\nu}_{r\,j\,L}(x)
\end{aligned}
\tag{6. 6}
$$

by using the asymptotic expansion of Bessel functions in (A. 2)-(A. 4).

First,

$$
{}^\flat\mathbf{Q}^{\alpha\,\nu}_{r\,j}(x) ~=~ \frac{1}{\pi}\int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{|r\xi|}\right)^{\frac{n-1}{2}+\alpha}\cos\left[2\pi|r\xi|-\left(\frac{n}{2}+\alpha-1\right)\frac{\pi}{2}-\frac{\pi}{4}\right]d\xi. \tag{6. 7}
$$

Next, there are finitely many

$$
\begin{aligned}
&{}^\flat\mathbf{Q}^{\alpha\,\nu}_{r\,j\,k}(x) ~=~ \\
&\quad(2\pi)^{-2k}\frac{\mathbf{a}_k}{\pi}\int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{|r\xi|}\right)^{\frac{n-1}{2}+\alpha+2k}\cos\left[2\pi|r\xi|-\left(\frac{n}{2}+\alpha-1\right)\frac{\pi}{2}-\frac{\pi}{4}\right]d\xi \\
&+(2\pi)^{-2k+1}\frac{\mathbf{b}_k}{\pi}\int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{|r\xi|}\right)^{\frac{n-1}{2}+\alpha+2k-1}\sin\left[2\pi|r\xi|-\left(\frac{n}{2}+\alpha-1\right)\frac{\pi}{2}-\frac{\pi}{4}\right]d\xi
\end{aligned}
\tag{6. 8}
$$

where $\mathbf{a}_k,\mathbf{b}_k$ are constants given in (A. 2) for $k=1,2,\ldots,L$.

Finally, the remainder term

$$
\begin{aligned}
{}^\flat\mathbf{R}^{\alpha\ \nu}_{r\ j\ L}(x) \ &= \ \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}x\cdot\xi}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\mathbf{H}^\alpha\Big(2\pi|r\xi|\Big)d\xi, \\
\partial^\gamma_\xi\mathbf{H}^\alpha(\rho) \ &= \ \mathbf{O}\left(\rho^{-2L-\frac{1}{2}+|\gamma|}\right), \qquad \rho\longrightarrow\infty
\end{aligned}
\tag{6. 9}
$$

where the implied constant is bounded by $\mathfrak{B}_{\gamma\ \mathbf{Re}\alpha}e^{\mathbf{c}|\mathbf{Im}\alpha|}$.

Recall φ^ν_j defined in (3. 14). We find $\partial_\xi\varphi^\nu_j(\xi) = 0$ whenever $\left|\frac{\xi}{|\xi|}-\xi^\nu_j\right| \leq 2^{-j/2}$. A direct computation shows $\left|\partial^\gamma_\xi\left[\varphi^\nu_j(\xi)\right]\right| \leq \mathfrak{B}_\gamma 2^{|\gamma|j/2}$ for every multi-index γ whenever $\frac{1}{3}<|\xi|\leq 3$.

We have $|r|\approx 2^j$ if $\lambda_{m-1}\leq|r|<\lambda_m$. An $(n+1)$-fold integration by parts *w.r.t* ξ inside (6. 9) gives

$$
\left|{}^\flat\mathbf{R}^{\alpha\ \nu}_{r\ j\ L}(x)\right| \ \leq \ \mathfrak{B}_{\mathbf{Re}\alpha}\ e^{\mathbf{c}|\mathbf{Im}\alpha|}\left(\frac{1}{1+|x|}\right)^{n+1} 2^{-\left[2L+\frac{1}{2}-(n+1)\right]j}. \tag{6. 10}
$$

Moreover, consider

$$
{}^\flat\mathbf{R}^{\alpha}_{r\ j\ L}(x) \ = \ \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{R}^{\alpha\ \nu}_{r\ j\ L}(x).
$$

Recall **Remark 1.3**. There are at most $\mathfrak{B}2^{\left(\frac{n-1}{2}\right)j}$ many $\xi^\nu_j\in\mathbb{S}^{n-1}$. For L chosen sufficiently large, both ${}^\flat\mathbf{R}^{\alpha\ \nu}_{r\ j\ L}$ and ${}^\flat\mathbf{R}^{\alpha}_{r\ j\ L}$ are integrable functions.

Let $\mathbf{Re}\alpha>0$ and $0<\mathbf{Re}\beta<\frac{1}{2}$. We aim to show

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbb{R}^n}\left|{}^\flat\mathbf{Q}^{\alpha\ \nu}_{r\ j}(x)\right|\left[1-\Psi^{\alpha\ \nu}_{j\ m}(x)\right]dx \ \leq \ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\ e^{\mathbf{c}|\mathbf{Im}\alpha|}\ 2^{-Nj}2^{-\varepsilon j}, \qquad N\geq 0 \tag{6. 11}
$$

and

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbb{R}^n}\Psi^{\alpha\ \nu}_{j\ m}(x)\left|{}^\flat\mathbf{Q}^{\alpha\ \nu}_{r\ j}(x)\right|dx \ \leq \ \mathfrak{B}_{\mathbf{Re}\alpha}\ e^{\mathbf{c}|\mathbf{Im}\alpha|}\ 2^{-\varepsilon j} \tag{6. 12}
$$

for some $\varepsilon=\varepsilon(\mathbf{Re}\alpha)>0$ whenever $\lambda_{m-1}\leq|r|<\lambda_m$.

Remark 6.1 *Later, it should be clear that (6. 11)-(6. 12) hold if* ${}^\flat\mathbf{Q}^{\alpha\ \nu}_{r\ j}$ *are replaced by* ${}^\flat\mathbf{Q}^{\alpha\ \nu}_{r\ j\ k}$. *Furthermore, an extra term* $2^{-j[2k-1]}$ *appears on the right hand side of these inequalities:*

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbb{R}^n}\left|{}^\flat\mathbf{Q}^{\alpha\ \nu}_{r\ j\ k}(x)\right|\left[1-\Psi^{\alpha\ \nu}_{j\ m}(x)\right]dx \ \leq \ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\ e^{\mathbf{c}|\mathbf{Im}\alpha|}\ 2^{-Nj}2^{-\varepsilon j}2^{-j[2k-1]}, \qquad N\geq 0
$$

and

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbb{R}^n}\Psi^{\alpha\ \nu}_{j\ m}(x)\left|{}^\flat\mathbf{Q}^{\alpha\ \nu}_{r\ j\ k}(x)\right|dx \ \leq \ \mathfrak{B}_{\mathbf{Re}\alpha}\ e^{\mathbf{c}|\mathbf{Im}\alpha|}\ 2^{-\varepsilon j}2^{-j[2k-1]}.
$$

From (6. 11)-(6. 12) and **Remark 6.1**, we conclude (6. 4)-(6. 5).

7 Proof of Proposition Two

Let $\mathbf{Re}\alpha > 0$ and $0 < \mathbf{Re}\beta < \frac{1}{2}$. Recall ${}^\flat\mathbf{Q}^{\alpha\,\nu}_{r\,j}(x)$ defined in (6. 7). Because of Euler's formulae, we replace the cosine function with $e^{2\pi\mathbf{i}|r\xi|}$ or $e^{-2\pi\mathbf{i}|r\xi|}$ multiplied by $\mathfrak{B}_{\mathbf{Re}\alpha}e^{\mathbf{c}|\mathbf{Im}\alpha|}$. This assertion leads us to study

$$\begin{array}{lcllcl} {}^\flat\mathbf{A}^{\alpha}_{r\,j}(x) &=& \displaystyle\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x), & {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x) &=& \displaystyle\int_{\mathbb{R}^n} e^{2\pi\mathbf{i}\left[x\cdot\xi-r|\xi|\right]}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{r|\xi|}\right)^{\frac{n-1}{2}+\alpha} d\xi, \\ {}^\flat\mathbf{B}^{\alpha}_{r\,j}(x) &=& \displaystyle\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{B}^{\alpha\,\nu}_{r\,j}(x), & {}^\flat\mathbf{B}^{\alpha\,\nu}_{r\,j}(x) &=& \displaystyle\int_{\mathbb{R}^n} e^{2\pi\mathbf{i}\left[x\cdot\xi+r|\xi|\right]}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{r|\xi|}\right)^{\frac{n-1}{2}+\alpha} d\xi \end{array} \tag{7. 1}$$

where $\lambda_{m-1} \le r < \lambda_m$ for $\lambda_m \in [2^{j-1}, 2^j]$ as before.

Note that $\mathbf{supp}\widehat{\phi} \subset \left\{\xi\in\mathbb{R}^n : \frac{1}{3} < |\xi| \le 3\right\}$ and $\mathbf{supp}\varphi^\nu_j \subset \Gamma^\nu_j = \left\{\xi\in\mathbb{R}^n : \left|\frac{\xi}{|\xi|} - \xi^\nu_j\right| < 2^{-j/2+1}\right\}$. We have

$$\begin{aligned} \left|{}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x)\right| &\le r^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]}\int_{\mathbf{supp}\widehat{\phi}\varphi^\nu_j}\left(\frac{1}{|\xi|}\right)^{\frac{n-1}{2}+\mathbf{Re}\alpha} d\xi \\ &\le \mathfrak{B}_{\mathbf{Re}\alpha}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j}2^{-\left(\frac{n-1}{2}\right)j}. \end{aligned} \tag{7. 2}$$

Recall $\Psi^{\alpha}_{j\,m}$ and $\Psi^{\alpha\,\nu}_{j\,m}$ defined in (3. 18). We find

$$\mathbf{vol}\ \mathbf{supp}\Psi^{\alpha\,\nu}_{j\,m} \lesssim 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j}. \tag{7. 3}$$

Recall **Remark 1.3**. There are at most a constant multiple of $2^{\left(\frac{n-1}{2}\right)j}$ many $\xi^\nu_j \in \mathbb{S}^{n-1}$. We have

$$\begin{aligned} &\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbb{R}^n}\Psi^{\alpha\,\nu}_{j\,m}(x)\left|{}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x)\right| dx \\ \le\ &\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}}\int_{\mathbf{supp}\Psi^{\alpha\,\nu}_{j\,m}}\left|{}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x)\right| dx \\ \le\ &\mathfrak{B}_{\mathbf{Re}\alpha}\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j}2^{-\left(\frac{n-1}{2}\right)j}\, 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j} \qquad \text{by (7. 2)-(7. 3)} \\ \le\ &\mathfrak{B}_{\mathbf{Re}\alpha}\, 2^{\left(\frac{n-1}{2}\right)j}2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j}2^{-\left(\frac{n-1}{2}\right)j}\, 2^{\left(\frac{n-1}{2}\right)j}2^{n\sigma j} \qquad \text{by } \textbf{Remark 1.3} \\ =\ &\mathfrak{B}_{\mathbf{Re}\alpha}\, 2^{-\left[\mathbf{Re}\alpha-n\sigma\right]j}. \end{aligned} \tag{7. 4}$$

Clearly, the same estimates in (7. 2)-(7. 4) apply to ${}^\flat\mathbf{B}^{\alpha}_{r\,j}(x)$. We obtain (6. 12) by choosing $n\sigma < \frac{1}{2}\mathbf{Re}\alpha$.

Lemma Two *Given $j > 0$ and $m = 1, \ldots, M$, we have*

$$\begin{aligned} \sum_{\nu:\xi_j^\nu \in \mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left| {}^\flat\mathbf{A}_{r\ j}^{\alpha\ \nu}(x) \right| \left[1 - \Psi_{j\ m}^{\alpha\ \nu}(x) \right] dx \ &\le\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-Nj} 2^{-\varepsilon j}, \qquad N \ge 0, \\ \sum_{\nu:\xi_j^\nu \in \mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left| {}^\flat\mathbf{B}_{r\ j}^{\alpha\ \nu}(x) \right| \left[1 - \Psi_{j\ m}^{\alpha\ \nu}(x) \right] dx \ &\le\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-Nj} 2^{-\varepsilon j}, \qquad N \ge 0 \end{aligned} \tag{7. 5}$$

for some $\varepsilon = \varepsilon(\mathbf{Re}\alpha) > 0$ whenever $\lambda_{m-1} \le r < \lambda_m$.

7.1 Proof of Lemma Two

Let ${}^\flat\mathbf{A}_{r\ j}^{\alpha}$, ${}^\flat\mathbf{A}_{r\ j}^{\alpha\ \nu}(x)$ defined in (7. 1). By taking $\xi \longrightarrow r^{-1}\xi$, we find

$$\begin{aligned} {}^\flat\mathbf{A}_{r\ j}^{\alpha}(x) \ &=\ \sum_{\nu:\xi_j^\nu \in \mathbb{S}^{n-1}} {}^\flat\mathbf{A}_{r\ j}^{\alpha\ \nu}(x), \\ {}^\flat\mathbf{A}_{r\ j}^{\alpha\ \nu}(x) \ &=\ r^{-n} \int_{\mathbb{R}^n} e^{2\pi \mathbf{i}\left[(r^{-1}x)\cdot\xi - |\xi| \right]} \varphi_j^\nu(\xi) \widehat{\phi}\left(r^{-1}\xi \right) \left(\frac{1}{|\xi|} \right)^{\frac{n-1}{2}+\alpha} d\xi. \end{aligned} \tag{7. 6}$$

Given $\xi_j^\nu \in \mathbb{S}^{n-1}$, $\mathbf{L}_\nu$ is an $n \times n$-orthogonal matrix with $\det \mathbf{L}_\nu = 1$ and

$$\mathbf{L}_\nu^T \xi_j^\nu = (1, 0)^T \in \mathbb{R} \times \mathbb{R}^{n-1}.$$

Let $\xi = \mathbf{L}_\nu \eta$ and $x = \mathbf{L}_\nu u$ in (7. 6). We have

$${}^\flat\mathbf{A}_{r\ j}^{\alpha\ \nu}(\mathbf{L}_\nu u) \ =\ r^{-n} \int_{\mathbb{R}^n} e^{2\pi \mathbf{i}\left[(r^{-1}u)\cdot\eta - |\eta| \right]} \varphi_j^\nu(\mathbf{L}_\nu \eta) \widehat{\phi}\left(r^{-1}\eta \right) \left(\frac{1}{|\eta|} \right)^{\frac{n-1}{2}+\alpha} d\eta. \tag{7. 7}$$

Note that φ_j^ν is defined in (3. 14). $\widehat{\phi}$ defined in (1. 9) is radially symmetric.

Remark 7.1 *Write $\eta = (\eta_1, \eta') \in \mathbb{R} \times \mathbb{R}^{n-1}$. We have $\frac{1}{3}r < \eta_1 < 3r$ and $|\eta'| \lesssim 2^{j/2}$ whenever $\varphi_j^\nu(\mathbf{L}_\nu\eta)\widehat{\phi}\left(r^{-1}\eta \right) \neq 0$.*

Consider

$$\begin{aligned} {}^\flat\mathbf{A}_{r\ j}^{\alpha\ \nu}(\mathbf{L}_\nu u) \ &=\ r^{-n} \int_{\mathbb{R}^n} e^{2\pi \mathbf{i}\left[(r^{-1}u)\cdot\eta - \eta_1 \right]} \Theta_{r\ j}^{\alpha\ \nu}(\eta) d\eta, \\ \Theta_{r\ j}^{\alpha\ \nu}(\eta) \ &=\ e^{-2\pi \mathbf{i}\left[|\eta| - \eta_1 \right]} \varphi_j^\nu(\mathbf{L}_\nu \eta) \widehat{\phi}\left(r^{-1}\eta \right) \left(\frac{1}{|\eta|} \right)^{\frac{n-1}{2}+\alpha}. \end{aligned} \tag{7. 8}$$

Remark 7.1 implies

$$\mathbf{vol}\left\{ \mathbf{supp}\Theta_{r\ j}^{\alpha\ \nu} \right\} \ \lesssim\ 2^j 2^{\left(\frac{n-1}{2}\right)j}. \tag{7. 9}$$

Recall $\lambda_{m-1} \le r < \lambda_m$ where $\lambda_m \in [2^{j-1}, 2^j] : 2^{\sigma j-1} \le \lambda_m - \lambda_{m-1} < 2^{\sigma j}$ where $0 < \sigma < \frac{1}{2}$ can be chosen sufficiently small. For brevity, write

$$\lambda \ = \ \lambda_m.$$

We aim to show

$$\left| \left(\lambda \partial_{\eta_1}\right)^N \left(\lambda^{\frac{1}{2}} \partial_{\eta'}^{\gamma}\right) \Theta^{\alpha\ \nu}_{r\ j}(\eta) \right| \ \le \ \mathfrak{B}_{N\ \gamma\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2} + \mathbf{Re}\alpha\right] j}, \qquad N \ge 0 \tag{7. 10}$$

for every multi-index γ.

1. Denote

$$\chi(\eta) \ = \ |\eta| - \eta_1, \qquad \eta_1 > 0.$$

We claim

$$\left| \partial_\eta^\gamma \chi(\eta) \right| \ \le \ \mathfrak{B}_\gamma\, 2^{-\gamma_1 j} 2^{-\left[|\gamma| - \gamma_1\right] j/2} \tag{7. 11}$$

for every multi-index γ whenever $\mathbf{L}_\nu \eta \in \Gamma_j^\nu \cap \left\{ \dfrac{1}{3} r < |\eta| \le 3r \right\}$.

Note that $\chi(\eta)$ is homogeneous of degree 1 in η: $\chi(\rho\eta) = \rho\chi(\eta)$, $\rho > 0$. We have

$$\left| \partial_\eta^\gamma \chi(\eta) \right| \ \le \ \mathfrak{B}_\gamma\, |\eta|^{1-|\gamma|} \tag{7. 12}$$

for every multi-index γ.

Indeed, $\partial_\eta^\gamma \chi(\rho\eta) = \rho^{|\gamma|} \partial_\xi^\gamma \chi(\xi)$ for $\xi = \rho\eta$. On the other hand, $\partial_\eta^\gamma \chi(\rho\eta) = \partial_\eta^\gamma \left[\rho\chi(\eta)\right] = \rho \partial_\eta^\gamma \chi(\eta)$. Choose $\rho = |\eta|^{-1}$. We find $\partial_\eta^\gamma \chi(\eta) = |\eta|^{1-|\gamma|} \partial_\xi^\gamma \chi(\xi)$ where $|\xi| = 1$.

Suppose $|\gamma| - \gamma_1 \ge 2$. From (7. 12), we find

$$\begin{aligned} \left| \partial_\eta^\gamma \chi(\eta) \right| \ &\le \ \mathfrak{B}_\gamma\, 2^{-\left(|\gamma|-1\right) j} \\ &= \ \mathfrak{B}_\gamma\, 2^{-\gamma_1 j} 2^{-\left[|\gamma| - \gamma_1 - 1\right] j} \\ &\le \ \mathfrak{B}_\gamma\, 2^{-\gamma_1 j} 2^{-\left[|\gamma| - \gamma_1\right] j/2}. \end{aligned} \tag{7. 13}$$

Suppose $|\gamma| - \gamma_1 = 0$ or 1. Denote $\eta^1 = (\eta_1, 0)^T \in \mathbb{R} \times \mathbb{R}^{n-1}$. A direct computation shows

$$\left(\nabla_\eta \chi\right)(\eta^1) \ = \ 0,$$

$$\left(\partial_{\eta_1}^N \nabla_{\eta'} \chi\right)(\eta^1) \ = \ \left(\nabla_{\eta'} \partial_{\eta_1}^N \chi\right)(\eta^1) \ = \ 0, \qquad N \ge 0.$$

By writing out the Taylor expansion of $\partial_{\eta_1}^N \chi(\eta)$ and $\nabla_{\eta'} \partial_{\eta_1}^N \chi(\eta)$ in the η'-subspace and using $\left(\nabla_{\eta'} \partial_{\eta_1}^N \chi\right)(\eta^1) = 0$, we have

$$\partial_{\eta_1}^N \chi(\eta) \ = \ \mathbf{O}\left(|\eta'|^2 |\eta|^{-N-1}\right), \qquad \nabla_{\eta'} \partial_{\eta_1}^N \chi(x, \eta) \ = \ \mathbf{O}\left(|\eta'| |\eta|^{-N-1}\right), \qquad N \ge 0. \tag{7. 14}$$

From (7. 14) and **Remark 7.1**, we conclude

$$\left| \partial_{\eta_1}^N \chi(\eta) \right| \ \le \ \mathfrak{B}_N 2^{-Nj}, \qquad \left| \nabla_{\eta'} \partial_{\eta_1}^N \chi(\eta) \right| \ \le \ \mathfrak{B}_N 2^{-Nj} 2^{-j/2}, \qquad N \ge 0.$$

2. For $\widehat{\phi}$ defined in (1. 9), we easily find

$$\left|\partial_\eta^\gamma\left\{\widehat{\phi}\left(r^{-1}\eta\right)\left(\frac{1}{|\eta|}\right)^{\frac{n-1}{2}+\alpha}\right\}\right| \leq \mathfrak{B}_{\gamma\,\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j} 2^{-|\gamma|j} \tag{7. 15}$$

for every multi-index γ.

3. Recall φ_j^ν defined in (3. 14). We have

$$\varphi_j^\nu(\mathbf{L}_\nu\eta) = \frac{\varphi\left[2^{j/2}\left|\frac{\eta}{|\eta|}-\eta_j^\nu\right|\right]}{\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}}\varphi\left[2^{j/2}\left|\frac{\eta}{|\eta|}-\eta_j^\nu\right|\right]}.$$

Observe that $\partial_\eta\varphi_j^\nu(\mathbf{L}_\nu\eta)=0$ whenever $\left|\frac{\eta}{|\eta|}-\eta_j^\nu\right|\leq 2^{-j/2}$. A direct computation shows

$$\begin{aligned}\frac{\partial}{\partial\eta_1}\frac{\eta_1}{|\eta|} &= \frac{1}{|\eta|}-\frac{\eta_1^2}{|\eta|^3} = \frac{|\eta'|^2}{|\eta|^3},\\ \frac{\partial}{\partial\eta_1}\frac{\eta_i}{|\eta|} &= -\frac{\eta_1\eta_i}{|\eta|^3}, \qquad i = 2,\ldots,n.\end{aligned} \tag{7. 16}$$

Because $\eta_1\approx 2^j$ and $\eta'\lesssim 2^{j/2}$ as shown in **Remark 7.1**, (7. 16) implies

$$\left|\frac{\partial}{\partial\eta_1}\frac{\eta}{|\eta|}\right| \approx 2^{j/2}|\eta|^{-2} \approx 2^{-j/2}|\eta|^{-1}. \tag{7. 17}$$

By carrying out the differentiation *w.r.t* $\eta=(\eta_1,\eta')$ and using (7. 17), we find

$$\left|\partial_{\eta_1}^N\partial_{\eta'}^\gamma\left[\varphi_j^\nu(\mathbf{L}_\nu\eta)\right]\right| \leq \mathfrak{B}_{N\,\gamma}\,|\eta|^{-N}2^{|\gamma|j/2}|\eta|^{-|\gamma|}, \qquad N\geq 0 \tag{7. 18}$$

for every multi-index γ.

By putting together the above estimates in **1-3**, we obtain (7. 10).

Define a differential operator

$$\mathbf{D}_\lambda f = Id+\lambda^2\partial_{\eta_1}^2 f+\lambda\Delta_{\eta'}f. \tag{7. 19}$$

Let ${}^\flat\mathbf{A}_{r\,j}^{\alpha\,\nu}(\mathbf{L}_\nu u)$ be given in (7. 8). An N-fold integration by parts *w.r.t* $\mathbf{D}_\lambda$ shows

$$\begin{aligned}{}^\flat\mathbf{A}_{r\,j}^{\alpha\,\nu}(\mathbf{L}_\nu u) &= (2\pi\mathbf{i})^{-2N}\left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\lambda\sum_{i=2}^n\left(r^{-1}u_i\right)^2\right]^{-N}\\ &\quad r^{-n}\int_{\mathbb{R}^n}e^{2\pi\mathbf{i}\left[\left(r^{-1}u\right)\cdot\eta-\eta_1\right]}\mathbf{D}_\lambda^N\Theta_{r\,j}^{\alpha\,\nu}(\eta)d\eta.\end{aligned} \tag{7. 20}$$

By using (7. 9)-(7. 10), we have

$$\left|\int_{\mathbb{R}^n}e^{2\pi\mathbf{i}\left[\left(r^{-1}u\right)\cdot\eta-\eta_1\right]}\mathbf{D}_\lambda^N\Theta_{r\,j}^{\alpha\,\nu}(\eta)d\eta\right| \leq \mathfrak{B}_{N\,\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j}2^j2^{\left(\frac{n-1}{2}\right)j}, \qquad N\geq 0. \tag{7. 21}$$

Recall $\Psi^{\alpha}_{j\,m}$, $\Psi^{\alpha\,\nu}_{j\,m}$ defined in (3. 18). We find

$$0 \le \Psi^{\alpha\,\nu}_{j\,m}(x) \le 1. \tag{7. 22}$$

Define

$$\widetilde{\Psi}^{\alpha\,\nu}_{j\,m}(x) = \begin{cases} 1, & \Psi^{\alpha\,\nu}_{j\,m}(x) = 1, \\ 0, & \Psi^{\alpha\,\nu}_{j\,m}(x) \neq 1. \end{cases} \tag{7. 23}$$

Note that $\widetilde{\Psi}^{\alpha\,\nu}_{j\,m}(\mathbf{L}_\nu u)$ is supported in the rectangle

$$\mathbf{R}^{\sigma}_{j\,\lambda} = \left\{ u \in \mathbb{R}^n \colon \lambda - 2^{\sigma j+1} \le u_1 \le \lambda + 2^{\sigma j+1},\ \ |u_i| \le 2^{\left[\frac{1}{2}+\sigma\right]j},\ \ i = 2, \ldots, n \right\}. \tag{7. 24}$$

Consider the first inequality in (7. 5). From (7. 22)-(7. 24), we have

$$\begin{aligned}
&\sum_{\nu:\xi^\nu_j \in \mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left| {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x) \right| \left[1 - \Psi^{\alpha\,\nu}_{j\,m}(x) \right] dx \\
&\le \sum_{\nu:\xi^\nu_j \in \mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left| {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x) \right| \left[1 - \widetilde{\Psi}^{\alpha\,\nu}_{j\,m}(x) \right] dx \\
&= \sum_{\nu:\xi^\nu_j \in \mathbb{S}^{n-1}} \int_{\mathbf{supp}\ 1-\widetilde{\Psi}^{\alpha\,\nu}_{j\,m}} \left| {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x) \right| dx \\
&= \sum_{\nu:\xi^\nu_j \in \mathbb{S}^{n-1}} \int_{\mathbb{R}^n \setminus \mathbf{supp} \widetilde{\Psi}^{\alpha\,\nu}_{j\,m}} \left| {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x) \right| dx \\
&= \sum_{\nu:\xi^\nu_j \in \mathbb{S}^{n-1}} \int_{\mathbb{R}^n \setminus \mathbf{R}^{\sigma}_{j\,\lambda}} \left| {}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(\mathbf{L}_\nu u) \right| du.
\end{aligned} \tag{7. 25}$$

Recall $\lambda_{m-1} \le r < \lambda$ ($\lambda = \lambda_m$) of which $\lambda \in [2^{j-1}, 2^j]$ and $2^{\sigma j-1} \le \lambda - \lambda_{m-1} < 2^{\sigma j}$. Moreover, $0 < \sigma = \sigma(\mathbf{Re}\alpha) < \frac{1}{2}$ can be chosen sufficiently small.

Let $\mathbf{R}^{\sigma}_{j\,\lambda}$ defined in (7. 24). Suppose $u_1 \notin [\lambda - 2^{\sigma j+1}, \lambda + 2^{\sigma j+1}]$. We find

$$\begin{aligned}
&\left(\lambda r^{-1}\right) u_1 - \lambda > \left[\lambda + 2^{\sigma j+1}\right] - \lambda, \ = 2^{\sigma j+1} \\
&\left(\lambda r^{-1}\right) u_1 - \lambda < \frac{\lambda}{\lambda - 2^{\sigma j}} \left[\lambda - 2^{\sigma j+1}\right] - \lambda \\
&\qquad = \lambda \left\{ \frac{\lambda - 2^{\sigma j+1}}{\lambda - 2^{\sigma j}} - 1 \right\} = \lambda \left[\frac{-2^{\sigma j+1} + 2^{\sigma j}}{\lambda - 2^{\sigma j}} \right] \\
&\qquad < -2^{\sigma j}.
\end{aligned} \tag{7. 26}$$

Recall **Remark 1.3**. There are at most a constant multiple of $2^{\left(\frac{n-1}{2}\right)j}$ many $\xi_j^\nu \in \mathbb{S}^{n-1}$. By using (7. 20)-(7. 21), we have

$$
\begin{aligned}
&\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} \int_{\mathbb{R}^n\setminus\mathbf{R}^\sigma_{j\,\lambda}} \left|{}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(\mathbf{L}_\nu u)\right| du \\
&\leq\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|} \sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} r^{-n} 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j}\, 2^j 2^{\left(\frac{n-1}{2}\right)j} \\
&\qquad \int_{\mathbb{R}^n\setminus\mathbf{R}^\sigma_{j\,\lambda}} \left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\lambda\sum_{i=2}^n\left(r^{-1}u_i\right)^2\right]^{-N} du \\
&\leq\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{\left(\frac{n-1}{2}\right)j}\, 2^{-nj} 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j}\, 2^j 2^{\left(\frac{n-1}{2}\right)j} \\
&\qquad \int_{\mathbb{R}^n\setminus\mathbf{R}^\sigma_{j\,\lambda}} \left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\lambda\sum_{i=2}^n\left(r^{-1}u_i\right)^2\right]^{-N} du \\
&=\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j} \int_{\mathbb{R}^n\setminus\mathbf{R}^\sigma_{j\,\lambda}} \left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\lambda\sum_{i=2}^n\left(r^{-1}u_i\right)^2\right]^{-N} du \\
&=\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j} r^{\frac{n-1}{2}} \qquad\qquad (7.\ 27)\\
&\quad \int_{\mathbb{R}^n\setminus\left\{\lambda-2^{\sigma j+1}\leq u_1\leq\lambda+2^{\sigma j+1},\ |u_i|\leq(2^jr^{-1})^{\frac{1}{2}}2^{\sigma j}\right\}} \left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\left(\lambda r^{-1}\right)\sum_{i=2}^n u_i^2\right]^{-N} du \\
&\qquad\qquad\qquad u_i\longrightarrow r^{\frac{1}{2}}u_i,\ i=2,\ldots,n \\
&\approx\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j} 2^{\left(\frac{n-1}{2}\right)j} \\
&\quad \int_{\mathbb{R}^n\setminus\left\{\lambda-2^{\sigma j+1}\leq u_1\leq\lambda+2^{\sigma j+1},\ |u_i|\leq(2^jr^{-1})^{\frac{1}{2}}2^{\sigma j}\right\}} \left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\left(\lambda r^{-1}\right)\sum_{i=2}^n u_i^2\right]^{-N} du \\
&\leq\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\left[\frac{n-1}{2}+\mathbf{Re}\alpha\right]j} 2^{\left(\frac{n-1}{2}\right)j} 2^{-2\sigma\left[N-n-1\right]j} \\
&\quad \int_{\mathbb{R}^n\setminus\left\{\lambda-2^{\sigma j+1}\leq |u_1|\leq\lambda+2^{\sigma j+1},\ |u_i|\leq(2^jr^{-1})^{\frac{1}{2}}2^{\sigma j}\right\}} \left[1+\left[\left(\lambda r^{-1}\right)u_1-\lambda\right]^2+\left(\lambda r^{-1}\right)\sum_{i=2}^n u_i^2\right]^{-n-1} du \\
&\qquad\qquad\qquad \text{by (7. 26) and } \lambda\approx r\approx 2^j \\
&=\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\mathbf{Re}\alpha j} 2^{-2\sigma\left[N-n-1\right]j} \int_{\mathbb{R}^n} \left[1+|u_1-r|^2+\sum_{i=2}^n u_i^2\right]^{-n-1} du.
\end{aligned}
$$

From (7. 25)-(7. 27), by taking N sufficiently large depending $\sigma=\sigma(\mathbf{Re}\alpha)>0$, we conclude

$$
\sum_{\nu:\xi_j^\nu\in\mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left|{}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}(x)\right| \left[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\right] dx\ \leq\ \mathfrak{B}_{N\ \mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\mathbf{Re}\alpha j} 2^{-Nj}, \qquad N\geq 0
$$

as desired.

Lastly, recall ${}^\flat\mathbf{B}^\alpha_{r\,j}(x) = \sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} {}^\flat\mathbf{B}^{\alpha\,\nu}_{r\,j}(x)$ from (7. 1). Write

$$
\begin{aligned}
{}^\flat\mathbf{B}^{\alpha\,\nu}_{r\,j}(x) &= \int_{\mathbb{R}^n} e^{2\pi\mathbf{i}\left[x\cdot\xi+r|\xi|\right]}\varphi^\nu_j(\xi)\widehat{\phi}(\xi)\left(\frac{1}{r|\xi|}\right)^{\frac{n-1}{2}+\alpha} d\xi \\
&= (-1)^n \int_{\mathbb{R}^n} e^{-2\pi\mathbf{i}\left[x\cdot\xi-r|\xi|\right]}\varphi^\nu_j(-\xi)\widehat{\phi}(\xi)\left(\frac{1}{r|\xi|}\right)^{\frac{n-1}{2}+\alpha} d\xi, \qquad \xi \longrightarrow -\xi.
\end{aligned}
\tag{7. 28}
$$

All regarding estimates from (7. 2) to (7. 27) remain valid if ${}^\flat\mathbf{A}^\alpha_{r\,j}$ and ${}^\flat\mathbf{A}^{\alpha\,\nu}_{r\,j}$ are replaced by ${}^\flat\mathbf{B}^\alpha_{r\,j}$ and ${}^\flat\mathbf{B}^{\alpha\,\nu}_{r\,j}$. We also have

$$
\sum_{\nu:\xi^\nu_j\in\mathbb{S}^{n-1}} \int_{\mathbb{R}^n} \left|{}^\flat\mathbf{B}^{\alpha\,\nu}_{r\,j}(x)\right|\left[1-\Psi^{\alpha\,\nu}_{j\,m}(x)\right]dx \leq \mathfrak{B}_{N\,\mathbf{Re}\alpha}\, e^{\mathbf{c}|\mathbf{Im}\alpha|}\, 2^{-\mathbf{Re}\alpha j}2^{-Nj}, \qquad N \geq 0.
$$

A Some estimates regarding Bessel functions

• For $a > -\frac{1}{2}, b \in \mathbb{R}$ and $\rho > 0$, a Bessel function $\mathbf{J}_{a+\mathbf{i}b}$ has an integral formula

$$
\mathbf{J}_{a+\mathbf{i}b}(\rho) = \frac{(\rho/2)^{a+\mathbf{i}b}}{\pi^{\frac{1}{2}}\Gamma\left(a+\frac{1}{2}+\mathbf{i}b\right)} \int_{-1}^{1} e^{\mathbf{i}\rho s}(1-s^2)^{a-\frac{1}{2}+\mathbf{i}b} ds. \tag{A. 1}
$$

• For every $a > -\frac{1}{2}, b \in \mathbb{R}$ and $\rho > 0$, we have

$$
\begin{aligned}
\mathbf{J}_{a+\mathbf{i}b}(\rho) &\sim \left(\frac{\pi\rho}{2}\right)^{-\frac{1}{2}} \cos\left[\rho-(a+\mathbf{i}b)\frac{\pi}{2}-\frac{\pi}{4}\right] \\
&+ \left(\frac{\pi\rho}{2}\right)^{-\frac{1}{2}} \sum_{k=1}^{\infty} \cos\left[\rho-(a+\mathbf{i}b)\frac{\pi}{2}-\frac{\pi}{4}\right]\mathbf{a}_k\rho^{-2k} + \sin\left[\rho-(a+\mathbf{i}b)\frac{\pi}{2}-\frac{\pi}{4}\right]\, \mathbf{b}_k\rho^{-2k+1}
\end{aligned}
\tag{A. 2}
$$

where

$$
\begin{aligned}
\mathbf{a}_k &= (-1)^k[a+\mathbf{i}b, 2k]2^{-2k}, \qquad \mathbf{b}_k = (-1)^{k+1}[a+\mathbf{i}b, 2k-1]2^{-2k+1}, \\
[a+\mathbf{i}b, m] &= \frac{\Gamma\left(\frac{1}{2}+a+\mathbf{i}b+m\right)}{m!\Gamma\left(\frac{1}{2}+a+\mathbf{i}b-m\right)}, \qquad m = 0,1,2,\ldots
\end{aligned}
\tag{A. 3}
$$

in the sense of that

$$
\begin{aligned}
&\left(\frac{d}{d\rho}\right)^{\ell}\Bigg\{\mathbf{J}_{a+\mathbf{i}b}(\rho) - \left(\frac{\pi\rho}{2}\right)^{-\frac{1}{2}} \cos\left[\rho-(a+\mathbf{i}b)\frac{\pi}{2}-\frac{\pi}{4}\right] \\
&\quad - \left(\frac{\pi\rho}{2}\right)^{-\frac{1}{2}} \sum_{k=1}^{N} \cos\left[\rho-(a+\mathbf{i}b)\frac{\pi}{2}-\frac{\pi}{4}\right]\mathbf{a}_k\rho^{-2k} + \sin\left[\rho-(a+\mathbf{i}b)\frac{\pi}{2}-\frac{\pi}{4}\right]\, \mathbf{b}_k\rho^{-2k+1}\Bigg\} \qquad \text{(A. 4)} \\
&= \mathbf{O}\left(\rho^{-2N-\frac{1}{2}}\right) \qquad N \geq 0, \qquad \ell \geq 0, \qquad \rho \longrightarrow \infty.
\end{aligned}
$$

The implied constant is bounded by $\mathfrak{B}_a e^{\mathbf{c}|b|}$.

By putting together (A. 1) and (A. 2)-(A. 4), we find a norm estimate in below.

• For every $a > -\frac{1}{2}$, $b \in \mathbb{R}$ and $\rho > 0$,

$$\left| \frac{1}{\rho^{a+\mathbf{i}b}} \mathbf{J}_{a+\mathbf{i}b}(\rho) \right| \leq \mathfrak{B}_a \left(\frac{1}{1+\rho} \right)^{\frac{1}{2}+a} e^{\mathbf{c}|b|}. \tag{A. 5}$$

Remark A.1 *(A. 5) is in fact true for every $a, b \in \mathbb{R}$ and $\rho > 0$.*

This is a consequence of using the following identity.

• For every $a, b \in \mathbb{R}$ and $\rho > 0$, we have

$$\mathbf{J}_{a-1+\mathbf{i}b}(\rho) = 2\left[\frac{a+\mathbf{i}b}{\rho}\right] \mathbf{J}_{a+\mathbf{i}b}(\rho) - \mathbf{J}_{a+1+\mathbf{i}b}(\rho). \tag{A. 6}$$

More discussion of Bessel functions can be found in the book of Watson [24].

Let $z \in \mathbb{C}$. Ω^z is a distribution defined by analytic continuation from

$$\mathbf{Re} z < 1, \qquad \Omega^z(x) = \pi^{-z}\Gamma^{-1}(1-z) \left(\frac{1}{1-|x|^2} \right)_+^z. \tag{A. 7}$$

• Ω^z can be equivalently defined by

$$\widehat{\Omega}^z(\xi) = \left(\frac{1}{|\xi|} \right)^{\frac{n}{2}-z} \mathbf{J}_{\frac{n}{2}-z}\Big(2\pi|\xi|\Big), \qquad z \in \mathbb{C}. \tag{A. 8}$$

Remark A.2 *From (A. 1), (A. 2)-(A. 4) and (A. 6), we conclude that $\widehat{\Omega}^z(\xi)$ for given $\xi \in \mathbb{R}^n$ is an analytic function of $z \in \mathbb{C}$.*

Denote $\omega_{n-2} = 2\pi^{\frac{n-1}{2}}\Gamma^{-1}\left(\frac{n-1}{2}\right)$ which is the area of $\mathbb{S}^{n-2}$. From (A. 7), we have

$$\begin{aligned}
\widehat{\Omega}^z(\xi) &= \pi^{-z}\Gamma^{-1}(1-z) \int_{|x|<1} e^{-2\pi\mathbf{i}x\cdot\xi} \left(\frac{1}{1-|x|^2} \right)^z dx \\
&= \pi^{-z}\Gamma^{-1}(1-z) \int_0^\pi \left\{ \int_0^1 e^{-2\pi\mathbf{i}|\xi|r\cos\vartheta}(1-r^2)^{-z} r^{n-1} dr \right\} \omega_{n-2} \sin^{n-2}\vartheta d\vartheta \\
&= \omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z) \int_{-1}^1 \left\{ \int_0^1 e^{2\pi\mathbf{i}|\xi|rs}(1-r^2)^{-z} r^{n-1} dr \right\} (1-s^2)^{\frac{n-3}{2}} ds \quad (-s = \cos\vartheta) \\
&= \omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z) \int_0^1 \left\{ \int_{-1}^1 e^{2\pi\mathbf{i}|\xi|rs}(1-s^2)^{\frac{n-3}{2}} ds \right\} (1-r^2)^{-z} r^{n-1} dr \\
&= \omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z) \int_0^1 \left\{ \int_{-1}^1 \cos(2\pi|\xi|rs)(1-s^2)^{\frac{n-3}{2}} ds \right\} (1-r^2)^{-z} r^{n-1} dr.
\end{aligned} \tag{A. 9}$$

Recall the Beta function identity:

$$\frac{\Gamma(z)\Gamma(w)}{\Gamma(z+w)} = \int_0^1 r^{z-1}(1-r)^{w-1}dr \tag{A. 10}$$

for every $\mathbf{Re}z > 0$ and $\mathbf{Re}w > 0$.

By writing out the Taylor expansion of the cosine function inside (A. 9), we find

$$\begin{aligned}
&\omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z)\int_0^1 \left\{\int_{-1}^1 \cos(2\pi|\xi|rs)(1-s^2)^{\frac{n-3}{2}}ds\right\}(1-r^2)^{-z}r^{n-1}dr \\
&= \omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z) \\
&\quad \sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\left\{\int_{-1}^1 s^{2k}(1-s^2)^{\frac{n-3}{2}}ds\right\}\left\{\int_0^1 r^{2k+n-1}(1-r^2)^{-z}dr\right\} \\
&= \frac{1}{2}\omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z) \\
&\quad \sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\left\{\int_0^1 t^{k+\frac{1}{2}-1}(1-t)^{\frac{n-1}{2}-1}dt\right\}\left\{\int_0^1 \rho^{k+\frac{n}{2}-1}(1-\rho)^{1-z-1}d\rho\right\} \qquad (\text{A. 11}) \\
&\qquad\qquad (t = s^2,\ \rho = r^2) \\
&= \frac{1}{2}\omega_{n-2}\pi^{-z}\Gamma^{-1}(1-z)\sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\frac{\Gamma\left(k+\frac{1}{2}\right)\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(k+\frac{n}{2}\right)}\frac{\Gamma\left(k+\frac{n}{2}\right)\Gamma(1-z)}{\Gamma\left(k+\frac{n}{2}+1-z\right)} \\
&\qquad\qquad \text{by (A. 10)} \\
&= \pi^{\frac{n-1}{2}-z}\sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\frac{\Gamma\left(k+\frac{1}{2}\right)}{\Gamma\left(k+\frac{n}{2}+1-z\right)}.
\end{aligned}$$

On the other hand, we have

$$\begin{aligned}
\left(\frac{1}{|\xi|}\right)^{\frac{n}{2}-z}\mathbf{J}_{\frac{n}{2}-z}\left(2\pi|\xi|\right) &= \pi^{\frac{n-1}{2}-z}\Gamma^{-1}\left(\tfrac{n+1}{2}-z\right)\int_{-1}^1 e^{2\pi\mathbf{i}|\xi|s}(1-s^2)^{\frac{n-1}{2}-z}ds \qquad \text{by (A. 1)} \\
&= \pi^{\frac{n-1}{2}-z}\Gamma^{-1}\left(\tfrac{n+1}{2}-z\right)\int_{-1}^1 \cos\left(2\pi|\xi|s\right)(1-s^2)^{\frac{n-1}{2}-z}ds \\
&= \pi^{\frac{n-1}{2}-z}\Gamma^{-1}\left(\tfrac{n+1}{2}-z\right)\sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\int_{-1}^1 s^{2k}(1-s^2)^{\frac{n-1}{2}-z}ds \\
&= \pi^{\frac{n-1}{2}-z}\Gamma^{-1}\left(\tfrac{n+1}{2}-z\right)\sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\int_0^1 \rho^{k+\frac{1}{2}-1}(1-\rho)^{\frac{n+1}{2}-z-1}d\rho \qquad (\rho = s^2) \\
&= \pi^{\frac{n-1}{2}-z}\sum_{k=0}^{\infty}(-1)^k\frac{(2\pi|\xi|)^{2k}}{(2k)!}\frac{\Gamma\left(k+\frac{1}{2}\right)}{\Gamma\left(k+\frac{n}{2}+1-z\right)} \qquad \text{by (A. 10).}
\end{aligned} \tag{A. 12}$$

B Fourier transform of Λ^α

We derive the formula of $\widehat{\Lambda}^\alpha(\xi,\tau)$ in (1. 8) by following the lines in p. 253 – 284, Chapter III of Gelfand and Shilov [10].

$\widehat{\mathrm{U}}^\alpha, \widehat{\mathrm{V}}^\alpha$ are distributions defined by analytic continuation from

$$\begin{aligned}\widehat{\mathrm{U}}^\alpha(\xi,\tau) &= \Gamma^{-1}(1-\alpha)\left(\frac{1}{\tau^2-|\xi|^2}\right)_-^\alpha,\\ \widehat{\mathrm{V}}^\alpha(\xi,\tau) &= \Gamma^{-1}(1-\alpha)\left(\frac{1}{\tau^2-|\xi|^2}\right)_+^\alpha, \qquad \mathbf{Re}\alpha<1.\end{aligned} \tag{B. 1}$$

Denote $\lambda(\alpha)=\frac{n+1}{2}-\alpha,\ \alpha\in\mathbb{C}$. Π^α, Λ^α are distributions defined by analytic continuation from

$$\begin{aligned}\Pi^\alpha(x,t) &= \Gamma^{-1}(1-\lambda(\alpha))\left(\frac{1}{t^2-|x|^2}\right)_-^{\lambda(\alpha)},\\ \Lambda^\alpha(x,t) &= \Gamma^{-1}(1-\lambda(\alpha))\left(\frac{1}{t^2-|x|^2}\right)_+^{\lambda(\alpha)}, \qquad \mathbf{Re}\lambda(\alpha)<1.\end{aligned} \tag{B. 2}$$

Let $a>0, b>0$ and

$$\begin{aligned}\rho(\xi,\tau,a,b) &= \left\{\left[\tau^2-|\xi|^2\right]^2+\left[a\tau^2+b|\xi|^2\right]^2\right\}^{\frac{1}{2}},\\ \cos\theta(\xi,\tau,a,b) &= \frac{\tau^2-|\xi|^2}{\rho(\xi,\tau,a,b)}.\end{aligned} \tag{B. 3}$$

For $\mathbf{Re}\alpha<\frac{n+1}{2}$ and $|\xi|\neq|\tau|$, we assert

$$\begin{aligned}\widehat{\mathrm{P}}^{\alpha\ a\ b}(\xi,\tau) &= \left[\tau^2-|\xi|^2+\mathbf{i}a\tau^2+\mathbf{i}b|\xi|^2\right]^{-\alpha}\\ &= \rho(\xi,\tau,a,b)^{-\alpha}e^{-\mathbf{i}\alpha\theta(\xi,\tau,a,b)},\\ \widehat{\mathrm{R}}^{\alpha\ a\ b}(\xi,\tau) &= \left[\tau^2-|\xi|^2-\mathbf{i}a\tau^2-\mathbf{i}b|\xi|^2\right]^{-\alpha}\\ &= \rho(\xi,\tau,a,b)^{-\alpha}e^{\mathbf{i}\alpha\theta(\xi,\tau,a,b)}.\end{aligned} \tag{B. 4}$$

$\widehat{\mathrm{P}}^{\alpha\ a\ b}$ and $\widehat{\mathrm{R}}^{\alpha\ a\ b}$ are distributions defined in $\mathbb{R}^{n+1}$ agree with (B. 4) whenever $|\xi|\neq|\tau|$. Define

$$\widehat{\mathrm{P}}^\alpha = \lim_{a\longrightarrow0,\ b\longrightarrow0}\widehat{\mathrm{P}}^{\alpha\ a\ b}, \qquad \widehat{\mathrm{R}}^\alpha = \lim_{a\longrightarrow0,\ b\longrightarrow0}\widehat{\mathrm{R}}^{\alpha\ a\ b}, \qquad \mathbf{Re}\alpha<\frac{n+1}{2}. \tag{B. 5}$$

For $\mathbf{Re}\lambda(\alpha) < \frac{n+1}{2}$ and $|x| \neq |t|$, we consider

$$\begin{aligned}
\Phi^{\alpha\, a\, b}(x,t) &= \left[t^2 - |x|^2 + \mathbf{i}at^2 + \mathbf{i}b|x|^2\right]^{-\lambda(\alpha)} \\
&= \rho(x,t,a,b)^{-\lambda(\alpha)} e^{-\mathbf{i}\lambda(\alpha)\theta(x,t,a,b)}, \\
\Psi^{\alpha\, a\, b}(x,t) &= \left[t^2 - |x|^2 - \mathbf{i}at^2 - \mathbf{i}b|x|^2\right]^{-\lambda(\alpha)} \\
&= \rho(x,t,a,b)^{-\lambda(\alpha)} e^{\mathbf{i}\lambda(\alpha)\theta(x,t,a,b)}.
\end{aligned} \tag{B. 6}$$

$\Phi^{\alpha\, a\, b}$ and $\Psi^{\alpha\, a\, b}$ are distributions defined in $\mathbb{R}^{n+1}$ agree with (B. 6) whenever $|x| \neq |t|$. Define

$$\Phi^{\alpha} = \lim_{a\longrightarrow 0,\, b\longrightarrow 0} \Phi^{\alpha\, a\, b}, \qquad \Psi^{\alpha} = \lim_{a\longrightarrow 0,\, b\longrightarrow 0} \Psi^{\alpha\, a\, b}, \qquad \mathbf{Re}\lambda(\alpha) < \tfrac{n+1}{2}. \tag{B. 7}$$

- $\widehat{\mathrm{P}}^{\alpha}$, $\widehat{\mathrm{R}}^{\alpha}$ are analytic for $\mathbf{Re}\alpha < \frac{n+1}{2}$.
- Φ^{α}, Ψ^{α} are analytic for $\mathbf{Re}\lambda(\alpha) < \frac{n+1}{2}$.

Regarding details can be found in 2.2, Chapter III of Gelfand and Shilov [10].

Define $\widehat{\mathrm{P}}^{\alpha}$ and $\widehat{\mathrm{R}}^{\alpha}$ by analytic continuation from (B. 5). Recall (B. 1) and (B. 3)-(B. 5). We find

$$\widehat{\mathrm{P}}^{\alpha} = \Gamma(1-\alpha)\left[e^{-\mathbf{i}\pi\alpha}\widehat{\mathrm{U}}^{\alpha} + \widehat{\mathrm{V}}^{\alpha}\right], \qquad \widehat{\mathrm{R}}^{\alpha} = \Gamma(1-\alpha)\left[e^{\mathbf{i}\pi\alpha}\widehat{\mathrm{U}}^{\alpha} + \widehat{\mathrm{V}}^{\alpha}\right]. \tag{B. 8}$$

Define Φ^{α} and Ψ^{α} by analytic continuation from (B. 7). Recall (B. 2), (B. 3) and (B. 6)-(B. 7). We have

$$\Phi^{\alpha} = \Gamma(1-\lambda(\alpha))\left[e^{-\mathbf{i}\pi\lambda(\alpha)}\Pi^{\alpha} + \Lambda^{\alpha}\right], \qquad \Psi^{\alpha} = \Gamma(1-\lambda(\alpha))\left[e^{\mathbf{i}\pi\lambda(\alpha)}\Pi^{\alpha} + \Lambda^{\alpha}\right]. \tag{B. 9}$$

For $\mathbf{Im}z \neq 0, \mathbf{Im}w \neq 0$ and $0 < \mathbf{Re}\lambda(\alpha) < \frac{n+1}{2}$, we assert

$$Q^{\alpha\, z\, w}(x,t) = \left\{\frac{1}{z|x|^2 + wt^2}\right\}^{\lambda(\alpha)}, \qquad (x,t) \neq (0,0).$$

Let $a > 0, b > 0$. From direct computation, we find

$$\begin{aligned}
\widehat{Q}^{\alpha\, \mathbf{i}a\, \mathbf{i}b}(\xi,\tau) &= (-\mathbf{i})^{\lambda(\alpha)} \iint_{\mathbb{R}^{n+1}} e^{-2\pi\mathbf{i}\left[x\cdot\xi + t\tau\right]} \left\{\frac{1}{a|x|^2 + bt^2}\right\}^{\lambda(\alpha)} dxdt \\
&= (-\mathbf{i})^{\lambda(\alpha)} \frac{1}{(\sqrt{a})^n \sqrt{b}} \iint_{\mathbb{R}^{n+1}} e^{-2\pi\mathbf{i}\left[x\cdot\xi/\sqrt{a} + t\tau/\sqrt{b}\right]} \left(\frac{1}{|x|^2 + t^2}\right)^{\lambda(\alpha)} dxdt \\
&\qquad\qquad x \longrightarrow x/\sqrt{a},\ t \longrightarrow t/\sqrt{b} \\
&= (-\mathbf{i})^{\lambda(\alpha)} \frac{\pi^{-\frac{n+1}{2}+2\lambda(\alpha)}}{(\sqrt{a})^n \sqrt{b}} \frac{\Gamma\left(\frac{n+1}{2} - \lambda(\alpha)\right)}{\Gamma(\lambda(\alpha))} \left\{\frac{1}{|\xi|^2/a + \tau^2/b}\right\}^{\frac{n+1}{2}-\lambda(\alpha)}, \qquad 0 < \mathbf{Re}\lambda(\alpha) < \tfrac{n+1}{2}
\end{aligned} \tag{B. 10}$$

whenever $(\xi,\tau) \neq (0,0)$.

The last equality in (B. 10) is derived by using

$$\int_{\mathbb{R}^{\mathbf{N}}} e^{-2\pi \mathbf{i} x\cdot\xi}|x|^{\gamma-\mathbf{N}}dx \;=\; \frac{\pi^{\frac{\mathbf{N}-\gamma}{2}}\Gamma\left(\frac{\gamma}{2}\right)}{\pi^{\frac{\gamma}{2}}\Gamma\left(\frac{\mathbf{N}-\gamma}{2}\right)}|\xi|^{-\gamma}, \qquad \xi \neq 0, \qquad 0 < \mathbf{Re}\gamma < \mathbf{N}.$$

Replace $a = -\mathbf{i}z$ and $b = -\mathbf{i}w$ inside (B. 10). We have

$$\begin{aligned} \widehat{Q}^{\alpha\, z\, w}(\xi,\tau) &= \iint_{\mathbb{R}^{n+1}} e^{-2\pi\mathbf{i}(x\cdot\xi+t\tau)}\left\{\frac{1}{z|x|^2+wt^2}\right\}^{\lambda(\alpha)}dxdt \\ &= \frac{\pi^{\lambda(\alpha)-\alpha}}{(\sqrt{z})^n\sqrt{w}}\frac{\Gamma(\alpha)}{\Gamma(\lambda(\alpha))}\left\{\frac{1}{|\xi|^2/z+\tau^2/w}\right\}^{\alpha}, \qquad (\xi,\tau)\neq(0,0). \end{aligned} \tag{B. 11}$$

Remark B.1 *(B. 11) is true for every* $\mathbf{Im}z > 0$ *and* $\mathbf{Im}w > 0$.

Let $w = \mathbf{i}b$ fixed. Both sides of (B. 11) are analytic whenever $\mathbf{Im}z > 0$. Moreover, they are equal at $z = \mathbf{i}a$ for every $a > 0$. By analytic continuation, this equality holds for $\mathbf{Im}z > 0$. Given z, $\mathbf{Im}z > 0$, a vice versa argument shows that (B. 11) is true for $\mathbf{Im}w > 0$.

On the other hand, we find

$$\widehat{Q}^{\alpha\, -\mathbf{i}a\, -\mathbf{i}b}(\xi,\tau) = (-\mathbf{i})^{-\lambda(\alpha)}\frac{\pi^{-\frac{n+1}{2}+2\lambda(\alpha)}}{(\sqrt{a})^n\sqrt{b}}\frac{\Gamma\left(\frac{n+1}{2}-\lambda(\alpha)\right)}{\Gamma(\lambda(\alpha))}\left\{\frac{1}{|\xi|^2/a+\tau^2/b}\right\}^{\frac{n+1}{2}-\lambda(\alpha)}, \tag{B. 12}$$

$$(\xi,\tau)\neq(0,0), \qquad 0<\mathbf{Re}\lambda(\alpha)<\tfrac{n+1}{2}$$

by carrying out the same estimate in (B. 10).

Replace $a = \mathbf{i}z$ and $b = \mathbf{i}w$ inside (B. 12), we obtain (B. 11) again. Furthermore, an analogue argument below **Remark B.1** shows (B. 11) hold for $\mathbf{Im}z < 0, \mathbf{Im}w < 0$.

Let

$$\rho(a) = \sqrt{1+a^2}, \qquad \cos\theta(a) = -1/\rho(a), \qquad \sin\theta(a) = a/\rho(a). \tag{B. 13}$$

Note that

$$(-1\pm\mathbf{i}a)^{\frac{n}{2}} = \rho(a)^{\frac{n}{2}}e^{\pm\mathbf{i}\theta\left(\frac{n}{2}\right)} \longrightarrow e^{\pm\mathbf{i}\pi\left(\frac{n}{2}\right)} \qquad \text{as} \qquad a\longrightarrow 0. \tag{B. 14}$$

Consider $z = -1 \pm \mathbf{i}a$ and $w = 1 \pm \mathbf{i}b$ inside (B. 11). We have

$$\begin{aligned} \widehat{Q}^{\alpha\, z\, w}(\xi,\tau) &\doteq \widehat{Q}^{\alpha\, a\, b}(\xi,\tau) \\ &= \iint_{\mathbb{R}^{n+1}} e^{-2\pi\mathbf{i}(x\cdot\xi+t\tau)}\left\{\frac{1}{(-1\pm\mathbf{i}a)|x|^2+(1\pm\mathbf{i}b)t^2}\right\}^{\lambda(\alpha)}dxdt \\ &= \frac{\pi^{\lambda(\alpha)-\alpha}}{(-1\pm\mathbf{i}a)^{\frac{n}{2}}(1\pm\mathbf{i}b)^{\frac{1}{2}}}\frac{\Gamma(\alpha)}{\Gamma(\lambda(\alpha))}\left\{\frac{(-1\pm\mathbf{i}a)(1\pm\mathbf{i}b)}{(1\pm\mathbf{i}b)|\xi|^2+(-1\pm\mathbf{i}a)\tau^2}\right\}^{\alpha} \end{aligned} \tag{B. 15}$$

for $(\xi,\tau)\neq(0,0)$ and $0<\mathbf{Re}\lambda(\alpha)<\frac{n+1}{2}$.

Define $\widehat{Q}^{\alpha\, a\, b}$, $a > 0, b > 0, 0 < \mathbf{Re}\lambda(\alpha) < \frac{n+1}{2}$ as a distribution in $\mathbb{R}^{n+1}$ agree with (B. 15) whenever $|\xi| \neq |\tau|$.

Recall (B. 4)-(B. 7). By using (B. 14) and taking $a \longrightarrow 0, b \longrightarrow 0$, we find

$$
\begin{aligned}
\widehat{\Phi}^{\alpha} &= \pi^{\lambda(\alpha)-\alpha} e^{-\mathbf{i}\pi\left(\frac{n}{2}\right)} \frac{\Gamma(\alpha)}{\Gamma(\lambda(\alpha))} \widehat{\mathrm{R}}^{\alpha}, \\
\widehat{\Psi}^{\alpha} &= \pi^{\lambda(\alpha)-\alpha} e^{\mathbf{i}\pi\left(\frac{n}{2}\right)} \frac{\Gamma(\alpha)}{\Gamma(\lambda(\alpha))} \widehat{\mathrm{P}}^{\alpha}, \qquad 0 < \mathbf{Re}\lambda(\alpha) < \tfrac{n+1}{2}.
\end{aligned} \tag{B. 16}
$$

Moreover, $\widehat{\mathrm{P}}^{\alpha}, \widehat{\mathrm{R}}^{\alpha}$ and $\Phi^{\alpha}, \Psi^{\alpha}$ can be given in terms of $\Pi^{\alpha}, \Lambda^{\alpha}$ and $\mathrm{U}^{\alpha}, \mathrm{V}^{\alpha}$ as (B. 8) and (B. 9). A direct computation shows

$$
\begin{aligned}
\widehat{\Lambda}^{\alpha}\left[e^{i\pi\lambda(\alpha)} - e^{-i\pi\lambda(\alpha)}\right] &= \Gamma^{-1}(1-\lambda(\alpha))\left[e^{\mathbf{i}\pi\lambda(\alpha)}\widehat{\Phi}^{\alpha} - e^{-\mathbf{i}\pi\lambda(\alpha)}\widehat{\Psi}^{\alpha}\right] \\
&= \Gamma^{-1}(1-\lambda(\alpha))\Gamma^{-1}(\lambda(\alpha)) \\
&\quad \pi^{\lambda(\alpha)-\alpha}\Gamma(\alpha)\left[e^{-\mathbf{i}\pi\left[\frac{n}{2}-\lambda(\alpha)\right]}\widehat{\mathrm{R}}^{\alpha} - e^{\mathbf{i}\pi\left[\frac{n}{2}-\lambda(\alpha)\right]}\widehat{\mathrm{P}}^{\alpha}\right]
\end{aligned} \tag{B. 17}
$$

for $0 < \mathbf{Re}\lambda(\alpha) < \frac{n+1}{2}$.

Let $\frac{1}{2} < \mathbf{Re}\alpha < 1$. Note that $\lambda(\alpha) = \frac{n+1}{2} - \alpha \notin \mathbb{Z}$. From (B. 17), by using the identity $\Gamma^{-1}(1-z)\Gamma^{-1}(z) = \frac{\sin \pi z}{z}$ for $z \in \mathbb{C}$, we obtain

$$
\begin{aligned}
\widehat{\Lambda}^{\alpha} &= \pi^{\lambda(\alpha)-\alpha-1}\Gamma(\alpha)\frac{1}{2\mathbf{i}}\left\{-e^{\mathbf{i}\pi\left[\frac{n}{2}-\lambda(\alpha)\right]}\widehat{\mathrm{P}}^{\alpha} + e^{-\mathbf{i}\pi\left[\frac{n}{2}-\lambda(\alpha)\right]}\widehat{\mathrm{R}}^{\alpha}\right\} \\
&= \pi^{\lambda(\alpha)-\alpha-1}\Gamma(\alpha)\Gamma(1-\alpha) \\
&\quad \frac{1}{2\mathbf{i}}\left\{-e^{\mathbf{i}\pi\left[\frac{n}{2}-\lambda(\alpha)\right]}\left[e^{-\mathbf{i}\pi\alpha}\widehat{\mathrm{U}}^{\alpha} + \widehat{\mathrm{V}}^{\alpha}\right] + e^{-\mathbf{i}\pi\left[\frac{n}{2}-\lambda(\alpha)\right]}\left[e^{\mathbf{i}\pi\alpha}\widehat{\mathrm{U}}^{\alpha} + \widehat{\mathrm{V}}^{\alpha}\right]\right\} \qquad \text{by (B. 8)} \\
&= \pi^{\lambda(\alpha)-\alpha-1}\Gamma(\alpha)\Gamma(1-\alpha) \\
&\quad \frac{1}{2\mathbf{i}}\left\{-e^{\mathbf{i}\pi\left[\alpha-\frac{1}{2}\right]}\left[e^{-\mathbf{i}\pi\alpha}\widehat{\mathrm{U}}^{\alpha} + \widehat{\mathrm{V}}^{\alpha}\right] + e^{-\mathbf{i}\pi\left[\alpha-\frac{1}{2}\right]}\left[e^{\mathbf{i}\pi\alpha}\widehat{\mathrm{U}}^{\alpha} + \widehat{\mathrm{V}}^{\alpha}\right]\right\} \\
&= \pi^{\lambda(\alpha)-\alpha-1}\Gamma(\alpha)\Gamma(1-\alpha)\left\{\widehat{\mathrm{U}}^{\alpha} - \sin\pi\left(\alpha-\tfrac{1}{2}\right)\widehat{\mathrm{V}}^{\alpha}\right\}.
\end{aligned} \tag{B. 18}
$$

Lastly, $\widehat{\mathrm{U}}^{\alpha}$ and $\widehat{\mathrm{V}}^{\alpha}$ agree with $\widehat{\mathrm{U}}^{\alpha}(\xi,\tau)$ and $\widehat{\mathrm{V}}^{\alpha}(\xi,\tau)$ in (B. 1) whenever $|\xi| \neq |\tau|$. We have

$$
\widehat{\Lambda}^{\alpha}(\xi,\tau) = \pi^{\lambda(\alpha)-\alpha-1}\Gamma(\alpha)\left\{\left(\frac{1}{\tau^2-|\xi|^2}\right)_-^{\alpha} - \sin\pi\left(\alpha-\tfrac{1}{2}\right)\left(\frac{1}{\tau^2-|\xi|^2}\right)_+^{\alpha}\right\}, \qquad |\xi| \neq |\tau|.
$$

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